

Modular Forms

Henry Twiss

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1 Congruence Subgroups

All sums and products over a quotient group will be consider to be over a complete set of representatives and we identify cosets with corresponding representatives. If an integer a is taken modulo m , for any positive integer m , we write $\bar{a}$ for the inverse of a . Also, for any complex z we set

$$e(z) = e^{2\pi iz}.$$

1.1 Congruence Subgroups

To define congruence subgroups, we first need to introduce an associated group. We call $\mathrm{SL}_2(\mathbb{Z})$ the **modular group** which is the group of matrices with integer entries and of determinant 1. We will distinguish two matrices

$$S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \quad \text{and} \quad T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

The first is a reflection, being of order two, while the second has infinite order. The first result usually proved about the modular group is that it is generated by S and T .

Proposition 1.1.

$$\mathrm{SL}_2(\mathbb{Z}) = \langle S, T \rangle.$$

Proof. Let n be an integer and $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z})$. Short computation show

$$S\gamma = \begin{pmatrix} -c & -d \\ a & b \end{pmatrix} \quad \text{and} \quad T^n\gamma = \begin{pmatrix} a + nc & b + nd \\ c & d \end{pmatrix}.$$

In particular, S acts by interchanging the upper-left and lower-left entries of γ up to sign and T^n acts by adding an n multiple of the lower-left entry to the upper-left entry. We will demonstrate $\gamma \in \langle S, T \rangle$ by showing that an inverse exists.

If $c = 0$ then γ is the identity since $\det(\gamma) = 1$ so suppose $c \neq 0$. Write $a = qc + r$ for some integers q and r with $|r| < |c|$. A short computation shows

$$ST^{-q}\gamma = \begin{pmatrix} -c & -d \\ r & b - qd \end{pmatrix}.$$

The upper-left entry of this matrix strictly larger than the lower-left entry in absolute value. Therefore if we repeatedly apply this argument, it must terminate with the lower-left entry vanishing. But then we have reached the identity. Therefore γ has an inverse in $\langle S, T \rangle$. $\square$

We can now introduce congruence subgroups which are special subgroups of the modular group defined by congruence conditions on their entries. We begin with principal congruence subgroups. For a positive integer N , the **principal congruence subgroup** $\Gamma(N)$ of level N is defined by

$$\Gamma(N) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z}) : \begin{pmatrix} a & b \\ c & d \end{pmatrix} \equiv \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \pmod{N} \right\}.$$

As $\Gamma(N)$ is the kernel of the homomorphism

$$\varphi : \mathrm{SL}_2(\mathbb{Z}) \rightarrow \mathrm{SL}_2(\mathbb{Z}/N\mathbb{Z}) \quad \begin{pmatrix} a & b \\ c & d \end{pmatrix} \mapsto \begin{pmatrix} a \pmod{N} & b \pmod{N} \\ c \pmod{N} & d \pmod{N} \end{pmatrix},$$

it is a normal subgroup. A subgroup Γ of the modular group is said to be a **congruence subgroup** if it contains a principal congruence subgroup. The minimal such N for which $\Gamma(N) \subseteq \Gamma$ is called the **level** of Γ . Note that if $M \mid N$ then $\Gamma(N) \subseteq \Gamma(M)$. Thus if Γ is a congruence subgroup of level N then it contains the principal congruence subgroups $\Gamma(kN)$ for all positive integers k . This implies that congruence subgroups are preserved under intersection. In fact, if Γ_1 and Γ_2 are congruence subgroups of levels N_1 and N_2 , then $\Gamma_1 \cap \Gamma_2$ is guaranteed to be a congruence subgroup of level at most $[N_1, N_2]$. Also, it turns out that the aforementioned homomorphism is surjective.

Proposition 1.2. *Let N be a positive integer. Then the homomorphism*

$$\varphi : \mathrm{SL}_2(\mathbb{Z}) \rightarrow \mathrm{SL}_2(\mathbb{Z}/N\mathbb{Z}) \quad \begin{pmatrix} a & b \\ c & d \end{pmatrix} \mapsto \begin{pmatrix} a \pmod{N} & b \pmod{N} \\ c \pmod{N} & d \pmod{N} \end{pmatrix},$$

is surjective.

Proof.

By Proposition 1.2, $[\mathrm{SL}_2(\mathbb{Z}) : \Gamma(N)] = |\mathrm{SL}_2(\mathbb{Z}/N\mathbb{Z})|$. Since $\Gamma(N) \subseteq \Gamma$ and $\Gamma(N)$ has finite index in $\mathrm{SL}_2(\mathbb{Z})$ so does Γ . Accordingly, we set

$$d_\Gamma = [\mathrm{SL}_2(\mathbb{Z}) : \Gamma].$$

The subgroups

$$\Gamma_1(N) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z}) : \begin{pmatrix} a & b \\ c & d \end{pmatrix} \equiv \begin{pmatrix} 1 & * \\ 0 & 1 \end{pmatrix} \pmod{N} \right\},$$

and

$$\Gamma_0(N) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z}) : \begin{pmatrix} a & b \\ c & d \end{pmatrix} \equiv \begin{pmatrix} * & * \\ 0 & * \end{pmatrix} \pmod{N} \right\},$$

are particularly important and are congruence subgroups of level N . The latter subgroup is called the **Hecke congruence subgroup** of level N . Note that $\Gamma(N) \subseteq \Gamma_1(N) \subseteq \Gamma_0(N)$.

Remark 1.3. If Γ is a general congruence subgroup, it is useful to find a generating set for Γ in order to reduce results about Γ to that of the generators. This is usually achieved by performing some sort of Euclidean division argument on the entries of a matrix $\gamma \in \Gamma$ using the supposed generating set to construct the inverse for γ .

Now we discuss double cosets associated to congruence subgroups. We first a useful lemma which says that congruence subgroups are preserved under conjugation by elements of $\mathrm{GL}_2^+(\mathbb{Q})$ provided we restrict to those elements in $\mathrm{SL}_2(\mathbb{Z})$:

Lemma 1.4. *Let Γ be a congruence subgroup and let $\alpha \in \mathrm{GL}_2^+(\mathbb{Q})$. Then $\alpha^{-1}\Gamma\alpha \cap \mathrm{SL}_2(\mathbb{Z})$ is a congruence subgroup.*

Proof. Recall that if Γ is of level M then $\Gamma(kM) \subseteq \Gamma$ for every $k \geq 1$. Thus there is an integer $\tilde{N}$ such that $\Gamma(\tilde{N}) \subseteq \Gamma$, $\tilde{N}\alpha \in \mathrm{GL}_2^+(\mathbb{Z})$, and $\tilde{N}\alpha \in \mathrm{GL}_2^+(\mathbb{Z})$. Now let $N = \tilde{N}^3$ and notice that any $\gamma \in \Gamma(N)$ is of the form

$$\gamma = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + N \begin{pmatrix} k_1 & k_2 \\ k_3 & k_4 \end{pmatrix},$$

for $k_1, \dots, k_4 \in \mathbb{Z}$. Therefore $\Gamma(N) \subseteq I + N\mathrm{Mat}_2(\mathbb{Z})$. Thus

$$\alpha\Gamma(N)\alpha^{-1} \subseteq \alpha(I + N\mathrm{Mat}_2(\mathbb{Z}))\alpha^{-1} = I + \tilde{N}\mathrm{Mat}_2(\mathbb{Z}).$$

As every matrix in $\alpha\Gamma(N)\alpha^{-1}$ has determinant 1 and $\Gamma(\tilde{N}) \subseteq I + \tilde{N}\mathrm{Mat}_2(\mathbb{Z})$, it follows that $\alpha\Gamma(N)\alpha^{-1} \subseteq \Gamma(\tilde{N})$. As $\Gamma(\tilde{N}) \subseteq \Gamma$, we conclude

$$\Gamma(N) \subseteq \alpha^{-1}\Gamma\alpha,$$

and intersecting with $\mathrm{SL}_2(\mathbb{Z})$ completes the proof. $\square$

Note that by Lemma 1.4, if $\alpha^{-1}\Gamma\alpha \subseteq \mathrm{SL}_2(\mathbb{Z})$ then $\alpha^{-1}\Gamma\alpha$ is a congruence subgroup if Γ is. In fact, since the principal congruence subgroups are normal in $\mathrm{SL}_2(\mathbb{Z})$ we see that $\alpha^{-1}\Gamma\alpha$ is the same level as Γ . Moreover, since congruence subgroups are preserved under intersection, Lemma 1.4 further implies that $\alpha^{-1}\Gamma_1\alpha \cap \Gamma_2$ is a congruence subgroup for any two congruence subgroups Γ_1 and Γ_2 and any $\alpha \in \mathrm{GL}_2^+(\mathbb{Q})$. In this setting, consider the double coset

$$\Gamma_1\alpha\Gamma_2 = \{\gamma_1\alpha\gamma_2 : \gamma_1 \in \Gamma_1 \text{ and } \gamma_2 \in \Gamma_2\}.$$

Note that Γ_1 acts on $\Gamma_1\alpha\Gamma_2$ by left multiplication so that it decomposes into a disjoint union of orbits. That is,

$$\Gamma_1\alpha\Gamma_2 = \bigcup_{\beta \in \Gamma_1 \backslash \Gamma_1\alpha\Gamma_2} \Gamma_1\beta.$$

It turns out that this union is finite. To see this, we need to relate $\Gamma_1 \backslash \Gamma_1\alpha\Gamma_2$ to a quotient of Γ_2 with respect to the action of some subgroup. Consider the congruence subgroup $\alpha^{-1}\Gamma_1\alpha \cap \Gamma_2$ which is clearly a subgroup of Γ_2 . As $\alpha^{-1}\Gamma_1\alpha \cap \Gamma_2$ acts on Γ_2 by left multiplication we again have a space $(\alpha^{-1}\Gamma_1\alpha \cap \Gamma_2) \backslash \Gamma_2$. We first show how these spaces are related:

Lemma 1.5. *Let Γ_1 and Γ_2 be congruence subgroups and let $\alpha \in \mathrm{GL}_2^+(\mathbb{Q})$. Then the left multiplication map*

$$\Gamma_2 \rightarrow \Gamma_1 \backslash \Gamma_1\alpha\Gamma_2 \quad \gamma_2 \mapsto \Gamma_1\alpha\gamma_2,$$

induces a bijection from $(\alpha^{-1}\Gamma_1\alpha \cap \Gamma_2) \backslash \Gamma_2$ to $\Gamma_1 \backslash \Gamma_1\alpha\Gamma_2$.

Proof. We first show that this map is $(\alpha^{-1}\Gamma_1\alpha \cap \Gamma_2)$ -invariant so that it induces a map on the quotient. Indeed, if $\gamma \in \alpha^{-1}\Gamma_1\alpha \cap \Gamma_2$ then $\gamma = \alpha^{-1}\gamma_1\alpha$ for some $\gamma_1 \in \Gamma_1$. Thus $\Gamma_1\alpha\gamma\gamma_2 = \Gamma_1\alpha\gamma_2$ proving that the map is $(\alpha^{-1}\Gamma_1\alpha \cap \Gamma_2)$ -invariant and hence induces a map on the quotient. We now show that the induced map is a bijection.

To prove injectivity, let γ_2 and γ'_2 be representatives of orbits in $(\alpha^{-1}\Gamma_1\alpha \cap \Gamma_2) \backslash \Gamma_2$ such that $\Gamma_1\alpha\gamma_2 = \Gamma_1\alpha\gamma'_2$. This implies $\alpha\gamma_2(\gamma'_2)^{-1} \in \Gamma_1\alpha$ and so $\gamma_2(\gamma'_2)^{-1} \in \alpha^{-1}\Gamma_1\alpha$. But we also have $\gamma_2(\gamma'_2)^{-1} \in \Gamma_2$ and these two facts together show that $\gamma_2(\gamma'_2)^{-1} \in \alpha^{-1}\Gamma_1\alpha \cap \Gamma_2$. Hence

$$(\alpha^{-1}\Gamma_1\alpha \cap \Gamma_2)\gamma_2 = (\alpha^{-1}\Gamma_1\alpha \cap \Gamma_2)\gamma'_2,$$

from which injectivity follows. For surjectivity, any representative β of an orbit in $\Gamma_1 \backslash \Gamma_1\alpha\Gamma_2$ is of the form $\beta = \gamma_1\alpha\gamma_2$ for some $\gamma_1 \in \Gamma_1$ and $\gamma_2 \in \Gamma_2$. Since Γ_1 is acting on $\Gamma_1\alpha\Gamma_2$ by left multiplication, β can be written as $\beta = \alpha\gamma'_2$ for some $\gamma'_2 \in \Gamma_2$. Then γ'_2 represents an orbit of $(\alpha^{-1}\Gamma_1\alpha \cap \Gamma_2) \backslash \Gamma_2$ mapping to the orbit represented by β . This shows that the induced map is surjective. $\square$

With this lemma in hand, we can prove that the disjoint union is finite:

Proposition 1.6. *Let Γ_1 and Γ_2 be congruence subgroups and let $\alpha \in \mathrm{GL}_2^+(\mathbb{Q})$. Then the disjoint union*

$$\Gamma_1\alpha\Gamma_2 = \bigcup_{\beta \in \Gamma_1 \backslash \Gamma_1\alpha\Gamma_2} \Gamma_1\beta,$$

is finite.

Proof. Observe that $\alpha^{-1}\Gamma_1\alpha \cap \Gamma_2$ acts on Γ_2 by left multiplication. By Lemma 1.5, the number of orbits of $\Gamma_1 \backslash \Gamma_1\alpha\Gamma_2$ is the same as the number of cosets of $(\alpha^{-1}\Gamma_1\alpha \cap \Gamma_2) \backslash \Gamma_2$ which is $[\Gamma_2 : \alpha^{-1}\Gamma_1\alpha \cap \Gamma_2]$. By Lemma 1.4, $\alpha^{-1}\Gamma_1\alpha \cap \mathrm{SL}_2(\mathbb{Z})$ is a congruence subgroup and hence has finite index in $\mathrm{SL}_2(\mathbb{Z})$. As $\Gamma_2 = \mathrm{SL}_2(\mathbb{Z}) \cap \Gamma_2$ and $\alpha^{-1}\Gamma_1\alpha \cap \Gamma_2 = \alpha^{-1}\Gamma_1\alpha \cap \mathrm{SL}_2(\mathbb{Z}) \cap \Gamma_2$, it follows that $[\Gamma_2 : \alpha^{-1}\Gamma_1\alpha \cap \Gamma_2] \leq d_{\alpha^{-1}\Gamma_1\alpha \cap \mathrm{SL}_2(\mathbb{Z})}$ completing the proof. $\square$

1.2 Modular Curves and Fundamental Domains

Recall that $\mathrm{GL}_2^+(\mathbb{R})$ acts on the Riemann sphere $\hat{\mathbb{C}}$ by Möbius transformations. Explicitly, any $\alpha = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{GL}_2^+(\mathbb{R})$ acts on $z \in \hat{\mathbb{C}}$ by

$$\alpha z = \frac{az + b}{cz + d},$$

where $\alpha\infty = \frac{a}{c}$ and $\alpha(-\frac{d}{c}) = \infty$. Moreover, recall that this action is a group action, is invariant under scalar multiplication, and acts by automorphisms of $\hat{\mathbb{C}}$. Also note that α and $-\alpha$ act the same on $\hat{\mathbb{C}}$. Now observe

$$\mathrm{Im}(\alpha z) = \mathrm{Im}\left(\frac{az + b}{cz + d}\right) = \mathrm{Im}\left(\frac{ac|z|^2 + adz + bc\bar{z} + bd}{|cz + d|^2}\right) = \det(\alpha) \frac{\mathrm{Im}(z)}{|cz + d|^2}, \quad (1)$$

where in the last equality we have used that $\mathrm{Im}(\bar{z}) = -\mathrm{Im}(z)$ and that $\det(\alpha) = ad - bc$. Since $\det(\alpha) > 0$ and $|cz + d|^2 > 0$, α preserves the sign of the imaginary part of z . So α preserves the upper half-space $\mathbb{H}$, the lower half-space $\overline{\mathbb{H}}$, and the extended real line $\hat{\mathbb{R}}$ respectively. Also note that α restricts to an automorphism on these subspaces since Möbius transformations are automorphisms. In particular, $\mathrm{SL}_2(\mathbb{R})$ acts on $\hat{\mathbb{C}}$ by Möbius transformations and preserves $\mathbb{H}$, $\hat{\mathbb{R}}$, and $\overline{\mathbb{H}}$ respectively. For $\alpha \in \mathrm{SL}_2(\mathbb{R})$, Equation (1) becomes the simplified identity

$$\mathrm{Im}(\alpha z) = \frac{\mathrm{Im}(z)}{|cz + d|^2}. \quad (2)$$

In particular, $\mathrm{SL}_2(\mathbb{Z})$ acts on $\mathbb{C}$ by Möbius transformations and preserves $\mathbb{H}$, $\hat{\mathbb{R}}$, and $\overline{\mathbb{H}}$ respectively. Actually, as any $\gamma \in \mathrm{SL}_2(\mathbb{Z})$ has integer entries, it also follows that $\mathrm{SL}_2(\mathbb{Z})$ preserves the extended rationals $\hat{\mathbb{Q}}$. More generally, certain actions of subgroups of $\mathrm{SL}_2(\mathbb{R})$ also play important roles. A **Fuchsian group** is any discrete subgroup of $\mathrm{SL}_2(\mathbb{R})$ that acts properly discontinuously on $\mathbb{H}$:

Proposition 1.7. *The modular group is a Fuchsian group.*

Proof. $\mathrm{SL}_2(\mathbb{Z})$ is discrete in $\mathrm{SL}_2(\mathbb{R})$ because $\mathbb{Z}$ is discrete in $\mathbb{R}$ (recall that the topology on $\mathrm{SL}_2(\mathbb{R})$ is the subset topology of the metric topology on $\mathbb{R}^4$). So it remains to show that $\mathrm{SL}_2(\mathbb{Z})$ acts properly discontinuously on $\mathbb{H}$. Let $z \in \mathbb{H}$ and $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z})$. Then Equation (2) gives

$$\mathrm{Im}(\gamma z) = \frac{y}{(cx + d)^2 + (cy)^2}.$$

Since $\det(\gamma) = 1$ we cannot have $c = d = 0$. As $y \neq 0$, this chain of equalities shows that $|cz + d|^2$ is bounded away from zero and there are finitely many pairs (c, d) such that

$|cz + d|^2 \leq M$ for any $M > 0$. In particular, there are finitely many $\gamma \in \mathrm{SL}_2(\mathbb{Z})$ for which $\mathrm{Im}(\gamma z) \leq \mathrm{Im}(z) + 1$ say. Since there are only finitely many such γ , we can find a sufficiently small neighborhood U_z about z such that $\gamma U_z \cap U_z$ is empty except for $\gamma = I$. This proves $\mathrm{SL}_2(\mathbb{Z})$ acts properly discontinuously on $\mathbb{H}$ which completes the proof. $\square$

Note that Proposition 1.7 immediately implies that any subgroup of the modular group is also Fuchsian. In particular, all congruence subgroups are Fuchsian. A **modular curve** is a quotient $\Gamma \backslash \mathbb{H}$ of the upper half-space $\mathbb{H}$ by the action of a congruence subgroup Γ . Since Γ is Fuchsian and acts on $\mathbb{H}$ by automorphisms ?? implies that $\Gamma \backslash \mathbb{H}$ is also connected Hausdorff (recall that $\mathbb{H}$ is connected Hausdorff). In particular, $\Gamma \backslash \mathbb{H}$ admits the fundamental domain

$$\mathcal{F} = \left\{ z \in \mathbb{H} : |\mathrm{Re}(z)| \leq \frac{1}{2} \text{ and } |z| \geq 1 \right\},$$

as the following proposition shows:

Proposition 1.8. *$\mathcal{F}$ is a fundamental domain for $\mathrm{SL}_2(\mathbb{Z})$.*

Proof. Recall from Proposition 1.1 that $\mathrm{SL}_2(\mathbb{Z}) = \langle S, T \rangle$. Clearly $\mathcal{F}$ is closed. We first show any point in $\mathbb{H}$ is $\mathrm{SL}_2(\mathbb{Z})$ -equivalent to a point in $\mathcal{F}$. So let $z \in \mathbb{H}$ and $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z})$. Then Equation (2) gives

$$\mathrm{Im}(\gamma z) = \frac{y}{(cx + d)^2 + (cy)^2}.$$

Since $\det(\gamma) = 1$ we cannot have $c = d = 0$. As $y \neq 0$, this chain of equalities shows that $|cz + d|^2$ is bounded away from zero and there are finitely many pairs (c, d) such that $|cz + d|^2 \leq M$ for any $M > 0$. Therefore there exists $\gamma_0 \in \mathrm{SL}_2(\mathbb{Z})$ that minimizes $|cz + d|^2$ and hence maximizes $\mathrm{Im}(\gamma_0 z)$. In particular,

$$\mathrm{Im}(S\gamma_0 z) = \frac{\mathrm{Im}(\gamma_0 z)}{|\gamma_0 z|^2} \leq \mathrm{Im}(\gamma_0 z).$$

The inequality above implies $|\gamma_0 z| \geq 1$. Since $\mathrm{Im}(T^n \gamma_0 z) = \mathrm{Im}(\gamma_0 z)$ for all $n \in \mathbb{Z}$, repeating the argument above with $T^n \gamma_0$ in place of γ_0 , we see that $|T^n \gamma_0 z| \geq 1$. But T shifts the real part by 1 so we can choose n such that $|\mathrm{Re}(T^n \gamma_0 z)| \leq \frac{1}{2}$. Therefore $T^n \gamma_0 \in \mathrm{SL}_2(\mathbb{Z})$ sends z into $\mathcal{F}$ as desired. We will now show that no two distinct points in $\mathcal{F}$ are $\mathrm{SL}_2(\mathbb{Z})$ -equivalent. Since $\mathrm{SL}_2(\mathbb{Z})$ acts on $\mathbb{H}$ by automorphisms, by Proposition 1.1 it suffices to show that S and T map the interior of $\mathcal{F}$ outside of $\mathcal{F}$. This is clear for T since it maps the left boundary line $\{z \in \mathbb{H} : \mathrm{Re}(z) = -\frac{1}{2} \text{ and } |z| \geq 1\}$ to the right boundary line $\{z \in \mathbb{H} : \mathrm{Re}(z) = \frac{1}{2} \text{ and } |z| \geq 1\}$ and every other point of $\mathcal{F}$ is mapped to the right of this line. For S , note that it maps the semicircle $\{z \in \mathbb{H} : |z| = 1\}$ to itself (although not identically) and maps ∞ to zero. Since Möbius transformations send circles to circles and lines to lines it follows that every interior point of $\mathcal{F}$ is taken to a point in the interior of the region enclosed by the semicircle $\{z \in \mathbb{H} : |z| = 1\}$. This finishes the proof. $\square$

The region $\mathcal{F}$, shaded in Figure 1, is called the **standard fundamental domain**.

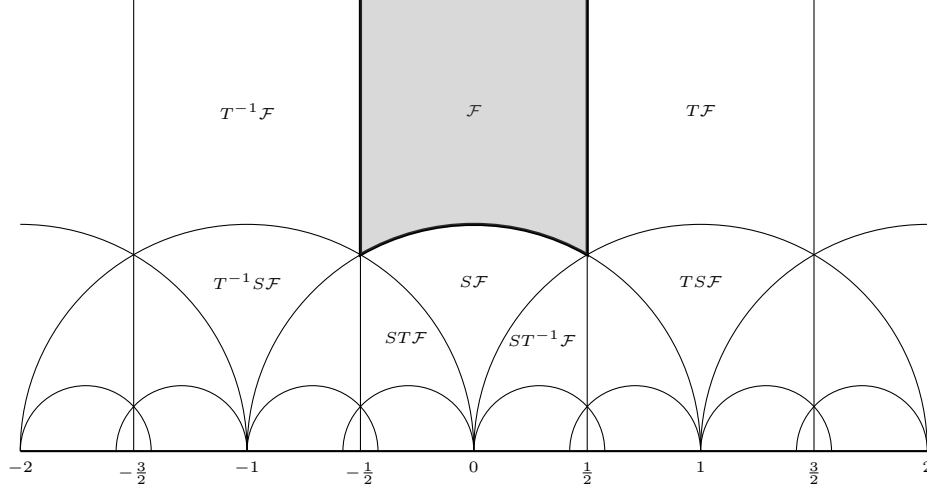


Figure 1: The standard fundamental domain for $\mathrm{SL}_2(\mathbb{Z}) \backslash \mathbb{H}$.

Remark 1.9. Any $z \in \mathcal{F}$ satisfies $\mathrm{Im}(z) \geq \frac{\sqrt{3}}{2}$. To see this, observe that any $z \in \mathcal{F}$ occurs on the arc of a circle $|z| = r$ for some $r \geq 1$ subject to $|\mathrm{Re}(z)| \leq \frac{1}{2}$. Letting $z = x + y$, y is minimized on the arc when $r = 1$. That is, when $\sqrt{x^2 + y^2} = 1$ for $|x| \leq \frac{1}{2}$. Solving for y gives $y = \sqrt{1 - x^2}$ which is minimized at $x = \pm \frac{1}{2}$ giving $y = \frac{\sqrt{3}}{2}$. This corresponds to the points $-\frac{1}{2} + \frac{\sqrt{3}}{2}$ and $\frac{1}{2} + \frac{\sqrt{3}}{2}$ which are ω and $1 + \omega$ where $\omega = e^{\frac{2\pi i}{3}}$. They are the vertices of $\mathcal{F}$.

Figure 1 also displays how this fundamental domain changes under the actions of the generators of $\mathrm{SL}_2(\mathbb{Z})$ as in Proposition 1.1. A fundamental domain for any other modular curve can be built from the standard fundamental domain using representatives of orbits in $\Gamma \backslash \mathrm{SL}_2(\mathbb{Z})$ as the following proposition shows:

Proposition 1.10. *Let Γ be any congruence subgroup. Then*

$$\mathcal{F}_\Gamma = \bigcup_{\gamma \in \Gamma \backslash \mathrm{SL}_2(\mathbb{Z})} \gamma \mathcal{F},$$

is a fundamental domain for $\Gamma \backslash \mathbb{H}$.

Proof. Clearly $\mathcal{F}_\Gamma$ is closed because the $\gamma \mathcal{F}$ are (as $\mathcal{F}$ is). We first show that every point in $\mathbb{H}$ is Γ -equivalent to a point in $\mathcal{F}_\Gamma$. From the definition of $\mathcal{F}_\Gamma$, we have

$$\bigcup_{\gamma' \in \Gamma} \gamma' \mathcal{F}_\Gamma = \bigcup_{\gamma' \in \Gamma} \bigcup_{\gamma \in \Gamma \backslash \mathrm{SL}_2(\mathbb{Z})} \gamma' \gamma \mathcal{F} = \bigcup_{\gamma \in \mathrm{SL}_2(\mathbb{Z})} \gamma \mathcal{F} = \mathbb{H},$$

where the second equality holds because $\gamma' \gamma$ runs over all the elements of $\mathrm{SL}_2(\mathbb{Z})$ (as γ' and γ run over Γ and $\Gamma \backslash \mathrm{SL}_2(\mathbb{Z})$ respectively) and the last equality holds because $\mathcal{F}$ is the standard fundamental domain. This shows that every point in $\mathbb{H}$ is Γ -equivalent to a point in $\mathcal{F}_\Gamma$. We will now show that no two distinct interior points of $\mathcal{F}_\Gamma$ are Γ -equivalent. Suppose, to the contrary, that $z_1, z_2 \in \mathcal{F}_\Gamma$ are distinct interior points which are Γ -equivalent. In particular,

$z_1 \in \gamma_1 \mathcal{F}$ and $z_2 \in \gamma_2 \mathcal{F}$ for some $\gamma_1, \gamma_2 \in \Gamma \backslash \mathrm{SL}_2(\mathbb{Z})$ and there is a $\gamma \in \Gamma$ such that $\gamma z_1 = z_2$. Then γ_1 and γ_2 belong to the same orbit in $\Gamma \backslash \mathrm{SL}_2(\mathbb{Z})$. Whence $\gamma_1 = \gamma_2$ and implying $z_1, z_2 \in \gamma_1 \mathcal{F} \cap \gamma_2 \mathcal{F}$. As they are interior points of $\mathcal{F}_\Gamma$ and $\mathcal{F}$ is the standard fundamental domain, this forces γ to be the identity so that $z_1 = z_2$ which is a contradiction. Hence no two distinct interior points of $\mathcal{F}_\Gamma$ are Γ -equivalent. This completes the proof. $\square$

It is now possible to say something about the stabilizer subgroups of points in $\mathbb{H}$ with respect to the action of a congruence subgroup Γ . Let Γ_z denote the stabilizer subgroup of $z \in \mathbb{H}$. Then $\Gamma_{\gamma z} = \gamma \Gamma_z \gamma^{-1}$ for any $\gamma \in \Gamma$ and it follows that all of the stabilizer subgroups are determined by those for $z \in \mathcal{F}_\Gamma$. Moreover, since $\mathcal{F}_\Gamma$ is a fundamental domain, Γ_z is trivial if z is in the interior of $\mathcal{F}_\Gamma$. So any point with a nontrivial stabilizer subgroups must lie on the boundary of $\mathcal{F}_\Gamma$. We can say more. Since Γ is a subgroup of $\mathrm{SL}_2(\mathbb{Z})$, Γ_z is a subgroup of $(\mathrm{SL}_2(\mathbb{Z}))_z$. We now determine the stabilizer subgroups of the modular group:

Proposition 1.11. *Let $z \in \mathcal{F}$. Then*

$$(\mathrm{SL}_2(\mathbb{Z}))_z = \begin{cases} \langle S \rangle & \text{if } z = i, \\ \langle ST \rangle & \text{if } z = \omega, \\ \langle TS \rangle & \text{if } z = 1 + \omega, \\ \langle -I \rangle & \text{if } z \neq i, \omega, 1 + \omega, \end{cases}$$

where $\omega = e^{\frac{2\pi i}{3}}$. Moreover, $(\mathrm{SL}_2(\mathbb{Z}))_i$ is of order 4 while $(\mathrm{SL}_2(\mathbb{Z}))_\omega$ and $(\mathrm{SL}_2(\mathbb{Z}))_{1+\omega}$ are of order 6.

Proof. Let $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z})$. Now $\frac{az+b}{cz+d} = z$ implies

$$cz^2 + (d - a)z - b = 0. \quad (3)$$

Taking $z = i$ in Equation (3) gives

$$-c + (d - a)i - b = 0.$$

Equating the real and imaginary parts shows that $b = -c$ and $d = a$. Then $\gamma = \begin{pmatrix} a & -c \\ c & a \end{pmatrix}$ and $a^2 + c^2 = 1$. Therefore we have $a = 1$ and $c = 0$, $a = -1$ and $c = 0$, $a = 0$ and $c = 1$, or $a = 0$ and $c = -1$. These four cases correspond to the matrices I , $-I$, S , and $-S$ respectively. Since $S^2 = -I$, it follows that $(\mathrm{SL}_2(\mathbb{Z}))_i = \langle S \rangle$ and is of order 4. Taking $z = \omega$ in Equation (3) instead gives

$$c\omega^2 + (d - a)\omega - b = 0.$$

Recall tht $m_\omega(x) = x^2 + x + 1$ is the minimal polynomial for ω over $\mathbb{Q}$ (its roots are ω and ω^2 since the sum of all n -th roots of unity is zero). So $\omega^2 + \omega + 1 = 0$ which is to say $\omega^2 = -\omega - 1$. Substituting this identity into the one above yields

$$(d - a - c)\omega - (b + c) = 0.$$

By minimality of $m_\omega(x)$, ω and 1 are linearly independent over $\mathbb{Q}$ so we may equate coefficients to see that $d = a + c$ and $b = -c$. Then $\gamma = \begin{pmatrix} a & -c \\ c & a+c \end{pmatrix}$ and $a^2 + ac + c^2 = 1$. Completing

the square shows $(a + c)^2 - ac = 1$. Therefore we have $a = 1$ and $c = 0$, $a = -1$ and $c = 0$, $a = 0$ and $c = 1$, $a = 0$ and $c = -1$, $a = -1$ and $c = 1$, or $a = 1$ and $c = -1$. These six cases correspond to the matrices I , $-I$, ST , $(ST)^4$, $(ST)^2$, and $(ST)^5$ respectively. As $(ST)^3 = -I$, it follows that $(\mathrm{SL}_2(\mathbb{Z}))_\omega = \langle ST \rangle$ and is of order 6. Now suppose $z = 1 + \omega$. Then $z = T\omega$ so that $(\mathrm{SL}_2(\mathbb{Z}))_{1+\omega} = T(\mathrm{SL}_2(\mathbb{Z}))_\omega T^{-1} = \langle TS \rangle$ and is of order 6. It remains to show that if $z \neq i, \omega, 1 + \omega$ then $(\mathrm{SL}_2(\mathbb{Z}))_z = \langle I \rangle$. Equivalently, will show the contrapositive which is to say that if $(\mathrm{SL}_2(\mathbb{Z}))_z \neq \langle -I \rangle$, then $z = i, \omega, 1 + \omega$. Moreover, if z lies in the interior of $\mathcal{F}$ then its stabilizer subgroup is necessarily trivial since $\mathcal{F}$ is a fundamental domain. Therefore we may additionally assume z lies on the boundary of $\mathcal{F}$. Now suppose for such z , $\gamma \neq \pm I$. We first claim $c > 0$. For if $c = 0$, $a = d = \pm 1$ and thus $\gamma = \begin{pmatrix} \pm 1 & b \\ 0 & \pm 1 \end{pmatrix}$. But then $z \pm b = z$ which is only true if $b = 0$. This would imply $\gamma = \pm I$ which is a contradiction. Hence $c \neq 0$ as claimed. It follows that z satisfies a quadratic equation with integer coefficients given in Equation (3). The discriminant is

$$(d - a)^2 + 4bc = a^2 - 2ad + d^2 - 4bc = a^2 + 2ad + d^2 - 4 = (a + d)^2 - 4,$$

where we have used the fact that $ad - bc = 1$. Since this quadratic has integer coefficients and z is complex, the discriminant must be negative. So

$$(a + d)^2 - 4 < 0,$$

which implies that $a + d = 0, 1, -1$. Now suppose $a + d = 0$. Then $d = -a$, the discriminant is -4 , and the quadratic formula gives

$$z = \frac{2a \pm \sqrt{-4}}{2c} = \frac{a \pm i}{c}.$$

As $\mathrm{Im}(z) \geq \frac{\sqrt{3}}{2}$ (by Remark 1.9) and $\mathrm{Im}(z) = \pm \frac{1}{c}$, we must have $c = \pm 1$. But then $\mathrm{Re}(z) = a$, and as z belongs to the boundary of $\mathcal{F}$, this forces $a = 0$. Hence $z = i$. Now suppose $a + d = 1$. Then $d = 1 - a$, the discriminant is -3 , and the quadratic formula gives

$$z = \frac{2a - 1 \pm \sqrt{-3}}{2c} = \frac{2a - 1 \pm \sqrt{3}i}{2c}.$$

As $\mathrm{Im}(z) \geq \frac{3}{2}$ (by Remark 1.9) and $\mathrm{Im}(z) = \pm \frac{\sqrt{3}}{2c}$, we must have $c = \pm 1$. But then $\mathrm{Re}(z) = \pm(a - \frac{1}{2})$, and as z belongs to the boundary of $\mathcal{F}$, this forces $a = 0, 1$. Hence $z = \omega, 1 + \omega$. Now suppose $a + d = -1$. Then $d = -1 - a$, the discriminant is -3 , and the quadratic formula gives

$$z = \frac{2a + 1 \pm \sqrt{-3}}{2c} = \frac{2a + 1 \pm \sqrt{3}i}{2c}.$$

As $\mathrm{Im}(z) \geq \frac{3}{2}$ (by Remark 1.9) and $\mathrm{Im}(z) = \pm \frac{\sqrt{3}}{2c}$, we must have $c = \pm 1$. But then $\mathrm{Re}(z) = \pm(a + \frac{1}{2})$, and as z belongs to the boundary of $\mathcal{F}$, this forces $a = 0, -1$. Hence $z = 1 + \omega, \omega$. This completes the proof. $\square$

It follows from Proposition 1.11 and our previous remarks that Γ_z is necessarily trivial if z is not $\mathrm{SL}_2(\mathbb{Z})$ -equivalent to i , ω , or $1 + \omega$. Moreover, if z is $\mathrm{SL}_2(\mathbb{Z})$ -equivalent to i , ω , or $1 + \omega$ respectively, then Γ_z is trivial unless at least one of the matrices $\pm S$, $\pm ST$, or $\pm TS$ belongs to Γ respectively. In the first case Γ_z is of order dividing 4 while in the second and third cases it is of order dividing 6. In particular, Γ_z is always finite.

1.3 Cusps

Let Γ be a congruence subgroup. Notice that $\mathcal{F}_\Gamma$ is not compact as it doesn't contain the point ∞ of points of $\mathbb{Q}$. (see Figure 1 and Proposition 1.10). However, if we consider $\mathcal{F}_\Gamma \cup \hat{\mathbb{Q}}$ then it would appear that this space is compact. The elements of $\hat{\mathbb{Q}}$ correspond to cusps which we now define. Note that any $\gamma \in \Gamma$ preserves $\hat{\mathbb{Q}}$ (because $\mathrm{SL}_2(\mathbb{Z})$ preserves $\hat{\mathbb{Q}}$). Therefore Γ acts on $\hat{\mathbb{Q}}$ by Möbius transformations. A **cuspidal class** of $\Gamma \backslash \mathbb{H}$ is a class of $\Gamma \backslash \hat{\mathbb{Q}}$. There is only one cuspidal class for the modular group since its action is transitive. Indeed, $\hat{\mathbb{Q}}$ is exactly the orbit of ∞ . This is clear for ∞ . If $\frac{a}{c} \in \mathbb{Q}$ with $(a, c) = 1$, Bézout's identity provides the existence of integers b and d such that $ad - bc = 1$. Then $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z})$ is such that $\gamma\infty = \frac{a}{c}$. As Γ has finite index in the modular group and $\hat{\mathbb{Q}}$ are all $\mathrm{SL}_2(\mathbb{Z})$ -equivalent (as we have just shown), there can only be finitely many cuspidal classes and the number of such is at most the index of Γ . In particular, the orbit of ∞ is a cuspidal class of $\Gamma \backslash \mathbb{H}$. A **cusp** is a representative of a cuspidal class. We denote cusps by gothic characters $\mathfrak{a}, \mathfrak{b}, \mathfrak{c}, \dots$.

Remark 1.12. It turns out that cusps can be represented as the points needed to make a fundamental domain $\mathcal{F}_\Gamma$ compact as a subset of $\hat{\mathbb{C}}$. To see this, suppose $\mathfrak{a}$ is a limit point of $\mathcal{F}_\Gamma$ that does not belong to $\mathcal{F}_\Gamma$. Then $\mathfrak{a} \in \hat{\mathbb{R}}$. As $\mathcal{F}_\Gamma$ is a union of images of $\mathcal{F}$ by Proposition 1.10 and since $\mathrm{SL}_2(\mathbb{Z})\infty = \hat{\mathbb{Q}}$, we find that $\mathfrak{a} \in \hat{\mathbb{Q}}$ which means $\mathfrak{a}$ is a cusp. In the case of the standard fundamental domain $\mathcal{F}$, the only cusp is ∞ .

As Γ acts on $\hat{\mathbb{Q}}$, let $\Gamma_{\mathfrak{a}}$ denote the stabilizer subgroup of the cusp $\mathfrak{a}$. For the ∞ cusp, we can describe Γ_∞ explicitly. If $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma$ stabilizes ∞ then necessarily $c = 0$ and since $\det(\gamma) = 1$ we must have $a = d = 1$. Therefore $\gamma = \begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix}$ for some $b \in \mathbb{Z}$ and γ acts on $\mathbb{H}$ by translation by b . Let $h_\infty \geq 1$ be the smallest positive integer such that $\gamma_\infty = \begin{pmatrix} 1 & h_\infty \\ 0 & 1 \end{pmatrix} \in \Gamma_\infty$. Then Γ_∞ is an infinite cyclic group generated by γ_∞ . Now, as discussed above, let $\sigma_{\mathfrak{a}} \in \mathrm{SL}_2(\mathbb{Z})$ be such that $\sigma_{\mathfrak{a}}\infty = \mathfrak{a}$. We call $\sigma_{\mathfrak{a}}$ a **scaling matrix** for $\mathfrak{a}$. Note that $\sigma_{\mathfrak{a}}$ is determined up to composition on the right by an element of Γ_∞ . If $\mathfrak{a} = \infty$, then we may take $\sigma_{\mathfrak{a}} = I$ and in this case we suppress the dependence of the scaling matrix. Scaling matrices are useful because they allow us to transfer information at the cusp $\mathfrak{a}$ to the cusp at ∞ .

Remark 1.13. Suppose $\mathfrak{a}'$ is a cusp that belongs to the same cuspidal class at $\mathfrak{a}$. Then $\mathfrak{a}' = \gamma\mathfrak{a}$ for some $\gamma \in \Gamma$. It follows that $\gamma\sigma_{\mathfrak{a}}$ is a scaling matrix for the cusp $\mathfrak{a}'$. So $\Gamma\sigma_{\mathfrak{a}}\Gamma_\infty$ only depends upon the cuspidal class of $\mathfrak{a}$. Moreover, $\Gamma_{\mathfrak{a}'} = \gamma\Gamma_{\mathfrak{a}}\gamma^{-1}$ and $\Gamma_{\mathfrak{a}'} = \gamma\sigma_{\mathfrak{a}}\Gamma_\infty(\gamma\sigma_{\mathfrak{a}})^{-1} \cap \Gamma$.

If $\sigma_{\mathfrak{a}}$ is a scaling matrix for the cusp $\mathfrak{a}$, then $\Gamma_{\mathfrak{a}} = \sigma_{\mathfrak{a}}\Gamma_\infty\sigma_{\mathfrak{a}}^{-1} \cap \Gamma$ and therefore $\Gamma_{\mathfrak{a}}$ is an infinite cyclic group as well. It follows that the generator $\gamma_{\mathfrak{a}}$ of $\Gamma_{\mathfrak{a}}$ is of the form

$$\gamma_{\mathfrak{a}} = \sigma_{\mathfrak{a}} \begin{pmatrix} 1 & h_{\mathfrak{a}} \\ 0 & 1 \end{pmatrix} \sigma_{\mathfrak{a}}^{-1},$$

for some $h_{\mathfrak{a}} \geq 1$. Accordingly, we call $h_{\mathfrak{a}}$ the **width** of the cusp $\mathfrak{a}$. Equivalently, $h_{\mathfrak{a}}$ is the shift of the smallest translation belonging to $\sigma_{\mathfrak{a}}^{-1}\Gamma\sigma_{\mathfrak{a}}$. Also observe that $h_{\mathfrak{a}}$ does not depend on the choice of scaling matrix. We will set

$$\gamma_\infty^{(h_{\mathfrak{a}})} = \begin{pmatrix} 1 & h_{\mathfrak{a}} \\ 0 & 1 \end{pmatrix} \quad \text{and} \quad \Gamma_\infty^{(h_{\mathfrak{a}})} = \left\{ \begin{pmatrix} 1 & nh_{\mathfrak{a}} \\ 0 & 1 \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z}) : n \in \mathbb{Z} \right\},$$

so that

$$\Gamma_\infty^{(h_a)} = \langle \gamma_\infty^{(h_a)} \rangle \quad \text{and} \quad \Gamma_a = \sigma_a \Gamma_\infty^{(h_a)} \sigma_a^{-1}.$$

In particular, $\Gamma_\infty^{(h_\infty)} = \Gamma_\infty$ and $\gamma_\infty^{(h_\infty)} = \gamma_\infty$. If Γ is of level N then N is the smallest positive integer such that $\Gamma(N) \subseteq \Gamma$ so that $\begin{pmatrix} 1 & N \\ 0 & 1 \end{pmatrix}$ is the smallest translation guaranteed to belong to Γ . However, it is not necessarily the smallest translation in general. As every nontrivial subgroup of an infinite cyclic group is infinite cyclic, it follows that $h_a \mid N$. We say Γ is **reduced** at $\mathfrak{a}$ if $h_a = 1$. For example, $\Gamma(N)$, $\Gamma_1(N)$, and $\Gamma_0(N)$ are all reduced at ∞ . The width of the cusps are related to the index d_Γ of Γ in $\mathrm{SL}_2(\mathbb{Z})$ in the following manner:

Proposition 1.14. *For any congruence subgroup Γ and any cusp $\mathfrak{a}$ of $\Gamma \backslash \mathbb{H}$, there are h_a many cosets of $\Gamma \backslash \mathrm{SL}_2(\mathbb{Z})$ whose elements are scaling matrices for cusps in the cusp class of $\mathfrak{a}$. In particular,*

$$\sum_{\mathfrak{a} \in \Gamma \backslash \hat{\mathbb{Q}}} h_a = d_\Gamma.$$

Proof. Let σ_a be a scaling matrix for a cusp $\mathfrak{a}$. It follows from Remark 1.13 that the cusp classes of $\Gamma \backslash \mathbb{H}$ are in bijection with the double cosets of $\Gamma \backslash \mathrm{SL}_2(\mathbb{Z}) / \Gamma_\infty$ where a cusp class $\Gamma \sigma_a$ corresponds to the double coset $\Gamma \sigma_a \Gamma_\infty$ under this bijection. As every element of $\mathrm{SL}_2(\mathbb{Z})$ is a scaling matrix for some cusp, consider the surjective map

$$\phi : \Gamma \backslash \mathrm{SL}_2(\mathbb{Z}) \rightarrow \Gamma \backslash \mathrm{SL}_2(\mathbb{Z}) / \Gamma_\infty \quad \Gamma \sigma_a \mapsto \Gamma \sigma_a \Gamma_\infty.$$

We will compute the size of the fibers of ϕ . Suppose $\Gamma \sigma_a$ and $\Gamma \sigma_{a'}$ are two cosets of $\Gamma \backslash \mathrm{SL}_2(\mathbb{Z})$ such that $\Gamma \sigma_a \Gamma_\infty = \Gamma \sigma_{a'} \Gamma_\infty$. Then $\sigma_{a'} = \gamma \sigma_a \gamma_\infty^n$ for some $\gamma \in \Gamma$ and $n \in \mathbb{Z}$. Now $\Gamma \sigma_a = \Gamma \sigma_{a'}$ if and only if $\Gamma \sigma_a = \Gamma \sigma_a \gamma_\infty^n$ which holds if and only if $\gamma_\infty^n \in \sigma_a^{-1} \Gamma \sigma_a$. As $\gamma_\infty^n \in \Gamma_\infty$, our previous containment is equivalent to $\gamma_\infty^n \in \sigma_a^{-1} \Gamma \sigma_a \cap \Gamma_\infty = \sigma_a^{-1} \Gamma_a \sigma_a = \Gamma_\infty^{(h_a)}$. This means $h_a \mid n$. So the cosets $\Gamma \sigma_a$ and $\Gamma \sigma_{a'}$ are equal if and only if $h_a \mid n$. This means there are h_a distinct fibers of $\Gamma \sigma_a \Gamma_\infty$. In other words, there are h_a many cosets of $\Gamma \backslash \mathrm{SL}_2(\mathbb{Z})$ whose elements are scaling matrices for cusps in the cusp class of $\mathfrak{a}$. This proves the first statement so it remains to prove the second. As ϕ is a surjection, the order of $\Gamma \backslash \mathrm{SL}_2(\mathbb{Z})$ is equal to the sum of the fibers which means

$$d_\Gamma = \sum_{\mathfrak{a} \in \Gamma \backslash \hat{\mathbb{Q}}} h_a.$$

This completes the proof. □

Let $\mathfrak{a}$ and $\mathfrak{b}$ be cusps of $\Gamma \backslash \mathbb{H}$ with scaling matrices σ_a and σ_b respectively. When investigating modular forms, it will be useful to have a double coset decomposition for sets of the form $\sigma_a^{-1} \Gamma \sigma_b$. To state this decomposition, we require some notation. Let $\delta_{\mathfrak{a}, \mathfrak{b}} = 1, 0$ according to whether $\mathfrak{a}$ and $\mathfrak{b}$ are in the same cusp class or not. Equivalently, $\delta_{\mathfrak{a}, \mathfrak{b}} = 1, 0$ if $\Gamma \mathfrak{a} = \Gamma \mathfrak{b}$ or not. We can now state and prove the **Bruhat decomposition** for $\sigma_a^{-1} \Gamma \sigma_b$:

Theorem (Bruhat decomposition). *Let Γ be any congruence subgroup and let $\mathfrak{a}$ and $\mathfrak{b}$ be cusps of $\Gamma \backslash \mathbb{H}$ with scaling matrices σ_a and σ_b respectively. Then we have the disjoint decomposition*

$$\sigma_a^{-1} \Gamma \sigma_b = \delta_{\mathfrak{a}, \mathfrak{b}} \Omega_\infty \cup \left(\bigcup_{\substack{c \in \mathbb{Z} - \{0\} \\ d \pmod{ch_b}}} \Omega_{c/d} \right),$$

where

$$\Omega_\infty = \Gamma_\infty^{(h_a)} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \Gamma_\infty^{(h_b)} \quad \text{and} \quad \Omega_{c/d} = \Gamma_\infty^{(h_a)} \omega_{c/d} \Gamma_\infty^{(h_b)},$$

with $\omega_{c/d} = \begin{pmatrix} a & * \\ c & d \end{pmatrix} \in \sigma_a^{-1} \Gamma \sigma_b$ and $c \in \mathbb{Z} - \{0\}$ if such a matrix exists otherwise $\Omega_{c/d}$ is empty. In the former case, a and d are determined modulo ch_a and ch_b respectively and $\Omega_{c/d}$ does not depend on the entry a of $\omega_{c/d}$.

Proof. We first show $\sigma_a^{-1} \Gamma \sigma_b$ is invariant under multiplication from the left or right by elements of $\Gamma_\infty^{(h_a)} = \sigma_a^{-1} \Gamma_a \sigma_a$ and $\Gamma_\infty^{(h_b)} = \sigma_b^{-1} \Gamma_b \sigma_b$ respectively. Indeed,

$$\Gamma_\infty^{(h_a)} \sigma_a^{-1} \Gamma \sigma_b \Gamma_\infty^{(h_b)} = \sigma_a^{-1} \Gamma_a \Gamma \Gamma_b \sigma_b = \sigma_a^{-1} \Gamma \sigma_b,$$

as desired. In particular, this shows that Ω_∞ and $\Omega_{c/d}$ are subsets of $\sigma_a^{-1} \Gamma \sigma_b$. Now we prove that Ω_∞ is nonempty if and only if $\delta_{a,b} = 1$. Indeed, if $\omega \in \Omega_\infty$ then $\omega \in \sigma_a^{-1} \Gamma \sigma_b$ and so $\omega = \sigma_a^{-1} \gamma \sigma_b$ for some $\gamma \in \Gamma$. It follows that

$$\gamma b = \sigma_a \omega \sigma_b^{-1} b = \sigma_a \omega \infty = \sigma_a \infty = a.$$

This shows $\delta_{a,b} = 1$. Conversely, suppose $\delta_{a,b} = 1$. Then $\sigma_a^{-1} \Gamma \sigma_b$ contains $\sigma_a^{-1} \Gamma \sigma_a = \sigma_b^{-1} \Gamma \sigma_b$. It follows that $h_a = h_b$ and $\Gamma_\infty^{(h_a)} = \Gamma_\infty^{(h_b)}$. In particular, this proves Ω_∞ is nonempty. So Ω_∞ is nonempty if and only if $\delta_{a,b} = 1$. Every other element of $\sigma_a^{-1} \Gamma \sigma_b$ belongs to one of the double cosets $\Omega_{c/d}$ with $c \in \mathbb{Z} - \{0\}$. The relation

$$\gamma_\infty^{(h_a)} \omega_{c/d} \gamma_\infty^{(h_b)} = \begin{pmatrix} 1 & h_a \\ 0 & 1 \end{pmatrix} \begin{pmatrix} a & * \\ c & d \end{pmatrix} \begin{pmatrix} 1 & h_b \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} a + ch_a & * \\ c & d + ch_b \end{pmatrix},$$

shows that $\Omega_{c/d}$ is generated uniquely by a matrix of the form $\omega_{c/d} = \begin{pmatrix} a & * \\ c & d \end{pmatrix} \in \sigma_a^{-1} \Gamma \sigma_b$ with a and d are determined modulo ch_a and ch_b respectively. To see that $\Omega_{c/d}$ does not depend on the entry a of $\omega_{c/d}$, let $\omega'_{c/d} = \begin{pmatrix} a' & * \\ c & d \end{pmatrix} \in \sigma_a^{-1} \Gamma \sigma_b$. We may write $\omega_{c/d} = \sigma_a^{-1} \gamma \sigma_b$ and $\omega'_{c/d} = \sigma_a^{-1} \gamma' \sigma_b$ for some $\gamma, \gamma' \in \Gamma$. The relations

$$\omega_{c/d} (\omega'_{c/d})^{-1} = \begin{pmatrix} a & * \\ c & d \end{pmatrix} \begin{pmatrix} d & * \\ -c & a \end{pmatrix} = \begin{pmatrix} * & * \\ 0 & * \end{pmatrix} \quad \text{and} \quad \omega_{c/d} (\omega'_{c/d})^{-1} = \sigma_a^{-1} \gamma (\gamma')^{-1} \sigma_a,$$

show that $\gamma (\gamma')^{-1} = \sigma_a \omega_{c/d} (\omega'_{c/d})^{-1} \sigma_a^{-1} \in \Gamma_a$. Therefore $\omega_{c/d} (\omega'_{c/d})^{-1} \in \Gamma_\infty^{(h_a)}$ which is to say that $\omega_{c/d} = \left(\gamma_\infty^{(h_a)} \right)^n \omega'_{c/d}$ for some $n \in \mathbb{Z}$. But then $\Gamma_\infty^{(h_a)} \omega_{c/d} \Gamma_\infty^{(h_b)} = \Gamma_\infty^{(h_a)} \omega'_{c/d} \Gamma_\infty^{(h_b)}$ as desired. $\square$

Notice that the Bruhat decomposition for $\sigma_a^{-1} \Gamma \sigma_b$ implies

$$\Gamma_\infty^{(h_a)} \backslash \sigma_a^{-1} \Gamma \sigma_b = \delta_{a,b} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \Gamma_\infty^{(h_b)} \cup \left(\bigcup_{\substack{c \in \mathbb{Z} - \{0\} \\ d \pmod{ch_b}}} \omega_{c/d} \Gamma_\infty^{(h_b)} \right),$$

where it is understood that the coset $\omega_{c/d} \Gamma_\infty^{(h_b)}$ is empty if the double coset $\Omega_{c/d}$ is too. This shows that every coset of $\Gamma_\infty^{(h_a)} \backslash \sigma_a^{-1} \Gamma \sigma_b$ other than $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \Gamma_\infty^{(h_b)}$ corresponds to a unique pair

(c, d) with $c \in \mathbb{Z} - \{0\}$ and d taken modulo ch_b . Of course, this correspondence need not be surjective since many of the double cosets $\Omega_{c/d}$ may be empty. To track such pairs for which $\Omega_{c/d}$ is nonempty, let $\mathcal{C}_{a,b}$ and $\mathcal{D}_{a,b}(c)$ be the sets

$$\mathcal{C}_{a,b} = \left\{ c \in \mathbb{Z} - \{0\} : \begin{pmatrix} * & * \\ c & * \end{pmatrix} \in \sigma_a^{-1} \Gamma \sigma_b \right\} \quad \text{and} \quad \mathcal{D}_{a,b}(c) = \left\{ d \in \mathbb{Z}/ch_b\mathbb{Z} : \begin{pmatrix} * & * \\ c & d \end{pmatrix} \in \sigma_a^{-1} \Gamma \sigma_b \right\}.$$

Then $\mathcal{C}_{a,b}$ and $\mathcal{D}_{a,b}(c)$ run over precisely the values of c and d such that $\Omega_{c/d}$ is nonempty.

Remark 1.15. The Bruhat decomposition for $\sigma_a^{-1} \Gamma \sigma_b$ implies

$$\Gamma_\infty^{(h_a)} \backslash \sigma_a^{-1} \Gamma \sigma_b = \delta_{a,b} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \Gamma_\infty^{(h_b)} \cup \left(\bigcup_{c \in \mathcal{C}_{a,b}, d \in \mathcal{D}_{a,b}(c)} \omega_{c/d} \Gamma_\infty^{(h_b)} \right),$$

where none of the cosets $\omega_{c/d} \Gamma_\infty^{(h_b)}$ are empty. In particular, every $\gamma \in \Gamma_\infty^{(h_a)} \backslash \sigma_a^{-1} \Gamma \sigma_b - \delta_{a,b} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \Gamma_\infty^{(h_b)}$ is of the form $\gamma = \begin{pmatrix} * & * \\ c & d+ch_b\ell \end{pmatrix}$ for some $\ell \in \mathbb{Z}$ and where (c, d) is a pair with $c \in \mathcal{C}_{a,b}$ and $d \in \mathcal{D}_{a,b}(c)$.

We will now introduce Kloosterman and Salié sums associated to cusps. We begin with the Kloosterman sums. Let $\mathfrak{a}$ and $\mathfrak{b}$ be cusps of $\Gamma \backslash \mathbb{H}$. Then for any $c \in \mathcal{C}_{a,b}$ and $m, n \in \mathbb{Z}$, the **generalized Kloosterman sum** $K_{a,b}(m, n, c)$ relative to $\mathfrak{a}$ and $\mathfrak{b}$ is defined by

$$K_{a,b}(m, n, c) = \sum_{\substack{d \in \mathcal{D}_{a,b}(c) \\ \begin{pmatrix} a & * \\ c & d \end{pmatrix} \in \sigma_a^{-1} \Gamma \sigma_b}} e^{2\pi i \left(\frac{ma}{ch_a} + \frac{nd}{ch_b} \right)}.$$

This sum is well-defined since a is determined modulo ch_a by the Bruhat decomposition. If $m = n = 0$ we trivially see by the Bruhat decomposition for $\sigma_a^{-1} \Gamma \sigma_b$ that

$$K_{a,b}(0, 0, c) = |\mathcal{D}_{a,b}|.$$

The Salié sums are defined in a similar manner. Let $\mathfrak{a}$ and $\mathfrak{b}$ be cusps of $\Gamma \backslash \mathbb{H}$. Then for any $c \in \mathcal{C}_{a,b}$, $m, n \in \mathbb{Z}$, and Dirichlet character χ with conductor $q \mid c$, the **generalized Salié sum** $S_{\chi,a,b}(m, n, c)$ relative to $\mathfrak{a}$ and $\mathfrak{b}$ is defined by

$$S_{\chi,a,b}(m, n, c) = \sum_{\substack{d \in \mathcal{D}_{a,b}(c) \\ \begin{pmatrix} a & * \\ c & d \end{pmatrix} \in \sigma_a^{-1} \Gamma \sigma_b}} \chi(d) e^{2\pi i \left(\frac{ma}{ch_a} + \frac{nd}{ch_b} \right)}.$$

This sum is well-defined since a is determined modulo ch_a by the Bruhat decomposition. To obtain the usual Kloosterman and Salié sums, consider the case when $\Gamma = \Gamma_0(N)$ and $\mathfrak{a} = \mathfrak{b} = \infty$. Take $\sigma_a = \sigma_b = I$ and notice that $h_a = h_b = 1$. The Bruhat decomposition for $\Gamma_0(N)$ shows that in $\omega_{c/d} = \begin{pmatrix} a & * \\ c & d \end{pmatrix}$ the entries a and d are determined modulo c and so we may take $a = \bar{d}$ because $\det(\omega_{c/d}) = 1$. Also, $\mathcal{D}_{\infty,\infty}(c) = (\mathbb{Z}/c\mathbb{Z})^*$ because every matrix of the form $\gamma = \begin{pmatrix} * & * \\ c & d \end{pmatrix} \in \text{SL}_2(\mathbb{Z})$ with $(c, d) = 1$ and $c \equiv 0 \pmod{N}$ belongs to $\Gamma_0(N)$ (by Bézout's identity). Whence

$$K_{\infty,\infty}(n, m, c) = K(n, m, c) \quad \text{and} \quad S_{\chi,\infty,\infty}(n, m, c) = S_\chi(n, m, c),$$

as desired. In these cases, we will suppress the dependencies upon $\mathfrak{a}$ and $\mathfrak{b}$.

1.4 The Hyperbolic Measure

We will also need to integrate over $\Gamma \backslash \mathbb{H}$. In order to do this, we require a measure on $\mathbb{H}$. Our choice of measure will be the **hyperbolic measure** $d\mu$ given by

$$d\mu = d\mu(z) = \frac{dx dy}{y^2}.$$

The most important property about the hyperbolic measure is that it is $\mathrm{GL}_2^+(\mathbb{R})$ -invariant:

Proposition 1.16. *The hyperbolic measure $d\mu$ is $\mathrm{GL}_2^+(\mathbb{R})$ -invariant on $\mathbb{H}$.*

Proof. Let $\alpha = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{GL}_2^+(\mathbb{R})$. We must show that

$$\mu(\alpha z) = \mu(z).$$

Since αz is a Möbius transformation, which is holomorphic, it satisfies the Cauchy-Riemann equations. So for the change of variables $z \mapsto \alpha z$, the absolute value of the determinant of the Jacobian matrix of this transformation is the absolute value of the square of the derivative of αz . This is $\frac{\det(\alpha)^2}{|cz+d|^4}$ because the derivative of αz is $\frac{\det(\alpha)}{(cz+d)^2}$. From this fact and Equation (1), we have

$$\mu(\alpha z) = \frac{\det(\alpha)^2}{|cz+d|^4} \frac{dx dy}{\det(\alpha)^2 \frac{y^2}{|cz+d|^4}} = \frac{dx dy}{y^2} = \mu(z). \quad \square$$

As this result will be used so frequently any time we integrate, we will not mention it explicitly. Another particularly important implication is that if Γ is a congruence subgroup then $d\mu$ is Γ -invariant. One of the reasons this is useful is because we can apply the unfolding/folding method to many integrals. The most common instance is when we are integrating the sum $\sum_{\gamma \in \Gamma_\infty \backslash \Gamma} f(\gamma z)$ of some holomorphic function $f(z)$ over a fundamental domain $\mathcal{F}_\Gamma$ for $\Gamma \backslash \mathbb{H}$. Indeed, $\mathbb{H} = \bigcup_{\gamma \in \Gamma} \gamma \mathcal{F}_\Gamma$ and so $\Gamma_\infty \backslash \mathbb{H} = \bigcup_{\gamma \in \Gamma_\infty \backslash \Gamma} \gamma \mathcal{F}_\Gamma$. Since $\mathcal{F}_\Gamma$ is a fundamental domain, the conditions of the unfolding/folding method are satisfied and it follows that

$$\int_{\mathcal{F}_\Gamma} \sum_{\gamma \in \Gamma_\infty \backslash \Gamma} f(\gamma z) d\mu = \int_{\Gamma_\infty \backslash \mathbb{H}} f(z) d\mu,$$

provided either side is absolutely convergent. It is also worth highlighting another fact. Let $\alpha \in \mathrm{GL}_2^+(\mathbb{Q})$ and suppose $\alpha^{-1}\Gamma\alpha \subseteq \mathrm{SL}_2(\mathbb{Z})$ so that $\alpha^{-1}\Gamma\alpha$ is a congruence subgroup by Lemma 1.4. As α acts as an automorphism of $\mathbb{H}$, this action induces a bijection between $\alpha^{-1}\Gamma\alpha \backslash \mathbb{H}$ and $\Gamma \backslash \mathbb{H}$ and hence between the fundamental domains $\mathcal{F}_{\alpha^{-1}\Gamma\alpha}$ and $\mathcal{F}_\Gamma$. Thus the change of variables $z \mapsto \alpha z$ transforms the fundamental domain $\mathcal{F}_\Gamma$ into $\mathcal{F}_{\alpha^{-1}\Gamma\alpha}$. Therefore

$$\int_{\mathcal{F}_\Gamma} f(z) d\mu = \int_{\mathcal{F}_{\alpha^{-1}\Gamma\alpha}} f(\alpha z) d\mu,$$

provided either side is bounded. Now let us discuss the volume of $\Gamma \backslash \mathbb{H}$. We define the **volume** V_Γ of $\Gamma \backslash \mathbb{H}$ by

$$V_\Gamma = \int_{\mathcal{F}_\Gamma} d\mu.$$

In other words, V_Γ is the measure of the fundamental domain $\mathcal{F}_\Gamma$ with respect to $d\mu$. Also, if $\mathcal{F}_\Gamma = \mathcal{F}$ we write $V_\Gamma = V$. Since the integrand is Γ -invariant, V_Γ is independent of the choice of fundamental domain. Using Proposition 1.8, we have

$$V = \int_{\mathcal{F}} d\mu = \int_{-\frac{1}{2}}^{\frac{1}{2}} \int_{\sqrt{1-x^2}}^{\infty} \frac{dy dx}{y^2} = \int_{-\frac{1}{2}}^{\frac{1}{2}} \frac{1}{\sqrt{1-x^2}} dx = 2 \arcsin\left(\frac{1}{2}\right) = \frac{\pi}{3}.$$

Therefore V is finite. There is also a simple relation between V_Γ and d_Γ given by

$$V_\Gamma = d_\Gamma V, \tag{4}$$

which follows immediately from Proposition 1.10. Moreover, V_Γ is finite for every congruence subgroup Γ by Equation (4) and that congruence subgroups have finite index in the modular group. A particularly nice application of this fact is that any integral of the form

$$\int_{\mathcal{F}_\Gamma} f(z) d\mu,$$

is absolutely convergent provided $f(z)$ is bounded. That is, bounded functions are absolutely convergent over $\mathcal{F}_\Gamma$ with respect to $d\mu$. Moreover, we have a useful lemma:

Lemma 1.17. *Let Γ be a congruence subgroup and $\alpha \in \mathrm{GL}_2^+(\mathbb{Q})$. If $\alpha^{-1}\Gamma\alpha \subseteq \mathrm{SL}_2(\mathbb{Z})$ then $V_{\alpha^{-1}\Gamma\alpha} = V_\Gamma$ and $d_{\alpha^{-1}\Gamma\alpha} = d_\Gamma$.*

Proof. The first statement follows from the chain

$$V_{\alpha^{-1}\Gamma\alpha} = \int_{\mathcal{F}_{\alpha^{-1}\Gamma\alpha}} d\mu = \int_{\mathcal{F}_\Gamma} d\mu = V_\Gamma,$$

where the middle equality is justified by making the change of variables $z \mapsto \alpha^{-1}z$. The second statement is now immediate from Equation (4). $\square$

2 The Theory of Modular Forms

Throughout,

2.1 Modular Forms

Define $j(\alpha, z)$ by

$$j(\alpha, z) = (cz + d),$$

for all $\alpha = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{GL}_2^+(\mathbb{Q})$ and $z \in \mathbb{H}$. With this notation, Equation (1) becomes

$$\mathrm{Im}(\alpha z) = \det(\alpha) \frac{\mathrm{Im}(z)}{|j(\alpha, z)|^2} \tag{5}$$

There is a very useful property that $j(\alpha, z)$ satisfies. To state it, let $\alpha' = \begin{pmatrix} a' & b' \\ c' & d' \end{pmatrix} \in \mathrm{GL}_2^+(\mathbb{Q})$. Then

$$\alpha\alpha' = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} a' & b' \\ c' & d' \end{pmatrix} = \begin{pmatrix} a'a + b'c & a'b + b'd \\ c'a + d'c & c'b + d'd \end{pmatrix},$$

and we have

$$\begin{aligned}
j(\alpha', \alpha z)j(\alpha, z) &= \left(c' \frac{az + b}{cz + d} + d' \right) (cz + d) \\
&= (c'(az + b) + d'(cz + d)) \\
&= (c'a + d'c)z + c'b + d'd) \\
&= j(\alpha'\alpha, z).
\end{aligned}$$

The relation

$$j(\alpha'\alpha, z) = j(\alpha', \alpha z)j(\alpha, z),$$

is called the **cocycle condition** for $j(\alpha, z)$. For any integer $k \geq 1$ and any $\alpha \in \mathrm{GL}_2^+(\mathbb{Q})$ we define the **slash operator** $|_k\alpha : C(\mathbb{H}) \rightarrow C(\mathbb{H})$ to be the linear operator given by

$$(f|_k\alpha)(z) = \det(\alpha)^{\frac{k}{2}} j(\alpha, z)^{-k} f(\alpha z).$$

Note that for $\gamma \in \mathrm{SL}_2(\mathbb{Z})$, the slash operator takes the simpler form

$$(f|_k\gamma)(z) = j(\gamma, z)^{-k} f(\gamma z).$$

The cocycle condition implies that the slash operator is multiplicative. Indeed, if $\alpha, \alpha' \in \mathrm{GL}_2^+(\mathbb{Q})$ then

$$\begin{aligned}
((f|_k\alpha')|_k\alpha)(z) &= \det(\alpha)^{\frac{k}{2}} j(\alpha, z)^{-k} (f|_k\alpha')(\alpha z) \\
&= \det(\alpha'\alpha)^{\frac{k}{2}} j(\alpha', \alpha z)^{-k} j(\alpha, z)^{-k} f(\alpha'\alpha z) \\
&= \det(\alpha'\alpha)^{\frac{k}{2}} j(\alpha'\alpha, z)^{-k} f(\alpha'\alpha z) && \text{cocycle condition} \\
&= (f|_k\alpha'\alpha)(z).
\end{aligned}$$

The slash operator also respects multiplication. Let $f, g \in C(\mathbb{H})$ and $k, \ell \geq 1$. Then for any $\alpha \in \mathrm{GL}_2^+(\mathbb{Q})$, we have

$$\begin{aligned}
((fg)|_{k+\ell}\alpha)(z) &= \det(\alpha)^{\frac{k+\ell}{2}} j(\alpha, z)^{-(k+\ell)} (fg)(\alpha z) \\
&= \left(\det(\alpha)^{\frac{k}{2}} j(\alpha, z)^{-k} f(\alpha z) \right) \left(\det(\alpha)^{\frac{\ell}{2}} j(\alpha, z)^{-\ell} g(\alpha z) \right) \\
&= ((f|_k\alpha)(z))((g|_\ell\alpha)(z)).
\end{aligned}$$

If an operator commutes with the slash operators $|_k\alpha$ for every $\alpha \in \mathrm{SL}_2(\mathbb{Q})$, we say that it is **invariant**. Now let Γ be a congruence subgroup of level N and let $\nu : \Gamma \rightarrow S^1$ be a finite order homomorphism such that $\nu(\gamma_\infty) = 1$. We will now introduce modular forms. We say that a function $f : \mathbb{H} \rightarrow \mathbb{C}$ is **modular form** on $\Gamma \backslash \mathbb{H}$ of **weight** k , **level** N , and **nebentypus** ν if the following properties are satisfied:

- (i) f is holomorphic on $\mathbb{H}$.
- (ii) $(f|_k\gamma)(z) = \nu(\gamma)f(z)$ for all $\gamma \in \Gamma$.
- (iii) $(f|_k\gamma)(z) = O(1)$ for all $\gamma \in \mathrm{SL}_2(\mathbb{Z})$ (or equivalently $(f|_k\alpha)(z) = O(1)$ for all $\alpha \in \mathrm{GL}_2^+(\mathbb{Q})$).

We say f is a (modular) **cuspidal form** if the additional property is satisfied:

(iv) For all cusps $\mathfrak{a}$ and any $y > 0$, we have

$$\int_0^{h_{\mathfrak{a}}} (f|_k \sigma_{\mathfrak{a}})(x + iy) dx = 0.$$

Property (ii) is called the **modularity condition** and we say f is **modular**. In particular, f is a function on $\mathcal{F}_{\Gamma}$. The modularity condition can equivalently be expressed as

$$f(\gamma z) = \nu(\gamma) j(\gamma, z)^k f(z).$$

Moreover, we say that the nebentypus ν is **singular** at a cusp $\mathfrak{a}$ if $\nu(\gamma_{\mathfrak{a}}) = 1$. Equivalently, $\nu(\gamma) = 1$ for every $\gamma \in \Gamma_{\mathfrak{a}}$. Necessarily, a nebentypus must be singular at ∞ . Property (iii) is called the **growth condition** for modular forms and we say f is **holomorphic at the cusps**. By modularity, it suffices to verify the growth condition on a complete set scaling matrices for the cusp classes. Therefore we only have to check this condition for finitely many matrices since the number of cusp classes is finite. To see the equivalence in the growth condition, we will require a lemma.

Lemma 2.1. *Every $\alpha \in \mathrm{GL}_2^+(\mathbb{Q})$ can be written as $\alpha = \gamma\eta$ for some $\gamma \in \mathrm{SL}_2(\mathbb{Z})$ and $\eta \in \mathrm{GL}_2^+(\mathbb{Q})$ with $\eta = \begin{pmatrix} * & * \\ 0 & * \end{pmatrix}$.*

Proof. Let $r \geq 1$ be such that $r\alpha = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{GL}_2^+(\mathbb{Z})$. If $c = 0$ then $a \neq 0$ (otherwise α is not invertible) and so $(a, c) \geq 1$. Set $a' = \frac{a}{(a, c)}$ and $c' = \frac{c}{(a, c)}$ so that $a', c' \in \mathbb{Z}$ with $(a', c') = 1$. Then there exists $\gamma \in \mathrm{SL}_2(\mathbb{Z})$ with $\gamma^{-1} = \begin{pmatrix} * & * \\ -c' & a' \end{pmatrix}$. Moreover, $\gamma^{-1}r\alpha = \begin{pmatrix} * & * \\ 0 & * \end{pmatrix} \in \mathrm{GL}_2^+(\mathbb{Z})$. The claim is complete upon setting $\eta = \gamma^{-1}\alpha$. $\square$

This lemma implies that we may write $\alpha = \gamma\eta$ for some $\gamma \in \mathrm{SL}_2(\mathbb{Z})$ and $\eta \in \mathrm{GL}_2^+(\mathbb{Q})$ with $\eta = \begin{pmatrix} * & * \\ 0 & * \end{pmatrix}$. This decomposition together with the cocycle condition gives

$$j(\alpha, z) = j(\gamma, \eta z) j(\eta, z).$$

As $j(\eta, z)$ is a constant, it follows that $(f|_k \alpha)(z) = O(1)$ for all $\alpha \in \mathrm{GL}_2^+(\mathbb{Q})$ which proves the forward implication. The reverse implication is trivial since $\mathrm{SL}_2(\mathbb{Z}) \subset \mathrm{GL}_2^+(\mathbb{Q})$. Now let $\sigma_{\mathfrak{a}}$ be a scaling matrix for the $\mathfrak{a}$ cusp and recall that $\sigma_{\mathfrak{a}}^{-1}\Gamma\sigma_{\mathfrak{a}}$ is a congruence subgroup of the same level as Γ . Let $\gamma \in \sigma_{\mathfrak{a}}^{-1}\Gamma\sigma_{\mathfrak{a}}$ so that $\gamma = \sigma_{\mathfrak{a}}^{-1}\gamma'\sigma_{\mathfrak{a}}$ for some $\gamma' \in \Gamma$. Then modularity gives

$$((f|_k \sigma_{\mathfrak{a}})|_k \gamma)(z) = (f|_k \sigma_{\mathfrak{a}} \gamma)(z) = (f|_k \gamma' \sigma_{\mathfrak{a}})(z) = \nu(\gamma') (f|_k \sigma_{\mathfrak{a}})(z) = \nu_{\sigma_{\mathfrak{a}}}(\gamma) (f|_k \sigma_{\mathfrak{a}})(z),$$

where we have set

$$\nu_{\sigma_{\mathfrak{a}}}(\gamma) = \nu(\sigma_{\mathfrak{a}} \gamma \sigma_{\mathfrak{a}}^{-1}),$$

for all $\gamma \in \sigma_{\mathfrak{a}}^{-1}\Gamma\sigma_{\mathfrak{a}}$. From this it follows that $f|_k \sigma_{\mathfrak{a}}$ is a modular form on $\sigma_{\mathfrak{a}}^{-1}\Gamma\sigma_{\mathfrak{a}} \backslash \mathbb{H}$ of the same weight and level as f but with nebentypus $\nu_{\sigma_{\mathfrak{a}}}$ and is a cuspidal form if f is.

Remark 2.2. If f is a weight k modular form with nebentypus ν on $\Gamma \backslash \mathbb{H}$ and $-I \in \Gamma$ then modularity implies

$$f(z) = \nu(-I)(-1)^k f(z).$$

Therefore $\nu(-I) = (-1)^k$. Therefore ν cannot be trivial if k is odd. In particular, there are no nonzero modular forms of odd weight with trivial nebentypus on $\mathrm{SL}_2(\mathbb{Z})$.

Modular forms also admit Fourier series. First note that modularity and the definition of the nebentypus together imply

$$(f|_k \sigma_{\mathfrak{a}})(z + h_{\mathfrak{a}}) = (f|_k \sigma_{\mathfrak{a}})(\gamma_{\mathfrak{a}} z) = (f|_k \sigma_{\mathfrak{a}})(z).$$

So $f|_k \sigma_{\mathfrak{a}}$ is $h_{\mathfrak{a}}$ -periodic. This implies that we only need to verify the growth condition for f as $y \rightarrow \infty$. In any case, as $f|_k \sigma_{\mathfrak{a}}$ is also smooth, ?? (after scaling) shows that it admits a Fourier series given by

$$(f|_k \sigma_{\mathfrak{a}})(z) = \sum_{n \geq 0} a_{\mathfrak{a}}(y, n) e^{\frac{2\pi i n x}{h_{\mathfrak{a}}}},$$

where the sum is only over $n \geq 0$ because holomorphy at the cusps forces $f|_k \sigma_{\mathfrak{a}}$ to be bounded. Moreover, this Fourier series converges uniformly everywhere to $f|_k \sigma_{\mathfrak{a}}$. We can also simplify the Fourier coefficients $a_{\mathfrak{a}}(y, n)$. To see this, since $f|_k \sigma_{\mathfrak{a}}$ is holomorphic it satisfies the Cauchy-Riemann equations so that

$$\frac{1}{2} \left(\frac{\partial f|_k \sigma_{\mathfrak{a}}}{\partial x} + i \frac{\partial f|_k \sigma_{\mathfrak{a}}}{\partial y} \right) = 0.$$

Substituting in the Fourier series and equating coefficients we obtain the ODE

$$\frac{2\pi n}{h_{\mathfrak{a}}} a_{\mathfrak{a}}(y, n) + a'_{\mathfrak{a}}(y, n) = 0,$$

Solving this ODE by separation of variables, we see that there exists an $a_{\mathfrak{a}}(n)$ such that

$$a_{\mathfrak{a}}(y, n) = a_{\mathfrak{a}}(n) e^{-\frac{2\pi n y}{h_{\mathfrak{a}}}}.$$

The coefficients $a_{\mathfrak{a}}(n)$ are the only part of the Fourier series depending on the cusp $\mathfrak{a}$ and scaling matrix $\sigma_{\mathfrak{a}}$. Using these coefficients instead, f admits a **Fourier series** at the $\mathfrak{a}$ cusp given by

$$(f|_k \sigma_{\mathfrak{a}})(z) = \sum_{n \geq 0} a_{\mathfrak{a}}(n) e^{\frac{2\pi i n z}{h_{\mathfrak{a}}}},$$

with **Fourier coefficients** $a_{\mathfrak{a}}(n)$. If $\mathfrak{a} = \infty$ and $\sigma_{\mathfrak{a}} = I$, we will drop this dependence and in this case $f|_k \sigma_{\mathfrak{a}} = f$ with $a_{\mathfrak{a}}(n) = a(n)$. As we have already noted, this Fourier series converges uniformly everywhere to $f|_k \sigma_{\mathfrak{a}}$. Moreover, property (iv) implies that f is a cusp form if and only if $a_{\mathfrak{a}}(n) = 0$ for every cusp $\mathfrak{a}$. This does not depend upon the choice of scaling matrix used for $f|_k \sigma_{\mathfrak{a}}$ because

$$(f|_k \sigma_{\mathfrak{a}} \gamma_{\infty})(z) = (f|_k \sigma_{\mathfrak{a}})(z + h_{\infty}). \quad (6)$$

We can also easily derive a bound for the size of the Fourier coefficients of cusp forms. To see this, note that $\left| (f|_k \sigma_a)(z) \operatorname{Im}(z)^{\frac{k}{2}} \right|$ is $\sigma_a^{-1} \Gamma \sigma_a$ -invariant by modularity of $f|_k \sigma_a$, the cocycle condition, Equation (5), and that $|\chi(\gamma)| = 1$. Moreover, this function is bounded on $\mathcal{F}_{\sigma_a^{-1} \Gamma \sigma_a}$ because f is a cusp form. The $\sigma_a^{-1} \Gamma \sigma_a$ -invariance implies that $\left| (f|_k \sigma_a)(z) \operatorname{Im}(z)^{\frac{k}{2}} \right|$ is bounded on $\mathbb{H}$. From the definition of Fourier series, we then have

$$a_a(n) \operatorname{Im}(z)^{\frac{k}{2}} = \int_0^{h_a} (f|_k \sigma_a)(z) \operatorname{Im}(z)^{\frac{k}{2}} e^{-\frac{2\pi i n z}{h_a}} dx \ll \int_0^{h_a} e^{\frac{2\pi n y}{h_a}} dx \ll e^{\frac{2\pi n y}{h_a}}.$$

Upon setting $y = \frac{h_a}{n}$, we obtain

$$a_a(n) \ll n^{\frac{k}{2}}.$$

This bound is known as the **Hecke bound** for modular forms. It follows from the Hecke bound that

$$(f|_k \sigma_a)(z) = O\left(\sum_{n \geq 1} n^{\frac{k}{2}} e^{-\frac{2\pi n y}{h_a}}\right) = O\left(e^{-\frac{2\pi y}{h_a}}\right),$$

where the second equality holds because each term is of smaller order than the next so that the series is bounded by a constant times its first term. This implies that $(f|_k \sigma_a)(z)$ exhibits exponential decay. Accordingly, we say that f exhibits **exponential decay at the cusps**. This is stronger than simply knowing $f|_k \sigma_a$ is bounded on $\mathbb{H}$ and tends to zero as $z \rightarrow \infty$ (which follows because $f|_k \sigma_a$ is a cusp form and is h_a -periodic).

Remark 2.3. Note that the Hecke bound only holds for the Fourier coefficients of cusp forms. In fact, as we will see it is not even an optimal bound.

Many of the modular forms we will study will be over the congruence subgroups $\Gamma(N)$, $\Gamma_1(N)$, and $\Gamma_0(N)$. To collect all of these cases, we will let Γ be any of these congruence subgroups. There are a few important simplifications when working with Γ . The first, which we have already noted, is that Γ is reduced ∞ . Therefore all of our modular forms will be 1-periodic. Second is that the nebentypus can be induced by a Dirichlet character χ with conductor q dividing N . In particular, for any $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma$ we know that $c \equiv 0 \pmod{N}$. Therefore a is the inverse of d modulo N because $ad - bc = 1$ and moreover $(d, N) = 1$. Thus $(d, q) = 1$ and we can define a function $\nu_\chi : \Gamma \rightarrow S^1$ by

$$\nu_\chi(\gamma) = \chi(d).$$

If $\gamma' = \begin{pmatrix} a' & b' \\ c' & d' \end{pmatrix} \in \Gamma_0(N)$ then the relations

$$\gamma\gamma' \equiv \begin{pmatrix} a & * \\ 0 & d \end{pmatrix} \begin{pmatrix} a' & * \\ 0 & d' \end{pmatrix} = \begin{pmatrix} * & * \\ 0 & d'd \end{pmatrix} \pmod{N} \quad \text{and} \quad \gamma^{-1} = \begin{pmatrix} d & * \\ 0 & a \end{pmatrix} \pmod{N},$$

together show that

$$\nu_\chi(\gamma\gamma') = \nu_\chi(\gamma)\nu_\chi(\gamma') \quad \text{and} \quad \nu_\chi(\gamma^{-1}) = \nu_\chi(\gamma)^{-1}.$$

In other words, ν_χ is a homomorphism. It is necessarily of finite order because χ is and clearly $\nu_\chi(\gamma_\infty) = 1$. Therefore ν_χ satisfies the assumptions of a nebentypus. In the case f has nebentypus ν_χ , we simply say that f has **character** χ and write

$$\chi(\gamma) = \nu_\chi(\gamma).$$

2.2 Poincaré and Eisenstein Series

We will introduce two important classes of modular forms on $\Gamma(N)\backslash\mathbb{H}$, $\Gamma_1(N)\backslash\mathbb{H}$, and $\Gamma_0(N)\backslash\mathbb{H}$. These are the Poincaré and Eisenstein series. The latter of which will be a special case of the former. As we will only work with these congruence subgroups, let Γ denote such a subgroup. Let $m \geq 0$, $k \geq 3$, χ be a Dirichlet character with conductor $q \mid N$, and $\mathfrak{a}$ be a cusp of $\Gamma\backslash\mathbb{H}$. Also assume χ is singular at $\mathfrak{a}$. We define the m -th (modular) **Poincaré series** $P_{m,k,\chi,\mathfrak{a}}(z)$ of weight k with character χ on $\Gamma\backslash\mathbb{H}$ at the $\mathfrak{a}$ cusp by

$$P_{m,k,\chi,\mathfrak{a}}(z) = \sum_{\gamma \in \Gamma_{\mathfrak{a}} \backslash \Gamma} \bar{\chi}(\gamma) \left(e^{\frac{2\pi i m z}{h_{\mathfrak{a}}}} \Big|_k \sigma_{\mathfrak{a}}^{-1} \gamma \right) (z).$$

We call m the **index** of $P_{m,k,\chi,\mathfrak{a}}(z)$. If χ is the trivial character or $\mathfrak{a} = \infty$, we will drop these dependencies accordingly.

Remark 2.4. The reason why we restrict to $k \geq 3$ is because for $k = 0, 1, 2$ the Poincaré series need not converge (see ??).

We first verify that $P_{m,k,\chi,\mathfrak{a}}(z)$ is well-defined. It suffices to show that the summands are independent of the representatives γ and $\sigma_{\mathfrak{a}}$. To see that $\bar{\chi}(\gamma)$ is independent of γ , it suffices to verify independence by multiplication on the left by the generator $\gamma_{\mathfrak{a}}$ of $\Gamma_{\mathfrak{a}}$. Indeed,

$$\bar{\chi}(\gamma_{\mathfrak{a}}\gamma) = \bar{\chi}(\gamma_{\mathfrak{a}})\bar{\chi}(\gamma) = \bar{\chi}(\gamma),$$

since χ is singular at $\mathfrak{a}$. This verifies that $\bar{\chi}(\gamma)$ is independent of the representative γ . As the set of representatives for the scaling matrix $\sigma_{\mathfrak{a}}$ is $\sigma_{\mathfrak{a}}\Gamma_{\infty}$ and the set of representatives for γ is $\Gamma_{\mathfrak{a}}\gamma$, we see that the set of representatives for $\sigma_{\mathfrak{a}}^{-1}\gamma$ is $\Gamma_{\infty}\sigma_{\mathfrak{a}}^{-1}\Gamma_{\mathfrak{a}}\gamma$. But as $\Gamma_{\mathfrak{a}} = \sigma_{\mathfrak{a}}\Gamma_{\infty}^{(h_{\mathfrak{a}})}\sigma_{\mathfrak{a}}^{-1}$ and $h_{\infty} = 1$, this set of representatives is $\Gamma_{\infty}^{(h_{\mathfrak{a}})}\sigma_{\mathfrak{a}}^{-1}\gamma$ and therefore it suffices to verify independence of $\sigma_{\mathfrak{a}}^{-1}\gamma$ by multiplication on the left by the generator $\gamma_{\infty}^{(h_{\mathfrak{a}})}$ of $\Gamma_{\infty}^{(h_{\mathfrak{a}})}$. Since the slash operator is multiplicative, it suffices to show

$$\left(e^{\frac{2\pi i m z}{h_{\mathfrak{a}}}} \Big|_k \gamma_{\infty}^{(h_{\mathfrak{a}})} \right) (z) = e^{\frac{2\pi i m z}{h_{\mathfrak{a}}}}.$$

Indeed,

$$\left(e^{\frac{2\pi i m z}{h_{\mathfrak{a}}}} \Big|_k \gamma_{\infty}^{(h_{\mathfrak{a}})} \right) (z) = j(\gamma_{\infty}^{(h_{\mathfrak{a}})}, z)^{-k} e^{\frac{2\pi i m \gamma_{\infty}^{(h_{\mathfrak{a}})} z}{h_{\mathfrak{a}}}} = e^{\frac{2\pi i m (z + h_{\mathfrak{a}})}{h_{\mathfrak{a}}}} = e^{2\pi i m z}.$$

Therefore $P_{m,k,\chi,\mathfrak{a}}(z)$ is well-defined. To see that $P_{m,k,\chi,\mathfrak{a}}(z)$ is holomorphic on $\mathbb{H}$, first note that

$$\left| e^{\frac{2\pi i m \sigma_{\mathfrak{a}}^{-1} \gamma z}{h_{\mathfrak{a}}}} \right| = e^{-\frac{2\pi m \operatorname{Im}(\sigma_{\mathfrak{a}}^{-1} \gamma z)}{h_{\mathfrak{a}}}} < 1. \quad (7)$$

Then the Bruhat decomposition for $\sigma_{\mathfrak{a}}^{-1}\Gamma$ gives

$$P_{m,k,\chi,\mathfrak{a}}(z) \ll \sum_{(c,d) \in \mathbb{Z}^2 - \{0\}} \frac{1}{|cz + d|^k}.$$

As $k \geq 3$, this latter series is locally absolutely uniformly convergent for $z \in \mathbb{H}$ by ?? . Hence $P_{m,k,\chi,a}(z)$ does too and so it is holomorphic on $\mathbb{H}$. We now verify modularity for $P_{m,k,\chi,a}(z)$. This is just a computation:

$$\begin{aligned}
(P_{m,k,\chi,a}|_k \gamma)(z) &= \sum_{\gamma' \in \Gamma_a \backslash \Gamma} \bar{\chi}(\gamma') \left(e^{\frac{2\pi i m z}{h_a}} \Big|_k \sigma_a^{-1} \gamma' \gamma \right) (z) \\
&= \sum_{\gamma' \in \Gamma_a \backslash \Gamma} \bar{\chi}(\gamma' \gamma^{-1}) \left(e^{\frac{2\pi i m z}{h_a}} \Big|_k \sigma_a^{-1} \gamma' \right) (z) & \gamma' \mapsto \gamma' \gamma^{-1} \\
&= \chi(\gamma) \sum_{\gamma' \in \Gamma_a \backslash \Gamma} \bar{\chi}(\gamma') \left(e^{\frac{2\pi i m z}{h_a}} \Big|_k \sigma_a^{-1} \gamma' \right) (z) \\
&= \chi(\gamma) P_{m,k,\chi,a}(z).
\end{aligned}$$

To verify the growth condition, we will need a technical lemma:

Lemma 2.5. *Let $a, b > 0$ be reals and consider the half-strip*

$$S_{a,b} = \{z \in \mathbb{H} : |x| \leq a \text{ and } y \geq b\}.$$

Then there is a $\delta \in (0, 1)$ such that

$$|nz + m| \geq \delta |ni + m|,$$

for all $n, m \in \mathbb{Z}$ and all $z \in S_{a,b}$.

Proof. If $n = 0$ then any δ is sufficient. If $n \neq 0$ then the desired inequality is equivalent to

$$\left| \frac{z + \frac{m}{n}}{i + \frac{n}{m}} \right| \geq \delta.$$

So consider the function

$$f(z, r) = \left| \frac{z + r}{i + r} \right|,$$

for $z \in S_{a,b}$ and $r \in \mathbb{R}$. It suffices to show $f(z, r) \geq \delta$. As $z \in \mathbb{H}$, $z - r \neq 0$ so that $f(z, r)$ is continuous and positive on $S_{a,b} \times \mathbb{R}$. Now let $Y > b$ and consider the region

$$S_{a,b}^Y = \{z \in \mathbb{H} : |x| \leq a \text{ and } b \leq y \leq Y\}.$$

We claim that there exists a Y such that if $y > Y$ and $|x| > Y$ then $f(z, r)^2 > \frac{1}{4}$. Indeed, we compute

$$f(z, r)^2 = \frac{(z + r)(\bar{z} + r)}{(i + r)(-i + r)} = \frac{|z|^2 + 2xr + r^2}{1 + r^2} \geq \frac{y + r^2}{1 + r^2},$$

where in the inequality we have used the bound $|z|^2 \geq y$ and that x is bounded. Now $\frac{r^2}{1+r^2} \rightarrow 1$ as $r \rightarrow \pm\infty$ so there exists a Y such that $|r| > Y$ implies $\frac{r^2}{1+r^2} \geq \frac{1}{4}$. Then

$$\frac{y + r^2}{1 + r^2} \geq \frac{y}{1 + r^2} + \frac{r^2}{1 + r^2} \geq \frac{y}{1 + r^2} + \frac{1}{4} > \frac{1}{4}.$$

It follows that $f(z, r) > \frac{1}{2}$ outside of $S_{a,b}^Y \times [-Y, Y]$. But this latter region is compact and so $f(z, r)$ attains a minimum δ' on it. Setting $\delta = \min(\frac{1}{2}, \delta')$ completes the proof. $\square$

We can now verify the growth condition for $P_{m,k,\chi,\mathfrak{a}}(z)$. Let $\sigma_{\mathfrak{b}}$ be a scaling matrix for a cusp $\mathfrak{b}$. Then Equation (7), the cocycle condition, and Bruhat decomposition for $\sigma_{\mathfrak{a}}^{-1}\Gamma\sigma_{\mathfrak{b}}$ together give

$$(P_{m,k,\chi,\mathfrak{a}}|_k\sigma_{\mathfrak{b}})(z) \ll \sum_{(c,d) \in \mathbb{Z}^2 - \{\mathbf{0}\}} \frac{1}{|cz + d|^k}.$$

Now decompose this last sum as

$$\sum_{(c,d) \in \mathbb{Z}^2 - \{\mathbf{0}\}} \frac{1}{|cz + d|^k} = \sum_{d \neq 0} \frac{1}{d^k} + \sum_{c \neq 0} \sum_{d \in \mathbb{Z}} \frac{1}{|cz + d|^k} = 2 \sum_{d \geq 1} \frac{1}{d^k} + 2 \sum_{c \geq 1} \sum_{d \in \mathbb{Z}} \frac{1}{|cz + d|^k}.$$

Since the first sum is bounded, it suffices to show that the double sum is too as $z \rightarrow \infty$. To see this, let $y \geq 1$ and δ be as in Lemma 2.5. Then for any integer $N \geq 1$ we can write

$$\begin{aligned} \sum_{c \geq 1} \sum_{d \in \mathbb{Z}} \frac{1}{|cz + d|^k} &= \sum_{c+|d| \leq N} \frac{1}{|cz + d|^k} + \sum_{c+|d| > N} \frac{1}{|cz + d|^k} \\ &\leq \sum_{c+|d| \leq N} \frac{1}{|cz + d|^k} + \sum_{c+|d| > N} \frac{1}{(\delta|ci + d|)^k} \\ &\leq \sum_{c+|d| \leq N} \frac{1}{|cz + d|^k} + \frac{1}{\delta^k} \sum_{c+|d| > N} \frac{1}{|ci + d|^k}. \end{aligned}$$

As $\sum_{c \geq 1} \sum_{d \in \mathbb{Z}} \frac{1}{|ci + d|^k}$ converges by ??, the second sum tends to zero as $N \rightarrow \infty$. As for the first sum, it is finite and each term is bounded. Thus the double sum is bounded. This verifies the growth condition. We collect this work as a theorem:

Theorem 2.6. *Let $m \geq 0$, $k \geq 3$, χ be a Dirichlet character with conductor dividing the level, and $\mathfrak{a}$ be a cusp of $\Gamma \backslash \mathbb{H}$. Also assume χ is singular at $\mathfrak{a}$. The Poincaré series*

$$P_{m,k,\chi,\mathfrak{a}}(z) = \sum_{\gamma \in \Gamma_{\mathfrak{a}} \backslash \Gamma} \bar{\chi}(\gamma) \left(e^{\frac{2\pi i m z}{h_{\mathfrak{a}}}} \Big|_k \sigma_{\mathfrak{a}}^{-1} \gamma \right) (z),$$

is a weight k modular form with character χ on $\Gamma \backslash \mathbb{H}$.

For $m = 0$, we write $E_{k,\chi,\mathfrak{a}}(z) = P_{0,k,\chi,\mathfrak{a}}(z)$ and call $E_{k,\chi,\mathfrak{a}}(z)$ the (modular) **Eisenstein series** of weight k and character χ on $\Gamma \backslash \mathbb{H}$ at the $\mathfrak{a}$ cusp. It is defined by

$$E_{k,\chi,\mathfrak{a}}(z) = \sum_{\gamma \in \Gamma_{\mathfrak{a}} \backslash \Gamma} \bar{\chi}(\gamma) (1|_k \sigma_{\mathfrak{a}}^{-1} \gamma)(z).$$

If χ is the trivial character or $\mathfrak{a} = \infty$, we will drop these dependencies accordingly. In particular, we have already verified the following theorem:

Theorem 2.7. *Let $k \geq 3$, χ be Dirichlet character with conductor dividing the level, and $\mathfrak{a}$ be a cusp of $\Gamma \backslash \mathbb{H}$. Also assume χ is singular at $\mathfrak{a}$. The Eisenstein series*

$$E_{k,\chi,\mathfrak{a}}(z) = \sum_{\gamma \in \Gamma_{\mathfrak{a}} \backslash \Gamma} \bar{\chi}(\gamma) (1|_k \sigma_{\mathfrak{a}}^{-1} \gamma)(z),$$

is a weight k modular form with character χ on $\Gamma \backslash \mathbb{H}$.

We will now compute the Fourier series of the Poincaré series with positive index:

Proposition 2.8. *Let $\mathfrak{b}$ be a cusp of $\Gamma \backslash \mathbb{H}$ with scaling matrix $\sigma_{\mathfrak{b}}$. The Fourier series of $P_{m,k,\chi,\mathfrak{a}}(z)$ on $\Gamma \backslash \mathbb{H}$ at the $\mathfrak{b}$ cusp is given by*

$$(P_{m,k,\chi,\mathfrak{a}}|_k \sigma_{\mathfrak{b}})(z) = \sum_{t \geq 1} \left(\delta_{\mathfrak{a},\mathfrak{b}} \delta_{m,t} + 2\pi i^{-k} \left(\frac{\sqrt{th_{\mathfrak{a}}}}{\sqrt{mh_{\mathfrak{b}}}} \right)^{k-1} \sum_{c \in \mathcal{C}_{\mathfrak{a},\mathfrak{b}}} \frac{S_{\chi,\mathfrak{a},\mathfrak{b}}(m,t,c)}{ch_{\mathfrak{b}}} J_{k-1} \left(\frac{4\pi\sqrt{mt}}{c\sqrt{h_{\mathfrak{a}}h_{\mathfrak{b}}}} \right) \right) e^{\frac{2\pi itz}{h_{\mathfrak{b}}}}.$$

Proof. We have

$$(P_{m,k,\chi,\mathfrak{a}}|_k \sigma_{\mathfrak{b}})(z) = \sum_{\gamma \in \Gamma_{\mathfrak{a}} \backslash \Gamma} \bar{\chi}(\gamma) j(\sigma_{\mathfrak{a}}^{-1} \gamma \sigma_{\mathfrak{b}}, z)^{-k} e^{\frac{2\pi im \sigma_{\mathfrak{a}}^{-1} \gamma \sigma_{\mathfrak{b}} z}{h_{\mathfrak{a}}}}.$$

Using the Bruhat decomposition (in particular Remark 1.15), this becomes

$$(P_{m,k,\chi,\mathfrak{a}}|_k \sigma_{\mathfrak{b}})(z) = \delta_{\mathfrak{a},\mathfrak{b}} e^{\frac{2\pi im z}{h_{\mathfrak{b}}}} + \sum_{\substack{c \in \mathcal{C}_{\mathfrak{a},\mathfrak{b}}, d \in \mathcal{D}_{\mathfrak{a},\mathfrak{b}}(c) \\ \begin{pmatrix} a & * \\ c & d \end{pmatrix} \in \sigma_{\mathfrak{a}}^{-1} \Gamma \sigma_{\mathfrak{b}} \\ \ell \in \mathbb{Z}}} \bar{\chi}(ch_{\mathfrak{b}}\ell + d) \frac{e^{\frac{2\pi im}{h_{\mathfrak{a}}} \left(\frac{a}{c} - \frac{1}{c^2 z + c^2 h_{\mathfrak{b}}\ell + cd} \right)}}{(cz + ch_{\mathfrak{b}}\ell + d)^k},$$

where we have used the fact that

$$\frac{az + b}{cz + (ch_{\mathfrak{b}}\ell + d)} = \frac{a}{c} - \frac{1}{c^2 z + c^2 h_{\mathfrak{b}}\ell + cd}.$$

As χ has conductor dividing N and $c \equiv 0 \pmod{N}$, we further have

$$(P_{m,k,\chi,\mathfrak{a}}|_k \sigma_{\mathfrak{b}})(z) = \delta_{\mathfrak{a},\mathfrak{b}} e^{\frac{2\pi im z}{h_{\mathfrak{b}}}} + \sum_{\substack{c \in \mathcal{C}_{\mathfrak{a},\mathfrak{b}}, d \in \mathcal{D}_{\mathfrak{a},\mathfrak{b}}(c) \\ \begin{pmatrix} a & * \\ c & d \end{pmatrix} \in \sigma_{\mathfrak{a}}^{-1} \Gamma \sigma_{\mathfrak{b}}}} \bar{\chi}(d) \sum_{\ell \in \mathbb{Z}} \frac{e^{\frac{2\pi im}{h_{\mathfrak{a}}} \left(\frac{a}{c} - \frac{1}{c^2 z + c^2 h_{\mathfrak{b}}\ell + cd} \right)}}{(cz + ch_{\mathfrak{b}}\ell + d)^k}.$$

Now let

$$I_{c,d}(z) = \sum_{\ell \in \mathbb{Z}} \frac{e^{\frac{2\pi im}{h_{\mathfrak{a}}} \left(\frac{a}{c} - \frac{1}{c^2 z + c^2 h_{\mathfrak{b}}\ell + cd} \right)}}{(cz + ch_{\mathfrak{b}}\ell + d)^k}.$$

We will apply the Poisson summation formula to $I_{c,d}(z)$. By the identity theorem it suffices to apply the Poisson summation formula for $z = iy$ with $y > 0$. Accordingly, let $f(x)$ be given by

$$f(x) = \frac{e^{\frac{2\pi im}{h_{\mathfrak{a}}} \left(\frac{a}{d} - \frac{1}{c^2 h_{\mathfrak{b}}x + cd + ic^2 y} \right)}}{(ch_{\mathfrak{b}}x + d + icy)^k}.$$

Then $f(x)$ is absolutely integrable on $\mathbb{R}$ because it exhibits polynomial decay of order $k > 1$. We compute its Fourier transform

$$(\mathcal{F}f)(t) = \int_{-\infty}^{\infty} f(x) e^{-2\pi itx} dx = \int_{-\infty}^{\infty} \frac{e^{\frac{2\pi im}{h_{\mathfrak{a}}} \left(\frac{a}{c} - \frac{1}{c^2 h_{\mathfrak{b}}x + cd + ic^2 y} \right)}}{(ch_{\mathfrak{b}}x + d + icy)^k} e^{-2\pi itx} dx.$$

Complexify the integral to get

$$\int_{\text{Im}(w)=0} \frac{e^{\frac{2\pi im}{h_a} \left(\frac{a}{c} - \frac{1}{c^2 h_b w + cd + ic^2 y} \right)}}{(ch_b w + d + icy)^k} e^{-2\pi it w} dw.$$

Now make the change of variables $w \mapsto w - \frac{d}{ch_b} - i \frac{y}{h_b}$ to obtain

$$e^{2\pi i \left(\frac{ma}{ch_a} + \frac{td}{ch_b} \right) - \frac{2\pi ty}{h_b}} \int_{\text{Im}(w)=\frac{y}{h_b}} \frac{e^{-\frac{2\pi im}{c^2 h_a h_b w}}}{(ch_b w)^k} e^{-2\pi it w} dw.$$

The integrand is meromorphic with a pole at $w = 0$. Moreover, we have

$$\frac{1}{(ch_b w)^k} \ll \frac{1}{|w|^k}, \quad e^{-\frac{2\pi im}{c^2 h_a h_b w}} \ll e^{-\frac{2\pi m \text{Im}(w)}{h_a h_b |cw|^2}}, \quad \text{and} \quad e^{-2\pi it w} \ll e^{2\pi ty}.$$

The first expression has polynomial decay, the second expression is bounded, and the third expression exhibits exponential decay if and only if $t < 0$ and is bounded if $t = 0$. So when $t \leq 0$ we may take the limit as $\text{Im}(w) \rightarrow \infty$ by shifting the line of integration and conclude that the integral vanishes. It remains to compute the integral for $t \geq 1$. To do this, make the change of variables $w \mapsto -\frac{w}{2\pi it}$ to the last integral to rewrite it as

$$\begin{aligned} \frac{1}{2\pi it} \int_{\left(\frac{2\pi ty}{h_b}\right)} \frac{e^{-\frac{4\pi^2 mt}{c^2 h_a h_b w}}}{\left(-\frac{ch_b w}{2\pi it}\right)^k} e^w dw &= \frac{1}{2\pi it} \int_{\left(\frac{2\pi ty}{h_b}\right)} \frac{e^{w - \frac{4\pi^2 mt}{c^2 h_a h_b w}}}{\left(-\frac{ch_b w}{2\pi it}\right)^k} dw \\ &= \frac{(-1)^k (2\pi it)^{k-1}}{(ch_b)^k} \int_{\left(\frac{2\pi ty}{h_b}\right)} \frac{e^{w - \frac{4\pi^2 mt}{c^2 h_a h_b w}}}{w^k} dw \\ &= \frac{(-1)^k (2\pi it)^{k-1}}{(ch_b)^k} \int_{-\infty}^{(0^+)} \frac{e^{w - \frac{4\pi^2 mt}{c^2 h_a h_b w}}}{w^k} dw \\ &= 2\pi i^{-k} \left(\frac{\sqrt{th_a}}{\sqrt{mh_b}} \right)^{k-1} \frac{1}{ch_b} J_{k-1} \left(\frac{4\pi \sqrt{mt}}{c\sqrt{h_a h_b}} \right), \end{aligned}$$

where in the second to last line we have homotoped the line of integration to a Hankel contour about the negative real axis and in the last line we have used the Schlöflin integral representation for the J -Bessel function (see ??). So in total we obtain

$$(\mathcal{F}f)(t) = \begin{cases} \left(2\pi i^{-k} \left(\frac{\sqrt{th_a}}{\sqrt{mh_b}} \right)^{k-1} \frac{e^{2\pi i \left(\frac{ma}{ch_a} + \frac{td}{ch_b} \right)}}{ch_b} J_{k-1} \left(\frac{4\pi \sqrt{mt}}{c\sqrt{h_a h_b}} \right) \right) e^{-2\pi ty} & \text{if } t \geq 1, \\ 0 & \text{if } t \leq 0. \end{cases}$$

The Poisson summation formula and the identity theorem together give

$$I_{c,r}(z) = \sum_{t \geq 1} \left(2\pi i^{-k} \left(\frac{\sqrt{th_a}}{\sqrt{mh_b}} \right)^{k-1} \frac{e^{2\pi i \left(\frac{ma}{ch_a} + \frac{td}{ch_b} \right)}}{ch_b} J_{k-1} \left(\frac{4\pi \sqrt{mt}}{c\sqrt{h_a h_b}} \right) \right) e^{\frac{2\pi it z}{h_b}}.$$

Plugging this back into the Poincaré series gives

$$(P_{m,k,\chi,a}|_k\sigma_b)(z) = \delta_{a,b}e^{\frac{2\pi imz}{h_b}} + \sum_{\substack{c \in \mathcal{C}_{a,b}, d \in \mathcal{D}_{a,b}(c) \\ \begin{pmatrix} a & * \\ c & d \end{pmatrix} \in \sigma_a^{-1}\Gamma\sigma_b}} \bar{\chi}(d) \sum_{t \geq 1} \left(2\pi i^{-k} \left(\frac{\sqrt{th_a}}{\sqrt{mh_b}} \right)^{k-1} \frac{e^{2\pi i \left(\frac{ma}{ch_a} + \frac{td}{ch_b} \right)}}{ch_b} J_{k-1} \left(\frac{4\pi\sqrt{mt}}{c\sqrt{h_a h_b}} \right) \right) e^{\frac{2\pi itz}{h_b}}.$$

By the Fubini-Tonelli theorem we may interchange the sums over t and c . Doing this and collecting the sums over a and d , we at last obtain

$$(P_{m,k,\chi,a}|_k\sigma_b)(z) = \sum_{t \geq 1} \left(\delta_{a,b}\delta_{m,t} + 2\pi i^{-k} \left(\frac{\sqrt{th_a}}{\sqrt{mh_b}} \right)^{k-1} \sum_{c \in \mathcal{C}_{a,b}} \frac{S_{\chi,a,b}(m,t,c)}{ch_b} J_{k-1} \left(\frac{4\pi\sqrt{mt}}{c\sqrt{h_a h_b}} \right) \right) e^{\frac{2\pi itz}{h_b}},$$

which is our desired Fourier series. $\square$

An immediate consequence of Proposition 2.8 is that the Poincaré series $P_{m,k,\chi,a}(z)$ with positive index are cusp forms. In a similar manner, we can obtain the Fourier series of the Eisenstein series too:

Proposition 2.9. *Let $\mathfrak{b}$ be a cusp of $\Gamma \backslash \mathbb{H}$ with scaling matrix σ_b . The Fourier series of $E_{k,\chi,a}(z)$ on $\Gamma \backslash \mathbb{H}$ at the $\mathfrak{b}$ cusp is given by*

$$(E_{k,\chi,a}|_k\sigma_b)(z) = \delta_{a,b} + \sum_{t \geq 1} \left(\frac{(2\pi)^k i^{-k} t^{k-1}}{(k-1)!} \sum_{c \in \mathcal{C}_{a,b}} \frac{S_{\chi}(0,t,c)}{(ch_b)^k} \right) e^{\frac{2\pi itz}{h_b}}.$$

Proof. We have

$$(E_{k,\chi,a}|_k\sigma_b)(z) = \sum_{\gamma \in \Gamma_a \backslash \Gamma} \bar{\chi}(\gamma) j(\sigma_a^{-1}\gamma\sigma_b, z)^{-k}.$$

Using the Bruhat decomposition (in particular Remark 1.15), this becomes

$$(E_{k,\chi,a}|_k\sigma_b)(z) = \delta_{a,b}e^{\frac{2\pi imz}{h_b}} + \sum_{\substack{c \in \mathcal{C}_{a,b}, d \in \mathcal{D}_{a,b}(c) \\ \begin{pmatrix} a & * \\ c & d \end{pmatrix} \in \sigma_a^{-1}\Gamma\sigma_b \\ \ell \in \mathbb{Z}}} \bar{\chi}(ch_b\ell + d) \frac{1}{(cz + ch_b\ell + d)^k}.$$

As χ has conductor dividing N and $c \equiv 0 \pmod{N}$, we further have

$$(E_{k,\chi,a}|_k\sigma_b)(z) = \delta_{a,b}e^{\frac{2\pi imz}{h_b}} + \sum_{\substack{c \in \mathcal{C}_{a,b}, d \in \mathcal{D}_{a,b}(c) \\ \begin{pmatrix} a & * \\ c & d \end{pmatrix} \in \sigma_a^{-1}\Gamma\sigma_b}} \bar{\chi}(d) \sum_{\ell \in \mathbb{Z}} \frac{1}{(cz + ch_b\ell + d)^k}.$$

Now let

$$I_{c,d}(z) = \sum_{\ell \in \mathbb{Z}} \frac{1}{(cz + ch_b\ell + d)^k}.$$

We will apply the Poisson summation formula to $I_{c,d}(z)$. By the identity theorem it suffices to apply the Poisson summation formula for $z = iy$ with $y > 0$. Accordingly, let $f(x)$ be given by

$$f(x) = \frac{1}{(ch_{\mathfrak{b}}x + d + icy)^k}.$$

Then $f(x)$ is absolutely integrable on $\mathbb{R}$ because it exhibits polynomial decay of order $k > 1$. We compute its Fourier transform

$$(\mathcal{F}f)(t) = \int_{-\infty}^{\infty} f(x)e^{-2\pi itx} dx = \int_{-\infty}^{\infty} \frac{e^{-2\pi itx}}{(ch_{\mathfrak{b}}x + d + icy)^k} dx.$$

Complexify the integral to get

$$\int_{\text{Im}(w)=0} \frac{e^{-2\pi itw}}{(ch_{\mathfrak{b}}w + d + icy)^k} dw.$$

Now make the change of variables $w \mapsto w - \frac{d}{ch_{\mathfrak{b}}} - i\frac{y}{h_{\mathfrak{b}}}$ to obtain

$$e^{2\pi i\left(\frac{td}{ch_{\mathfrak{b}}}\right) - \frac{2\pi ty}{h_{\mathfrak{b}}}} \int_{\text{Im}(w)=\frac{y}{h_{\mathfrak{b}}}} \frac{e^{-2\pi itw}}{(ch_{\mathfrak{b}}w)^k} dw.$$

The integrand is meromorphic with a pole at $w = 0$. Moreover, we have

$$\frac{1}{(ch_{\mathfrak{b}}w)^k} \ll \frac{1}{|w|^k} \quad \text{and} \quad e^{-2\pi itw} \ll e^{2\pi ty}.$$

The first expression has polynomial decay, the second expression is bounded, and the third expression exhibits exponential decay if and only if $t < 0$ and is bounded if $t = 0$. So when $t \leq 0$ we may take the limit as $\text{Im}(w) \rightarrow \infty$ by shifting the line of integration and conclude that the integral vanishes. It remains to compute the integral for $t \geq 1$. To do this, make the change of variables $w \mapsto -\frac{w}{2\pi it}$ to the last integral to rewrite it as

$$\frac{1}{2\pi it} \int_{\left(\frac{2\pi ty}{h_{\mathfrak{b}}}\right)} \frac{e^w}{\left(-\frac{ch_{\mathfrak{b}}w}{2\pi it}\right)^k} dw = \frac{(-1)^k (2\pi it)^{k-1}}{(ch_{\mathfrak{b}})^k} \int_{\left(\frac{2\pi ty}{h_{\mathfrak{b}}}\right)} \frac{e^w}{w^k} dw.$$

The integrand of the last integral has a pole of order k at $w = 0$. To find the residue, the Laurent series of $\frac{e^w}{w^k}$ is

$$\frac{e^w}{w^k} = \sum_{n \geq -k} \frac{w^n}{(k+n)!},$$

and thus the residue of the integrand is $\frac{1}{(k-1)!}$. It follows that if we shift the line of integration to $(-a)$ in our last integral, for some $a > 0$, we pass by this pole and obtain

$$\frac{(2\pi)^k i^{-k} t^{k-1}}{(k-1)!(ch_{\mathfrak{b}})^k} + \frac{(-1)^k (2\pi it)^{k-1}}{(ch_{\mathfrak{b}})^k} \int_{(-a)} \frac{e^w}{w^k} dw.$$

Moreover, we have

$$\frac{1}{w^k} \ll \frac{1}{|w|^k} \quad \text{and} \quad e^w \ll e^{\text{Re}(w)}.$$

The first expression has polynomial decay while the second expression exhibits exponential decay provided $\operatorname{Re}(w) < 0$. Therefore we make take the limit as $\operatorname{Re}(w) \rightarrow \infty$ by shifting the line of integration again and conclude that the remaining integral vanishes. Altogether, we have show that

$$(\mathcal{F}f)(t) = \begin{cases} \left(\frac{(2\pi)^{k_i-k} t^{k-1}}{(k-1)!(ch_{\mathfrak{b}})^k} \frac{e^{2\pi i \left(\frac{dt}{ch_{\mathfrak{b}}} \right)}}{(ch_{\mathfrak{b}})^k} \right) e^{-\frac{2\pi ty}{h_{\mathfrak{b}}}} & \text{if } t \geq 1, \\ 0 & \text{if } t \leq 0. \end{cases}$$

The Poisson summation formula and the identity theorem together give

$$I_{c,r}(z) = \sum_{t \geq 1} \left(\frac{(2\pi)^{k_i-k} t^{k-1}}{(k-1)!(ch_{\mathfrak{b}})^k} \frac{e^{2\pi i \left(\frac{dt}{ch_{\mathfrak{b}}} \right)}}{(ch_{\mathfrak{b}})^k} \right) e^{\frac{2\pi itz}{h_{\mathfrak{b}}}}.$$

Substituting this back into the Eisenstein series gives a form of the Fourier series:

$$(E_{k,\chi,\mathfrak{a}}|_k \sigma_{\mathfrak{b}})(z) = \delta_{\mathfrak{a},\mathfrak{b}} + \sum_{\substack{c \in \mathcal{C}_{\mathfrak{a},\mathfrak{b}}, d \in \mathcal{D}_{\mathfrak{a},\mathfrak{b}}(c) \\ \begin{pmatrix} a & * \\ c & d \end{pmatrix} \in \sigma_{\mathfrak{a}}^{-1} \Gamma \sigma_{\mathfrak{b}}}} \bar{\chi}(d) \sum_{t \geq 1} \left(\frac{(2\pi)^{k_i-k} t^{k-1}}{(k-1)!(ch_{\mathfrak{b}})^k} \frac{e^{2\pi i \left(\frac{dt}{ch_{\mathfrak{b}}} \right)}}{(ch_{\mathfrak{b}})^k} \right) e^{\frac{2\pi itz}{h_{\mathfrak{b}}}}.$$

By the Fubini-Tonelli theorem we may interchange the sums over t and c . Doing this and collecting the sums over a and d , we at last obtain

$$(E_{k,\chi,\mathfrak{a}}|_k \sigma_{\mathfrak{b}})(z) = \delta_{\mathfrak{a},\mathfrak{b}} + \sum_{t \geq 1} \left(\frac{(2\pi)^{k_i-k} t^{k-1}}{(k-1)!} \sum_{c \in \mathcal{C}_{\mathfrak{a},\mathfrak{b}}} \frac{S_{\chi}(0, t, c)}{(ch_{\mathfrak{b}})^k} \right) e^{\frac{2\pi itz}{h_{\mathfrak{b}}}},$$

which is our desired Fourier series. □

An interesting observation from Proposition 2.9 is that $E_{k,\chi,\mathfrak{a}}|_k \sigma_{\mathfrak{b}}$ is necessarily a cusp form unless $\mathfrak{a}$ is equivalent to $\mathfrak{b}$. Also, as Γ was one of the congruence subgroups $\Gamma(N)$, $\Gamma_1(N)$, or $\Gamma_0(N)$, Propositions 2.8 and 2.9 together show that there exist cusp and non-cusp modular forms on these congruence subgroups. In particular, there exist modular forms on the Hecke and principal congruence subgroups for every level $N \geq 1$.

2.3 Inner Product Spaces of Modular Forms

Let $\mathcal{M}_k(\Gamma, \nu)$ denote the complex vector space of all weight k modular forms with nebentypus ν on $\Gamma \backslash \mathbb{H}$ and $\mathcal{S}_k(\Gamma, \nu)$ denote the associated subspace of cusp forms. Moreover, if ν is trivial, we will suppress this dependence. Note that if Γ_1 and Γ_2 are two congruence subgroups such that $\Gamma_1 \subseteq \Gamma_2$ then we have the inclusion

$$\mathcal{M}_k(\Gamma_2, \nu) \subseteq \mathcal{M}_k(\Gamma_1, \nu),$$

and this respects the subspace of cusp forms. So in general, the smaller the congruence subgroup the more modular forms there are. We can also multiply modular forms as the following proposition shows:

Proposition 2.10. *If $f \in \mathcal{M}_k(\Gamma, \nu)$ and $g \in \mathcal{M}_\ell(\Gamma, \xi)$, then $fg \in \mathcal{M}_{k+\ell}(\Gamma, \nu\xi)$. Moreover, fg is a cusp form if either f or g is.*

Proof. Holomorphy of fg is clear. For modularity, let $\gamma \in \Gamma$. We compute

$$((fg)|_{k+\ell}\gamma)(z) = (f|_k\gamma)(z)(g|_\ell\gamma)(z) = \nu(\gamma)\xi(\gamma)f(z)g(z) = (\nu\xi)(\gamma)(fg)(z),$$

which proves modularity. For the growth condition, let $\sigma_{\mathfrak{a}}$ be a scaling matrix at the cusp $\mathfrak{a}$ of $\Gamma \backslash \mathbb{H}$. Then

$$((fg)|_{k+\ell}\sigma_{\mathfrak{a}})(z) = (f|_k\sigma_{\mathfrak{a}})(z)(g|_\ell\sigma_{\mathfrak{a}})(z),$$

and the growth condition follows from that of f and g . In particular, fg is a cusp form if either f or g is. $\square$

Our goal is to turn $\mathcal{S}_k(\Gamma, \nu)$ into a complex Hilbert space. To this end, for $f, g \in \mathcal{S}_k(\Gamma, \nu)$ define their **Petersson inner product** by

$$\langle f, g \rangle_\Gamma = \frac{1}{V_\Gamma} \int_{\mathcal{F}_\Gamma} f(z) \overline{g(z)} \operatorname{Im}(z)^k d\mu.$$

If the congruence subgroup is clear from context we will suppress the dependence upon Γ . Since f and g have exponential decay at the cusps, the integrand is bounded and thus is locally absolutely uniformly convergent because we are integrating over a region of finite volume. The integrand is also Γ -invariant so that the integral is independent of the choice of fundamental domain. These two facts together imply that the Petersson inner product is well-defined. We will continue to use this notation even if f and g do not belong to $\mathcal{S}_k(\Gamma, \nu)$ provided the integral is locally absolutely uniformly convergent. A basic property of the Petersson inner product is that it is invariant with respect to the slash operator:

Proposition 2.11. *For any $f, g \in \mathcal{S}_k(\Gamma, \nu)$ and $\gamma \in \operatorname{SL}_2(\mathbb{Z})$, we have*

$$\langle f|_k\gamma, g|_k\gamma \rangle_{\gamma^{-1}\Gamma\gamma} = \langle f, g \rangle_\Gamma.$$

Proof. This is just a computation:

$$\begin{aligned} \langle f|_k\gamma, g|_k\gamma \rangle_{\gamma^{-1}\Gamma\gamma} &= \frac{1}{V_{\gamma^{-1}\Gamma\gamma}} \int_{\mathcal{F}_{\gamma^{-1}\Gamma\gamma}} (f|_k\gamma)(z) \overline{(g|_k\gamma)(z)} \operatorname{Im}(z)^k d\mu \\ &= \frac{1}{V_\Gamma} \int_{\mathcal{F}_{\gamma^{-1}\Gamma\gamma}} (f|_k\gamma)(z) \overline{(g|_k\gamma)(z)} \operatorname{Im}(z)^k d\mu && \text{Lemma 1.17} \\ &= \frac{1}{V_\Gamma} \int_{\mathcal{F}_{\gamma^{-1}\Gamma\gamma}} |j(\gamma, z)|^{-2k} f(\gamma z) \overline{g(\gamma z)} \operatorname{Im}(z)^k d\mu \\ &= \frac{1}{V_\Gamma} \int_{\mathcal{F}_{\gamma^{-1}\Gamma\gamma}} f(\gamma z) \overline{g(\gamma z)} \operatorname{Im}(\gamma z)^k d\mu && \text{Equation (5)} \\ &= \frac{1}{V_\Gamma} \int_{\mathcal{F}_\Gamma} f(z) \overline{g(z)} \operatorname{Im}(z)^k d\mu && z \mapsto \gamma^{-1}z \\ &= \langle f, g \rangle_\Gamma. \end{aligned}$$

$\square$

To show that $\mathcal{S}_k(\Gamma, \nu)$ is a complex Hilbert space, we will need to first show that this space is finite dimensional. In fact, we will show that $\mathcal{M}_k(\Gamma, \nu)$ is finite dimensional. To do this, we will require an important formula relating the weight k , the index of Γ in $\mathrm{SL}_2(\mathbb{Z})$, and the zeros of any $f \in \mathcal{M}_k(\Gamma, \nu)$. We require some notation to state this formula. For any $z \in \mathcal{F}_\Gamma$, let $v_z(f)$ be the index of the first nonzero term in the Taylor series of f at z . Equivalently, $v_z(f)$ is the order of vanishing of f at z . Since f is holomorphic, $v_z(f) \geq 0$. Also, modularity implies

$$v_{\gamma z}(f) = v_z(f),$$

for any $\gamma \in \Gamma$. This means $v_z(f)$ is well-defined for $z \in \mathcal{F}_\Gamma$. For a cusp $\mathfrak{a}$ of $\Gamma \backslash \mathbb{H}$, we let $v_{\mathfrak{a}}(f)$ be the index of the first nonzero term in the Fourier series of f at the $\mathfrak{a}$ cusp. That is, $v_{\mathfrak{a}}(f)$ is the index of the first nonzero term in the Fourier series of $f|_k \sigma_{\mathfrak{a}}$. To show $v_{\mathfrak{a}}(f)$ is well-defined, we need to verify that it is independent of the choice of scaling matrix $\sigma_{\mathfrak{a}}$. Since scaling matrices are determined up to composition on the right by an element of Γ_∞ , it suffices to verify independence of $\sigma_{\mathfrak{a}}$ by multiplication on the right by the generator γ_∞ of Γ_∞ . Independence then follows from Equation (6). Also, modularity implies

$$v_{\mathfrak{a}'}(f) = v_{\mathfrak{a}}(f),$$

for any cusp $\mathfrak{a}'$ in the same cusp class as $\mathfrak{a}$ since then $\mathfrak{a}' = \gamma \mathfrak{a}$ for some $\gamma \in \Gamma$. We can now state and prove the **valence formula**:

Theorem (Valence formula). *Suppose $f \in \mathcal{M}_k(\Gamma, \nu)$ is nonzero. For $z \in \mathcal{F}_\Gamma$, let $\eta_\Gamma(z) = \frac{|\Gamma_z|}{2}$. Then*

$$\sum_{\mathfrak{a} \in \Gamma \backslash \hat{\mathbb{Q}}} v_{\mathfrak{a}}(f) + \sum_{z \in \mathrm{SL}_2(\mathbb{Z}) \backslash \mathcal{F}_\Gamma} \frac{v_z(f)}{\eta_\Gamma(z)} = \frac{kd_\Gamma}{24}.$$

Proof. Suppose $F \in \mathcal{M}_k(\mathrm{SL}_2(\mathbb{Z}))$ is nonzero. We will first prove the formula for F and then extend the result to f . As F is nonzero, its zeros are isolated. We claim that all of the zeros of F satisfy $\mathrm{Im}(z) < T$ for some $T > 0$. To see this, let the Fourier series of F be given by

$$F(z) = \sum_{n \geq 0} a_n e^{2\pi i n z}.$$

Since F is holomorphic at infinity it is bounded as $z \rightarrow \infty$. As $e^{2\pi i n z} \rightarrow 0$ as $z \rightarrow \infty$, this means that the function

$$G(q) = \sum_{n \geq 0} a_n q^n,$$

converges provided $0 \leq |q| < R$ for some $0 < R < 1$ and where $G(0) = \lim_{z \rightarrow \infty} F(z)$. Notice from this that the order of vanishing of G at 0 is exactly $v_\infty(F)$. In any case, $G(z)$ is defined on a compact subset within the unit disk and so its zeros there are finite and necessarily isolated. In particular, there exists an r with $0 < r \leq R$ such that $G(q)$ is nonzero for $0 < |q| < r$. But then $F(z)$ is nonzero for $e^{-2\pi n y} < r$ which is to say $y > -\frac{\log(r)}{2\pi y}$. Taking $T > -\frac{\log(r)}{2\pi y}$ proves the claim. It follows that every zero of F has at least one $\mathrm{SL}_2(\mathbb{Z})$ -equivalent zero ρ in $\mathcal{F} \cap \{z \in \mathbb{C} : \mathrm{Im}(z) < T\}$. Let $\omega = e^{\frac{2\pi i}{3}}$ and suppose $\rho \neq i, \omega, 1 + \omega$. If

this zero lies in the interior of $\mathcal{F}$ then there is one such representative (as $\mathcal{F}$ is a fundamental domain). If this zero lies on the boundary of $\mathcal{F}$ then there are two such representatives ρ and $\rho \pm 1$ if $\operatorname{Re}(\rho) = \mp \frac{1}{2}$ or ρ and $-\frac{1}{\rho}$ if $|\rho| = 1$. For our choice of T , let $\eta = \sum_{1 \leq i \leq 6} \eta_i$ be the contour in Figure 2. In this contour, η_2, η_4, η_5 , and η_6 contain indentations of radius ε around the pairs of possible zeros of F as well as the points i, ω , and $1 + \omega$. We choose ε sufficiently small and the paths symmetrically so that each zero ρ occurs exactly once inside η while the points i, ω , and $1 + \omega$ are excluded. As every point $z \in \mathcal{F} - \{i, \omega, 1 + \omega\}$ that is not a zero has $v_z(F) = 0$, the argument principle gives

$$\frac{1}{2\pi i} \int_{\eta} \frac{F'}{F}(z) dz = \sum_{z \in \operatorname{SL}_2(\mathbb{Z}) \setminus \mathcal{F}_{\Gamma}}^* v_z(F),$$

where the $*$ indicates that we are excluding the orbits of i and ω . As each term in the right-hand side is zero, we can further write

$$\frac{1}{2\pi i} \int_{\eta} \frac{F'}{F}(z) dz = \sum_{z \in \operatorname{SL}_2(\mathbb{Z}) \setminus \mathcal{F}_{\Gamma}}^* \frac{v_z(F)}{\eta_{\operatorname{SL}_2(\mathbb{Z})}(z)}. \quad (8)$$

Now define integrals

$$I_i(\varepsilon) = \int_{\eta_i} \frac{F'}{F}(z) dz,$$

for $1 \leq i \leq 8$, so that

$$\frac{1}{2\pi i} \int_{\eta} \frac{F'}{F}(z) dz = \sum_{1 \leq i \leq 8} I_i(\varepsilon). \quad (9)$$

We will compute the integral on the right-hand side of Equation (9). Since $\operatorname{SL}_2(\mathbb{Z})$ is reduced at infinity, $F(z+1) = F(z)$. Taking the logarithmic derivative, we also see that $\frac{F'}{F}(z+1) = \frac{F'}{F}(z)$. Whence

$$I_2(\varepsilon) + I_8(\varepsilon) = 0. \quad (10)$$

In other words, the integrals along η_2 and η_8 cancel. For the integral along η_1 , the change of variables $e^{2\pi iz} \mapsto q$ takes η_1 to the circle $|q| = e^{-2\pi T}$ transversed once negative orientation. Thus

$$I_1(\varepsilon) = -\frac{1}{2\pi i} \int_{|q|=e^{-2\pi T}} \frac{G'}{G}(q) dq = -v_{\infty}(F), \quad (11)$$

where the last equality follows by the argument principle since $G(q)$ is holomorphic and nonzero for $0 < |q| \leq e^{-2\pi T}$ and the order of vanishing of G at 0 is $v_{\infty}(F)$. The integrals along η_3 and η_7 are computed in a similar manner. Let $\theta_{\varepsilon} > 0$ be the angle such that η_3 and η_7 end and begin at angles $\frac{\pi}{6} + \theta_{\varepsilon}$ and $\frac{5\pi}{6} - \theta_{\varepsilon}$ relative to ω and $1 + \omega$ respectively. For η_3 , write $F(z) = (z - \omega)^{v_{\omega}(F)} F_{\omega}(z)$ where $F_{\omega}(z)$ holomorphic and nonzero at ω . Then

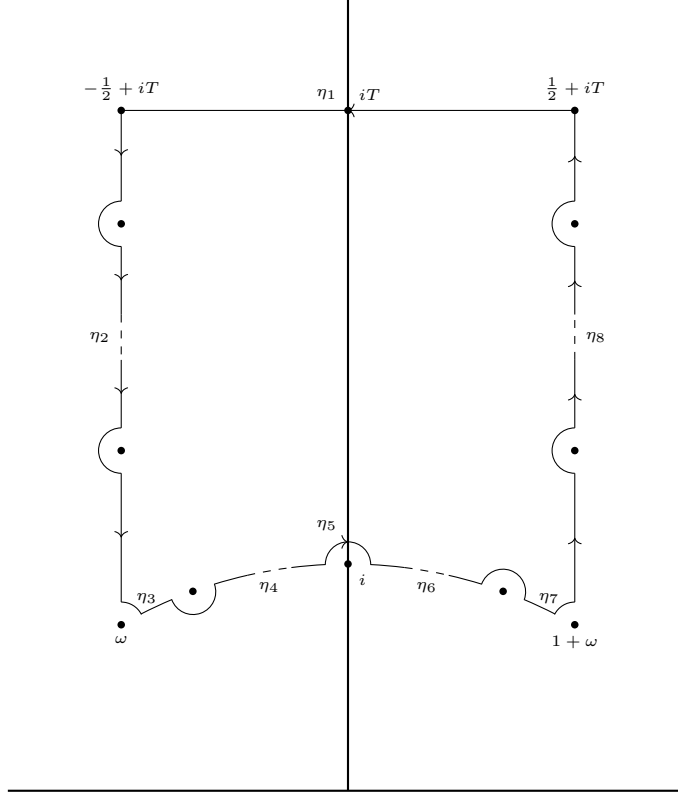


Figure 2: An indented standard fundamental domain contour.

$\frac{F'}{F}(z) = \frac{v_\omega(F)}{z-\omega} + \frac{F'_\omega}{F_\omega}(z)$ and making the change of variables $z \mapsto \omega + \varepsilon e^{i\theta}$ yields

$$\begin{aligned}
 I_3(\varepsilon) &= \frac{1}{2\pi i} \int_{\eta_4} \left(\frac{v_\omega(F)}{z-\omega} + \frac{F'_\omega}{F_\omega}(z) \right) dz \\
 &= -\frac{1}{2\pi} \int_{\frac{\pi}{6}+\theta_\varepsilon}^{\frac{\pi}{2}} v_\omega(F) d\theta - \frac{1}{2\pi} \int_{\frac{\pi}{6}+\theta_\varepsilon}^{\frac{\pi}{2}} \frac{F'_\omega}{F_\omega}(\omega + \varepsilon e^{i\theta}) \varepsilon e^{i\theta} d\theta \\
 &= -\frac{\frac{\pi}{3} - \theta_\varepsilon}{2\pi} v_\omega(F) - \frac{1}{2\pi} \int_{\frac{\pi}{6}+\theta_\varepsilon}^{\frac{\pi}{2}} \frac{F'_\omega}{F_\omega}(\omega + \varepsilon e^{i\theta}) \varepsilon e^{i\theta} d\theta.
 \end{aligned}$$

Taking the limit as $\varepsilon \rightarrow 0$, we see that the second integral vanishes while the first tends to $-\frac{v_\omega(F)}{6}$ because $\theta_\varepsilon \rightarrow 0$. By Proposition 1.11, this is $-\frac{v_\omega(F)}{2\eta_{\text{SL}_2(\mathbb{Z})}(\omega)}$. In short,

$$\lim_{\varepsilon \rightarrow 0} I_3(\varepsilon) = -\frac{v_\omega(F)}{2\eta_{\text{SL}_2(\mathbb{Z})}(\omega)}. \quad (12)$$

We argue similarly for η_7 . Write $F(z) = (z - (1 + \omega))^{v_{1+\omega}(F)} F_{1+\omega}(z)$ where $F_{1+\omega}(z)$ holomorphic and nonzero at $1 + \omega$. Then $\frac{F'}{F}(z) = \frac{v_{1+\omega}(F)}{z-(1+\omega)} + \frac{F'_{1+\omega}}{F_{1+\omega}}(z)$ and making the change of

variables $z \mapsto 1 + \omega + \varepsilon e^{i\theta}$ yields

$$\begin{aligned}
I_7(\varepsilon) &= \frac{1}{2\pi i} \int_{\eta_7} \left(\frac{v_{1+\omega}(F)}{z - (1 + \omega)} + \frac{F'_{1+\omega}(z)}{F_{1+\omega}} \right) dz \\
&= -\frac{1}{2\pi} \int_{\frac{\pi}{2}}^{\frac{5\pi}{6} - \theta_\varepsilon} v_{1+\omega}(F) d\theta - \frac{1}{2\pi} \int_{\frac{\pi}{2}}^{\frac{5\pi}{6} - \theta_\varepsilon} \frac{F'_{1+\omega}}{F_{1+\omega}} (1 + \omega + \varepsilon e^{i\theta}) \varepsilon e^{i\theta} d\theta \\
&= \frac{\frac{\pi}{3} - \theta_\varepsilon}{2\pi} v_{1+\omega}(F) - \frac{1}{2\pi} \int_{\frac{\pi}{2}}^{\frac{5\pi}{6} - \theta_\varepsilon} \frac{F'_{1+\omega}}{F_{1+\omega}} (1 + \omega + \varepsilon e^{i\theta}) \varepsilon e^{i\theta} d\theta.
\end{aligned}$$

Taking the limit as $\varepsilon \rightarrow 0$, the second integral vanishes while the first tends to $-\frac{v_{1+\omega}(F)}{6}$ because $\theta_\varepsilon \rightarrow 0$. By Proposition 1.11, this is $-\frac{v_{1+\omega}(F)}{2\eta_{\text{SL}_2(\mathbb{Z})}(1+\omega)}$. Thus

$$\lim_{\varepsilon \rightarrow 0} I_7(\varepsilon) = -\frac{v_{1+\omega}(F)}{\eta_{\text{SL}_2(\mathbb{Z})}(1+\omega)}. \quad (13)$$

For the integral along η_4 , write $F(z) = (z - i)^{v_i(F)} F_i(z)$ where $F_i(z)$ is holomorphic and nonzero at i . Then $\frac{F'}{F}(z) = \frac{v_i(F)}{z - i} + \frac{F'_i}{F_i}(z)$ and making the change of variables $z \mapsto i + \varepsilon e^{i\theta}$ yields

$$\begin{aligned}
I_5(\varepsilon) &= \frac{1}{2\pi i} \int_{\eta_5} \left(\frac{v_i(F)}{z - (i)} + \frac{F'_i(z)}{F_i} \right) dz \\
&= -\frac{1}{2\pi} \int_0^\pi v_i(F) d\theta - \frac{1}{2\pi} \int_0^\pi \frac{F'_i}{F_i} (i + \varepsilon e^{i\theta}) \varepsilon e^{i\theta} d\theta \\
&= -\frac{v_i(F)}{2} - \frac{1}{2\pi} \int_0^\pi \frac{F'_i}{F_i} (i + \varepsilon e^{i\theta}) \varepsilon e^{i\theta} d\theta.
\end{aligned}$$

Taking the limit as $\varepsilon \rightarrow 0$, the remaining integral vanishes. By Proposition 1.11, the first term is $-\frac{v_i(F)}{\eta_{\text{SL}_2(\mathbb{Z})}(i)}$. Thus

$$\lim_{\varepsilon \rightarrow 0} I_5(\varepsilon) = -\frac{v_i(F)}{\eta_{\text{SL}_2(\mathbb{Z})}(i)}. \quad (14)$$

For the integrals along η_4 and η_6 , let C_ε be the finite union of segments of η_4 that lie on the unit circle. Modularity implies $F(-\frac{1}{z}) = z^k F(z)$ because $S \in \text{SL}_2(\mathbb{Z})$. Taking the logarithmic derivative shows that $\frac{F'}{F}(-\frac{1}{z}) = \frac{k}{z} + \frac{F'}{F}(z)$. Whence

$$I_4(\varepsilon) + I_6(\varepsilon) = -\frac{1}{2\pi i} \int_{C_\varepsilon} \frac{k}{z} dz.$$

As $\varepsilon \rightarrow 0$, C_ε tends to the arc on the unit circle between radii $\frac{\pi}{2}$ and $\frac{2\pi}{3}$ with negative orientation. Upon making the change of variables $z \mapsto e^{i\theta}$, we find that

$$\lim_{\varepsilon \rightarrow 0} (I_4(\varepsilon) + I_6(\varepsilon)) = \frac{1}{2\pi} \int_{\frac{\pi}{2}}^{\frac{2\pi}{3}} k d\theta = \frac{k}{12}. \quad (15)$$

Taking the limit as $\varepsilon \rightarrow 0$ in Equation (9) and using Equations (10) to (15), we obtain

$$\lim_{\varepsilon \rightarrow 0} \frac{1}{2\pi i} \int_{\eta} \frac{F'}{F}(z) dz = \frac{k}{12} - v_{\infty}(F) - \frac{v_{\omega}(F)}{2\eta_{\mathrm{SL}_2(\mathbb{Z})}(\omega)} - \frac{v_{1+\omega}(F)}{2\eta_{\mathrm{SL}_2(\mathbb{Z})}(1+\omega)} - \frac{v_i(F)}{\eta_{\mathrm{SL}_2(\mathbb{Z})}(i)}. \quad (16)$$

Taking the limit as $\varepsilon \rightarrow 0$ in Equation (8) and using Equation (16) gives

$$v_{\infty}(F) + \frac{v_{\omega}(F)}{2\eta_{\mathrm{SL}_2(\mathbb{Z})}(\omega)} + \frac{v_{1+\omega}(F)}{2\eta_{\mathrm{SL}_2(\mathbb{Z})}(1+\omega)} + \frac{v_i(F)}{\eta_{\mathrm{SL}_2(\mathbb{Z})}(i)} + \sum_{z \in \mathrm{SL}_2(\mathbb{Z}) \setminus \mathcal{F}_{\Gamma}}^* \frac{v_z(F)}{\eta_{\mathrm{SL}_2(\mathbb{Z})}(z)} = \frac{k}{12}.$$

Since ω and $1 + \omega$ are $\mathrm{SL}_2(\mathbb{Z})$ -equivalent, $v_{\omega}(F) = v_{1+\omega}(F)$ and $\eta_{\mathrm{SL}_2(\mathbb{Z})}(\omega) = \eta_{\mathrm{SL}_2(\mathbb{Z})}(1 + \omega)$ by Proposition 1.11. Then our formula takes the form

$$v_{\infty}(F) + \sum_{z \in \mathrm{SL}_2(\mathbb{Z}) \setminus \mathcal{F}_{\Gamma}} \frac{v_z(F)}{\eta_{\mathrm{SL}_2(\mathbb{Z})}(z)} = \frac{k}{12},$$

which is precisely the valence formula for F . It remains to prove the formula for f . So let $f \in \mathcal{M}_k(\Gamma, \nu)$ be nonzero, let m be the order of ν , and define

$$F(z) = \prod_{\beta \in \Gamma \setminus \mathrm{SL}_2(\mathbb{Z})} ((f|_k \beta)(z))^m.$$

We claim that $F \in \mathcal{M}_{kmd_{\Gamma}}(\mathrm{SL}_2(\mathbb{Z}))$. First we must show that F is well-defined and therefore independent of the representatives β used. Indeed, if β and β' belong to the same orbit then $\beta'\beta^{-1} \in \Gamma$. Modularity and that ν is of order m together imply

$$((f|_k \beta')(z))^m = (((f|_k \beta' \beta^{-1})|_k \beta)(z))^m = ((f|_k \beta)(z))^m,$$

which proves that the F is well-defined. Since the product is finite (because Γ has finite index in $\mathrm{SL}_2(\mathbb{Z})$) holomorphy of F is clear. To prove modularity, let $\gamma \in \mathrm{SL}_2(\mathbb{Z})$. Because β runs over a complete set of representatives for $\Gamma \setminus \mathrm{SL}_2(\mathbb{Z})$ so does $\beta\gamma$. So each $\beta\gamma$ is of the form $\beta\gamma = \gamma_{\beta'}\beta'$ for some unique $\gamma_{\beta'} \in \Gamma$ and representative β' of $\Gamma \setminus \mathrm{SL}_2(\mathbb{Z})$. Using this fact and that ν is of order m , we compute

$$\begin{aligned} (F|_{kmd_{\Gamma}} \gamma)(z) &= \prod_{\beta \in \Gamma \setminus \mathrm{SL}_2(\mathbb{Z})} ((f|_k \beta \gamma)(z))^m \\ &= \prod_{\beta' \in \Gamma \setminus \mathrm{SL}_2(\mathbb{Z})} ((f|_k \gamma_{\beta'} \beta')(z))^m \\ &= \prod_{\beta' \in \Gamma \setminus \mathrm{SL}_2(\mathbb{Z})} ((f|_k \beta')(z))^m && \text{modularity} \\ &= F(z). \end{aligned}$$

This proves modularity of F . For the growth condition, let $\sigma_{\mathbf{a}}$ be a scaling matrix for a cusp $\mathbf{a}$ of $\Gamma_2 \setminus \mathbb{H}$. For any representative β , $\beta\sigma_{\mathbf{a}}$ takes ∞ to an element of $\hat{\mathbb{Q}}$ since $\beta \in \mathrm{GL}_2^+(\mathbb{Q})$. In other words, $\beta\sigma_{\mathbf{a}}\infty = \mathbf{b}$ for some cusp $\mathbf{b}$ of $\Gamma_1 \setminus \mathbb{H}$. Then

$$(F|_{k, \delta_{\Gamma}})(z) = \prod_{\beta \in \Gamma \setminus \mathrm{SL}_2(\mathbb{Z})} ((f|_k \beta \sigma_{\mathbf{a}})(z))^m,$$

and so the growth condition follows from that of f . This proves $F \in \mathcal{M}_{kmd_\Gamma}(\mathrm{SL}_2(\mathbb{Z}))$. Then the valence formula for F gives

$$v_\infty(F) + \sum_{z \in \mathrm{SL}_2(\mathbb{Z}) \backslash \mathcal{F}_\Gamma} \frac{v_z(F)}{\eta_{\mathrm{SL}_2(\mathbb{Z})}(z)} = \frac{km\delta_\Gamma}{12}. \quad (17)$$

By Proposition 1.14, the Fourier series of F is the product of the Fourier series of f at the $\mathfrak{a}$ cusp as $\mathfrak{a}$ runs over a complete set of cusps for $\Gamma \backslash \mathbb{H}$ each with multiplicity m . Since the Fourier series at the $\mathfrak{a}$ cusp is $h_\mathfrak{a}$ -periodic, the index of the first nonzero term in the Fourier series of F is

$$v_\infty(F) = \sum_{\mathfrak{a} \in \Gamma \backslash \hat{\mathbb{Q}}} \frac{mv_\mathfrak{a}(f)}{h_\mathfrak{a}}. \quad (18)$$

Lastly, by the orbit-stabilizer theorem **Todo: [xxx]** □

We can now prove that $\mathcal{M}_k(\Gamma, \nu)$ and $\mathcal{S}_k(\Gamma, \nu)$ are finite dimensional:

Theorem 2.12. *$\mathcal{M}_k(\Gamma, \nu)$ is finite dimensional. In particular, $\mathcal{S}_k(\Gamma, \nu)$ is finite dimensional too.*

Proof. As $\mathcal{S}_k(\Gamma, \nu)$ is a subspace of $\mathcal{M}_k(\Gamma, \nu)$, we only need to prove the first statement. Let $f \in \mathcal{M}_k(\Gamma, \nu)$. Recall that $v_\infty(f)$, $v_\mathfrak{a}(f)$ for $\mathfrak{a}$ a cusp of $\Gamma \backslash \mathbb{H}$, and $v_z(f)$ for $z \in \mathcal{F}_\Gamma$, are all nonnegative. In particular, $v_\infty(f) \geq 0$ and the valence formula implies that $v_\infty(f) \leq \frac{kd_\Gamma}{12}$. Let $n \geq \frac{kd_\Gamma}{12}$ and consider the linear map $F : \mathcal{M}_k(\Gamma, \nu) \rightarrow \mathbb{C}^n$ sending f to its first n Fourier coefficients. Then F is injective since $v_\infty(f) \leq n$ and hence $\mathcal{M}_k(\Gamma, \nu)$ is finite dimensional. □

We can now show that the Petersson inner product turns $\mathcal{S}_k(\Gamma, \nu)$ into a complex Hilbert space:

Proposition 2.13. *$\mathcal{S}_k(\Gamma, \nu)$ is a complex Hilbert space with respect to the Petersson inner product.*

Proof. Let $f, g \in \mathcal{S}_k(\Gamma, \nu)$. Linearity of the integral immediately implies that the Petersson inner product is linear on $\mathcal{S}_k(\Gamma, \nu)$. It is also positive definite since

$$\langle f, f \rangle = \frac{1}{V_\Gamma} \int_{\mathcal{F}_\Gamma} f(z) \overline{f(z)} \mathrm{Im}(z)^k d\mu = \frac{1}{V_\Gamma} \int_{\mathcal{F}_\Gamma} |f(z)|^2 \mathrm{Im}(z)^k d\mu \geq 0,$$

with equality if and only if f is identically zero. To see that it is conjugate symmetric, observe

$$\begin{aligned} \overline{\langle g, f \rangle} &= \overline{\frac{1}{V_\Gamma} \int_{\mathcal{F}_\Gamma} g(z) \overline{f(z)} \mathrm{Im}(z)^k d\mu} \\ &= \frac{1}{V_\Gamma} \int_{\mathcal{F}_\Gamma} \overline{g(z)} f(z) \mathrm{Im}(z)^k \overline{d\mu} \\ &= \frac{1}{V_\Gamma} \int_{\mathcal{F}_\Gamma} \overline{g(z)} f(z) \mathrm{Im}(z)^k d\mu & d\mu = \frac{dx dy}{y^2} \\ &= \frac{1}{V_\Gamma} \int_{\mathcal{F}_\Gamma} f(z) \overline{g(z)} \mathrm{Im}(z)^k d\mu \\ &= \langle f, g \rangle. \end{aligned}$$

So the Petersson inner product turns $\mathcal{S}_k(\Gamma, \nu)$ into a complex inner product space. Since $\mathcal{S}_k(\Gamma, \nu)$ is finite dimensional by Theorem 2.12, it follows immediately that $\mathcal{S}_k(\Gamma, \nu)$ is a complex Hilbert space. $\square$

We will now take $\Gamma = \Gamma_0(N)$ and $\nu = \chi$ for χ a Dirichlet character of conductor dividing N . Let $f \in \mathcal{S}_k(\Gamma, \nu)$ have Fourier coefficients $a_{\mathfrak{a}}(n)$ at the $\mathfrak{a}$ cusp. We define linear functionals $\phi_{m,k,\chi,\mathfrak{a}} : \mathcal{S}_k(\Gamma, \chi) \rightarrow \mathbb{C}$ by

$$\phi_{m,k,\chi,\mathfrak{a}}(f) = a_{\mathfrak{a}}(m).$$

Since $\mathcal{S}_k(\Gamma, \chi)$ is a finite dimensional complex Hilbert space, the Riesz representation theorem implies that there exists unique $v_{m,k,\chi,\mathfrak{a}} \in \mathcal{S}_k(\Gamma, \chi)$ such that

$$\langle f, v_{m,k,\chi,\mathfrak{a}} \rangle = \phi_{m,k,\chi,\mathfrak{a}}(f) = a_{\mathfrak{a}}(m).$$

We would like to know what these cusp forms are. It turns out that $v_{m,k,\chi,\mathfrak{a}}(z)$ will be the Poincaré series $P_{m,k,\chi,\mathfrak{a}}(z)$ of positive index up to a constant. To deduce this, recall the identity

$$\int_0^h e^{\frac{2\pi i(n-m)x}{h}} dx = h\delta_{n-m,0}, \quad (19)$$

for any $h \geq 1$ and $n, m \in \mathbb{Z}$. We will prove the following theorem:

Theorem 2.14. *Let $f \in \mathcal{S}_k(\Gamma, \chi)$ have Fourier coefficients $a_{\mathfrak{a}}(n)$ at the $\mathfrak{a}$ cusp of $\Gamma \backslash \mathbb{H}$. Then*

$$\langle f, P_{m,k,\chi,\mathfrak{a}} \rangle = \frac{\Gamma(k-1)h_{\mathfrak{a}}^k}{V_{\Gamma}(4\pi m)^{k-1}} a_{\mathfrak{a}}(m),$$

for all $m \geq 1$.

Proof. We compute the inner product as follows:

$$\begin{aligned}
\langle f, P_{m,k,\chi,\mathfrak{a}} \rangle &= \frac{1}{V_\Gamma} \int_{\mathcal{F}_\Gamma} f(z) \overline{P_{m,k,\chi,\mathfrak{a}}(z)} \text{Im}(z)^k d\mu \\
&= \frac{1}{V_\Gamma} \int_{\mathcal{F}_\Gamma} f(z) \sum_{\gamma \in \Gamma_{\mathfrak{a}} \backslash \Gamma} \chi(\gamma) \overline{\left(e^{\frac{2\pi i m z}{h_{\mathfrak{a}}}} \Big|_k \sigma_{\mathfrak{a}}^{-1} \gamma \right)}(z) \text{Im}(z)^k d\mu \\
&= \frac{1}{V_\Gamma} \int_{\mathcal{F}_\Gamma} f(z) \sum_{\gamma \in \Gamma_{\mathfrak{a}} \backslash \Gamma} \chi(\gamma) \overline{j(\sigma_{\mathfrak{a}}^{-1} \gamma, z)^{-k}} e^{-\frac{2\pi i m \overline{\sigma_{\mathfrak{a}}^{-1} \gamma z}}{h_{\mathfrak{a}}}} \text{Im}(z)^k d\mu \\
&= \frac{1}{V_\Gamma} \int_{\mathcal{F}_\Gamma} f(z) \sum_{\gamma \in \Gamma_{\mathfrak{a}} \backslash \Gamma} \chi(\gamma) j(\sigma_{\mathfrak{a}}^{-1} \gamma, z)^k e^{-\frac{2\pi i m \overline{\sigma_{\mathfrak{a}}^{-1} \gamma z}}{h_{\mathfrak{a}}}} \text{Im}(\sigma_{\mathfrak{a}}^{-1} \gamma z)^k d\mu && \text{Equation (5)} \\
&= \frac{1}{V_\Gamma} \int_{\mathcal{F}_\Gamma} \sum_{\gamma \in \Gamma_{\mathfrak{a}} \backslash \Gamma} \left(\frac{j(\sigma_{\mathfrak{a}}^{-1} \gamma, z)}{j(\gamma, z)} \right)^k f(\gamma z) e^{-\frac{2\pi i m \overline{\sigma_{\mathfrak{a}}^{-1} \gamma z}}{h_{\mathfrak{a}}}} \text{Im}(\sigma_{\mathfrak{a}}^{-1} \gamma z)^k d\mu && \text{modularity} \\
&= \frac{1}{V_\Gamma} \int_{\mathcal{F}_\Gamma} \sum_{\gamma \in \Gamma_{\mathfrak{a}} \backslash \Gamma} j(\sigma_{\mathfrak{a}}, \sigma_{\mathfrak{a}}^{-1} \gamma z)^{-k} f(\gamma z) e^{-\frac{2\pi i m \overline{\sigma_{\mathfrak{a}}^{-1} \gamma z}}{h_{\mathfrak{a}}}} \text{Im}(\sigma_{\mathfrak{a}}^{-1} \gamma z)^k d\mu && \text{cocycle condition} \\
&= \frac{1}{V_\Gamma} \int_{\mathcal{F}_{\sigma_{\mathfrak{a}}^{-1} \Gamma \sigma_{\mathfrak{a}}}} \sum_{\gamma \in \Gamma_{\mathfrak{a}} \backslash \Gamma} j(\sigma_{\mathfrak{a}}, \sigma_{\mathfrak{a}}^{-1} \gamma \sigma_{\mathfrak{a}} z)^{-k} f(\gamma \sigma_{\mathfrak{a}} z) e^{-\frac{2\pi i m \overline{\sigma_{\mathfrak{a}}^{-1} \gamma \sigma_{\mathfrak{a}} z}}{h_{\mathfrak{a}}}} \text{Im}(\sigma_{\mathfrak{a}}^{-1} \gamma \sigma_{\mathfrak{a}} z)^k d\mu \quad z \mapsto \sigma_{\mathfrak{a}} z \\
&= \frac{1}{V_\Gamma} \int_{\mathcal{F}_{\sigma_{\mathfrak{a}}^{-1} \Gamma \sigma_{\mathfrak{a}}}} \sum_{\gamma \in \Gamma_{\infty}^{(h_{\mathfrak{a}})} \backslash \sigma_{\mathfrak{a}}^{-1} \Gamma \sigma_{\mathfrak{a}}} j(\sigma_{\mathfrak{a}}, \gamma z)^{-k} f(\sigma_{\mathfrak{a}} \gamma z) e^{-\frac{2\pi i m \overline{\gamma z}}{h_{\mathfrak{a}}}} \text{Im}(\gamma z)^k d\mu \quad \gamma \mapsto \sigma_{\mathfrak{a}} \gamma \sigma_{\mathfrak{a}}^{-1} \\
&= \frac{1}{V_\Gamma} \int_{\mathcal{F}_{\sigma_{\mathfrak{a}}^{-1} \Gamma \sigma_{\mathfrak{a}}}} \sum_{\gamma \in \Gamma_{\infty}^{(h_{\mathfrak{a}})} \backslash \sigma_{\mathfrak{a}}^{-1} \Gamma \sigma_{\mathfrak{a}}} (f|_k \sigma_{\mathfrak{a}})(\gamma z) e^{-\frac{2\pi i m \overline{\gamma z}}{h_{\mathfrak{a}}}} \text{Im}(\gamma z)^k d\mu \\
&= \frac{1}{V_\Gamma} \int_{\Gamma_{\infty}^{(h_{\mathfrak{a}})} \backslash \mathbb{H}} (f|_k \sigma_{\mathfrak{a}})(z) e^{-\frac{2\pi i m \overline{z}}{h_{\mathfrak{a}}}} \text{Im}(z)^k d\mu && \text{unfolding.}
\end{aligned}$$

Substituting in the Fourier series of f at the $\mathfrak{a}$ cusp, we can finish the computation:

$$\begin{aligned}
\frac{1}{V_\Gamma} \int_{\Gamma_{\infty}^{(h_{\mathfrak{a}})} \backslash \mathbb{H}} (f|_k \sigma_{\mathfrak{a}})(z) e^{-\frac{2\pi i m \overline{z}}{h_{\mathfrak{a}}}} \text{Im}(z)^k d\mu &= \frac{1}{V_\Gamma} \int_0^\infty \int_0^{h_{\mathfrak{a}}} \sum_{n \geq 1} a_{\mathfrak{a}}(n) e^{\frac{2\pi i (n-m)x}{h_{\mathfrak{a}}}} e^{-\frac{2\pi (n+m)y}{h_{\mathfrak{a}}}} y^k \frac{dx dy}{y^2} \\
&= \frac{1}{V_\Gamma} \int_0^\infty \sum_{n \geq 1} \int_0^{h_{\mathfrak{a}}} a_{\mathfrak{a}}(n) e^{\frac{2\pi i (n-m)x}{h_{\mathfrak{a}}}} e^{-\frac{2\pi (n+m)y}{h_{\mathfrak{a}}}} y^k \frac{dx dy}{y^2} \quad \text{FTT} \\
&= \frac{h_{\mathfrak{a}}}{V_\Gamma} \int_0^\infty a_{\mathfrak{a}}(m) e^{-\frac{4\pi m y}{h_{\mathfrak{a}}}} y^k \frac{dy}{y^2},
\end{aligned}$$

where the last line follows because Equation (19) implies that the inner integral cuts off all

the terms except the diagonal $n = m$. Then

$$\begin{aligned}
\frac{h_a}{V_\Gamma} \int_0^\infty a_a(m) e^{-\frac{4\pi my}{h_a}} y^k \frac{dy}{y^2} &= \frac{h_a}{V_\Gamma} a_a(m) \int_0^\infty e^{-\frac{4\pi my}{h_a}} y^{k-1} \frac{dy}{y} \\
&= \frac{h_a}{V_\Gamma} a_a(m) \int_0^\infty e^{-\frac{4\pi my}{h_a}} y^{k-1} \frac{dy}{y} & y \mapsto \frac{h_a y}{4\pi m} \\
&= \frac{h_a^k}{V_\Gamma (4\pi m)^{k-1}} a_a(m) \int_0^\infty e^{-y} y^{k-1} \frac{dy}{y} \\
&= \frac{\Gamma(k-1) h_a^k}{V_\Gamma (4\pi m)^{k-1}} a_a(m).
\end{aligned}$$

This completes the proof. $\square$

It follows immediately from Theorem 2.14 that

$$v_{m,k,\chi,a}(z) = \frac{V_\Gamma (4\pi m)^{k-1}}{\Gamma(k-1) h_a^k} P_{m,k,\chi,a}(z).$$

Thus the Poincaré series $P_{m,k,\chi,a}(z)$ of positive index extract the Fourier coefficients of f at the $\mathfrak{a}$ cusp up to a constant. With Theorem 2.14 in hand we can prove the following result:

Theorem 2.15. *The Poincaré series $P_{m,k,\chi}(z)$ of positive index span $\mathcal{S}_k(\Gamma, \chi)$.*

Proof. Let $f \in \mathcal{S}_k(\Gamma, \chi)$ have Fourier coefficients $a(n)$. Since $\Gamma(k-1) \neq 0$, Theorem 2.14 implies $\langle f, P_{m,k,\chi,a} \rangle = 0$ if and only if $a(m) = 0$. So f is orthogonal to all the Poincaré series $P_{m,k,\chi}$ of positive index if and only if every Fourier coefficient $a(m)$ is zero. But this happens if and only if f is identically zero. $\square$

2.4 Double Coset Operators

We are ready to introduce a class of general operators depending upon double cosets. We will use these operators to define the diamond and Hecke operators. First, we define an operator closely related to the slash operator. For any integer $k \geq 1$ and any $\alpha \in \mathrm{GL}_2^+(\mathbb{Q})$ we define the **double slash operator** $\|_k \alpha : C(\mathbb{H}) \rightarrow C(\mathbb{H})$ to be the linear operator given by

$$(f \|_k \alpha)(z) = \det(\alpha)^{k-1} j(\alpha, z)^{-k} f(\alpha z).$$

If j is clear from context we will suppress this dependence accordingly. The relationship to the slash operator is clearly

$$(f \|_k \alpha) = \det(\alpha)^{\frac{k}{2}-1} (f|_k \alpha)(z). \quad (20)$$

The reason for the double slash operator is that it will be more easy to work with when discussing Fourier coefficients of modular forms. Note that the double slash operator is multiplicative because the determinant and the slash operator are. Now for any two congruence

subgroups Γ_1 and Γ_2 and any $\alpha \in \mathrm{GL}_2^+(\mathbb{Q})$, we define the **double coset operator** $[\Gamma_1\alpha\Gamma_2]_k$ on $\mathcal{S}_k(\Gamma_1)$ to be the linear operator given by

$$(f[\Gamma_1\alpha\Gamma_2]_k)(z) = \sum_{\beta \in \Gamma_1 \backslash \Gamma_1\alpha\Gamma_2} (f|_k\beta)(z).$$

This sum is finite by Proposition 1.6. It remains to check that $f[\Gamma_1\alpha\Gamma_2]_k$ is well-defined which amounts to showing that the sum is independent of the representatives β used. Indeed, if β and β' belong to the same orbit then $\beta'\beta^{-1} \in \Gamma_1$. Modularity then implies

$$(f|_k\beta')(z) = ((f|_k\beta'\beta^{-1})|_k\beta)(z) = (f|_k\beta)(z),$$

which proves that the $[\Gamma_1\alpha\Gamma_2]_k$ operator is well-defined. In fact, it also preserves modular forms:

Proposition 2.16. *For any two congruence subgroups Γ_1 and Γ_2 , $[\Gamma_1\alpha\Gamma_2]_k$ maps $\mathcal{S}_k(\Gamma_1)$ into $\mathcal{S}_k(\Gamma_2)$.*

Proof. Holomorphy is immediate since the sum in $f[\Gamma_1\alpha\Gamma_2]_k$ is finite by Proposition 1.6. For modularity, let $\gamma \in \Gamma_2$. Because β runs over a complete set of representatives for $\Gamma_1 \backslash \Gamma_1\alpha\Gamma_2$ so does $\beta\gamma$. So each $\beta\gamma$ is of the form $\beta\gamma = \gamma_{\beta'}\beta'$ for some unique $\gamma_{\beta'} \in \Gamma_1$ and representative β' of $\Gamma_1 \backslash \Gamma_1\alpha\Gamma_2$. In particular, $\det(\beta) = \det(\beta')$. Using these facts, we compute

$$\begin{aligned} (f[\Gamma_1\alpha\Gamma_2]_k|_k\gamma)(z) &= \sum_{\beta \in \Gamma_1 \backslash \Gamma_1\alpha\Gamma_2} \det(\beta)^{\frac{k}{2}-1} (f|_k\beta\gamma)(z) && \text{Equation (20)} \\ &= \sum_{\beta' \in \Gamma_1 \backslash \Gamma_1\alpha\Gamma_2} \det(\beta')^{\frac{k}{2}-1} (f|_k\gamma_{\beta'}\beta')(z) \\ &= \sum_{\beta' \in \Gamma_1 \backslash \Gamma_1\alpha\Gamma_2} \det(\beta')^{\frac{k}{2}-1} (f|_k\beta')(z) && \text{modularity} \\ &= \sum_{\beta' \in \Gamma_1 \backslash \Gamma_1\alpha\Gamma_2} (f|_k\beta')(z) \\ &= (f[\Gamma_1\alpha\Gamma_2]_k)(z) \end{aligned}$$

This proves modularity of $f[\Gamma_1\alpha\Gamma_2]_k$. For the growth condition, let $\sigma_{\mathfrak{a}}$ be a scaling matrix for a cusp $\mathfrak{a}$ of $\Gamma_2 \backslash \mathbb{H}$. For any representative β , $\beta\sigma_{\mathfrak{a}}$ takes ∞ to an element of $\hat{\mathbb{Q}}$ since $\beta \in \mathrm{GL}_2^+(\mathbb{Q})$. In other words, $\beta\sigma_{\mathfrak{a}}\infty = \mathfrak{b}$ for some cusp $\mathfrak{b}$ of $\Gamma_1 \backslash \mathbb{H}$. Then Equation (20) give

$$((f[\Gamma_1\alpha\Gamma_2]_k)|_k\sigma_{\mathfrak{a}})(z) = \sum_{\beta \in \Gamma_1 \backslash \Gamma_1\alpha\Gamma_2} \det(\beta)^{\frac{k}{2}-1} (f|_k\beta\sigma_{\mathfrak{a}})(z),$$

and so the growth condition follows from that of f . In particular, $f[\Gamma_1\alpha\Gamma_2]_k$ is a cusp form since f is. $\square$

The double coset operators are the most basic types of operators on modular forms. They are the building blocks needed to define the more important diamond and Hecke operators.

2.5 Diamond and Hecke Operators

The diamond and Hecke operators are special linear operators that are used to construct a linear theory of modular forms. They will also help us understand the Fourier coefficients. Throughout this discussion, we will obtain corresponding results for modular forms with nontrivial characters. We will discuss the diamond operator first. To define them, we need to consider both the congruence subgroups $\Gamma_1(N)$ and $\Gamma_0(N)$. Recall that $\Gamma_1(N) \subseteq \Gamma_0(N)$ and consider the map

$$\Gamma_0(N) \rightarrow (\mathbb{Z}/N\mathbb{Z})^* \quad \begin{pmatrix} a & b \\ c & d \end{pmatrix} \mapsto d \pmod{N},$$

(d is invertible modulo N since $c \equiv 0 \pmod{N}$ and $ad - bc = 1$). This is a surjective homomorphism and its kernel is exactly $\Gamma_1(N)$ so that $\Gamma_1(N)$ is a normal subgroup of $\Gamma_0(N)$ and $\Gamma_0(N)/\Gamma_1(N) \cong (\mathbb{Z}/N\mathbb{Z})^*$. Letting $\alpha = \begin{pmatrix} * & * \\ * & d \end{pmatrix} \in \Gamma_0(N)$ and $f \in \mathcal{S}_k(\Gamma_1)$, consider $(f[\Gamma_1(N)\alpha\Gamma_1(N)]_k)(z)$. This is only dependent upon the lower-right entry d of α taken modulo N . To see this, since $\Gamma_1(N)$ is normal in $\Gamma_0(N)$, we have $\alpha\Gamma_1(N) = \Gamma_1(N)\alpha$ so that $\Gamma_1(N)\alpha\Gamma_1(N) = \Gamma_1(N)\alpha$. This means there is only one orbit in $\Gamma_1(N)\backslash\Gamma_1(N)\alpha\Gamma_1(N)$. Therefore

$$(f[\Gamma_1(N)\alpha\Gamma_1(N)]_k)(z) = (f||_k\alpha)(z).$$

This induces an action of $\Gamma_0(N)$ on $\mathcal{S}_k(\Gamma_1)$, and since $\Gamma_1(N)$ acts trivially, this is really an action of $\Gamma_0(N)/\Gamma_1(N) \cong (\mathbb{Z}/N\mathbb{Z})^*$. In other words, we have an induced action that depends only upon the lower-right entry d of α taken modulo N . So for any d modulo N , we define the **diamond operator** $\langle d \rangle : \mathcal{S}_k(\Gamma_1(N)) \rightarrow \mathcal{S}_k(\Gamma_1(N))$ to be the linear operator given by

$$(\langle d \rangle f)(z) = (f||_k\alpha)(z),$$

for any $\alpha = \begin{pmatrix} * & * \\ * & d \end{pmatrix} \in \Gamma_0(N)$. Our discussion above has already shown that the diamond operators $\langle d \rangle$ are well-defined. Moreover, the diamond operators are also invertible with $\langle \bar{d} \rangle$ serving as an inverse and α^{-1} as a representative for the definition. Since

$$\begin{pmatrix} * & * \\ 0 & d \end{pmatrix} \begin{pmatrix} * & * \\ 0 & e \end{pmatrix} \equiv \begin{pmatrix} * & * \\ 0 & de \end{pmatrix} \pmod{N},$$

and the double slash operator is multiplicative, so are the diamond operators. One reason the diamond operators are useful is that they decompose $\mathcal{S}_k(\Gamma_1(N))$ into eigenspaces. For any Dirichlet character χ modulo N , we let

$$\mathcal{S}_k(N, \chi) = \{f \in \mathcal{S}_k(\Gamma_1(N)) : \langle d \rangle f = \chi(d)f \text{ for all } d \in (\mathbb{Z}/N\mathbb{Z})^*\},$$

be the χ -eigenspace. Also let $\mathcal{S}_k(N, \chi)$ be the corresponding subspace of cusp forms. Then $\mathcal{S}_k(\Gamma_1(N))$ admits a decomposition into these eigenspaces:

Proposition 2.17. *We have a direct sum decomposition*

$$\mathcal{S}_k(\Gamma_1(N)) = \bigoplus_{\chi \pmod{N}} \mathcal{S}_k(N, \chi).$$

Proof. The diamond operators give a representation of $\Gamma_0(N)/\Gamma_1(N) \cong (\mathbb{Z}/N\mathbb{Z})^*$ over $\mathcal{S}_k(\Gamma_1(N))$. Explicitly,

$$\Phi : (\mathbb{Z}/N\mathbb{Z})^* \times \mathcal{S}_k(\Gamma_1(N)) \rightarrow \mathcal{S}_k(\Gamma_1(N)) \quad (d, f) \mapsto \langle d \rangle f.$$

But any representation of a finite abelian group over $\mathbb{C}$ is completely reducible with respect to the characters of the group and every irreducible subrepresentation is 1-dimensional (see ??). Since the characters of $(\mathbb{Z}/N\mathbb{Z})^*$ are given by Dirichlet characters, the decomposition as a direct sum follows. $\square$

Proposition 2.17 shows that the diamond operators sieve modular forms on $\Gamma_1(N) \backslash \mathbb{H}$ with trivial character in terms of modular forms on $\Gamma_0(N) \backslash \mathbb{H}$ with nontrivial characters. In particular, $\mathcal{S}_k(N, \chi) = \mathcal{S}_k(\Gamma_0(N), \chi)$. So by Proposition 2.17, we have

$$\mathcal{S}_k(\Gamma_1(N)) = \bigoplus_{\chi \pmod{N}} \mathcal{S}_k(\Gamma_0(N), \chi).$$

This fact clarifies why it is necessary to consider modular forms with nontrivial characters. In light of Remark 2.2, note that $\mathcal{S}_k(N, \chi)$ is necessarily empty if k is odd and $\chi(-1) \neq -1$. So in the case of odd k , the decomposition in Proposition 2.17 restricts to that of odd characters. Now it is time to define the Hecke operators. For a prime p , we define the p -th **Hecke operator** $T_p : \mathcal{S}_k(\Gamma_1(N)) \rightarrow \mathcal{S}_k(\Gamma_1(N))$ to be the linear operator given by

$$(T_p f)(z) = \left(\left[\Gamma_1(N) \begin{pmatrix} 1 & 0 \\ 0 & p \end{pmatrix} \Gamma_1(N) \right]_k f \right)(z).$$

We will start discussing properties of the diamond and Hecke operators, but first we prove an important lemma classifying the double coset $\Gamma_1(N) \begin{pmatrix} 1 & 0 \\ 0 & p \end{pmatrix} \Gamma_1(N)$:

Lemma 2.18. *Let p be a prime. Then*

$$\Gamma_1(N) \begin{pmatrix} 1 & 0 \\ 0 & p \end{pmatrix} \Gamma_1(N) = \left\{ \gamma \in \mathrm{GL}_2^+(\mathbb{Z}) : \gamma \equiv \begin{pmatrix} 1 & * \\ 0 & p \end{pmatrix} \pmod{N} \text{ and } \det(\gamma) = p \right\}.$$

Proof. For the forward containment, it is clear that any element of $\Gamma_1(N) \begin{pmatrix} 1 & 0 \\ 0 & p \end{pmatrix} \Gamma_1(N)$ has determinant p and that

$$\begin{pmatrix} 1 & * \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & p \end{pmatrix} \begin{pmatrix} 1 & * \\ 0 & 1 \end{pmatrix} \equiv \begin{pmatrix} 1 & 0 \\ 0 & p \end{pmatrix} \pmod{N},$$

so the forward containment holds. For the reverse containment, let L be the set of 2×1 column vectors with entries in $\mathbb{Z}$, in particular $L \cong \mathbb{Z}^2$, and set

$$L_0 = \left\{ \begin{pmatrix} x \\ y \end{pmatrix} \in L : y \equiv 0 \pmod{N} \right\}.$$

Then $\mathrm{GL}_2^+(\mathbb{Z})$ acts on L_0 on the left by matrix multiplication. Now suppose $\gamma \in \mathrm{GL}_2^+(\mathbb{Z})$ is such that $\gamma \equiv \begin{pmatrix} 1 & * \\ 0 & p \end{pmatrix} \pmod{N}$ and $\det(\gamma) = p$. Clearly L_0 has index N in L . As the action

of γ on L_0 multiplies the lower entry by p , γL_0 has index p in L_0 . These two facts together imply

$$[L : \gamma L_0] = [L : L_0][L_0 : \gamma L_0] = Np.$$

As γL_0 is a finitely generated abelian group (L is a rank 2 free module over $\mathbb{Z}$ and γL_0 is a subgroup), the structure theorem for finitely generated abelian groups implies that there exists a basis u, v of L with $\det(u, v) = 1$, so $L = u\mathbb{Z} \oplus v\mathbb{Z}$, and positive integers m and n with $m \mid n$, $mn = Np$, and such that $\gamma L_0 = mu\mathbb{Z} \oplus nv\mathbb{Z}$. Moreover, as $m \mid n$ any element of γL_0 is zero modulo m . Now $\begin{pmatrix} 1 \\ 0 \end{pmatrix} \in L_0$ and by our choice of γ , we have $\begin{pmatrix} 1 \\ 0 \end{pmatrix} \in \gamma L_0$ as well. Thus $\begin{pmatrix} 1 \\ 0 \end{pmatrix} \equiv \begin{pmatrix} 0 \\ 0 \end{pmatrix} \pmod{m}$. This forces $m = 1$ and $n = Np$ so that $\gamma L_0 = u\mathbb{Z} \oplus Npv\mathbb{Z}$. As there are unique subgroups of index N and p between $\mathbb{Z}$ and $Np\mathbb{Z}$, namely $N\mathbb{Z}$ and $p\mathbb{Z}$ respectively, we have

$$L_0 = u\mathbb{Z} \oplus Nv\mathbb{Z}, \quad \gamma L = u\mathbb{Z} \oplus pv\mathbb{Z}, \quad \text{and} \quad \gamma L_0 = u\mathbb{Z} \oplus Npv\mathbb{Z}.$$

Now let $\gamma_1 = (u, v)$. Since $u \in L_0$, we have that $\gamma_1 \in \Gamma_0(N)$. Now set $\gamma_2 = \left(\gamma_1 \begin{pmatrix} 1 & 0 \\ 0 & p \end{pmatrix}\right)^{-1} \gamma$. Then $\gamma \in \text{GL}_2^+(\mathbb{Q})$ with $\det(\gamma_2) = 1$. It follows that

$$\gamma = \gamma_1 \begin{pmatrix} 1 & 0 \\ 0 & p \end{pmatrix} \gamma_2.$$

Let e_1 and e_2 be the standard basis vectors of L . Then $\gamma e_1 \in \gamma L_0$ implies that $\gamma e_1 = au + pcv$ with $a, c \in \mathbb{Z}$ and $N \mid c$. Similarly, $\gamma e_2 \in \gamma L$ implies that $\gamma e_2 = bu + pdv$ with $b, d \in \mathbb{Z}$. Letting $u = \begin{pmatrix} u_1 \\ u_2 \end{pmatrix}$ and $v = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix}$, these facts together imply that $\gamma_2 = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ and the above identity can be written in the form

$$\gamma = \begin{pmatrix} au_1 + pcv_1 & bu_1 + pdv_1 \\ au_2 + pcv_2 & bu_2 + pdv_2 \end{pmatrix} = \begin{pmatrix} u_1 & v_1 \\ u_2 & v_2 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & p \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix}.$$

As $\det(\gamma_2) = 1$ and $N \mid c$, we conclude that $\gamma_2 \in \Gamma_0(N)$. Hence $\gamma \in \Gamma_0(N) \begin{pmatrix} 1 & 0 \\ 0 & p \end{pmatrix} \Gamma_0(N)$. Moreover, as $\gamma_1, \gamma_2 \in \Gamma_0(N)$ it follows that they belong to $\Gamma_1(N)$ if u_1 or a is equivalent to 1 modulo N respectively. But as $\gamma \equiv \begin{pmatrix} 1 & * \\ 0 & p \end{pmatrix} \pmod{N}$ and $N \mid c$, we have $au_1 \equiv 1 \pmod{N}$. Thus if γ_1 or γ_2 belongs to $\Gamma_1(N)$ then they both do. So to complete the reverse containment it suffices to show

$$\Gamma_0(N) \begin{pmatrix} 1 & 0 \\ 0 & p \end{pmatrix} \Gamma_0(N) = \Gamma_1(N) \begin{pmatrix} 1 & 0 \\ 0 & p \end{pmatrix} \Gamma_0(N).$$

The reverse containment in this double coset equality is clear since $\Gamma_1(N) \subseteq \Gamma_0(N)$. For the forward containment, let $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma_0(N)$. Then we need to show that there exists a matrix $\delta \in \Gamma_1(N)$ such that

$$\gamma \begin{pmatrix} 1 & 0 \\ 0 & p \end{pmatrix} \Gamma_0(N) = \delta \begin{pmatrix} 1 & 0 \\ 0 & p \end{pmatrix} \Gamma_0(N).$$

Equivalently, there exists a matrix $\delta' \in \Gamma_1(N)$ (that is the inverse of δ) such that

$$\begin{pmatrix} 1 & 0 \\ 0 & p^{-1} \end{pmatrix} \delta' \gamma \begin{pmatrix} 1 & 0 \\ 0 & p \end{pmatrix} \in \Gamma_0(N).$$

If $p \mid N$, taking $\delta' = \begin{pmatrix} cd+1 & -1 \\ -cd & 1 \end{pmatrix} \in \Gamma_1(N)$ gives

$$\begin{pmatrix} 1 & 0 \\ 0 & p^{-1} \end{pmatrix} \delta' \gamma \begin{pmatrix} 1 & 0 \\ 0 & p \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & p^{-1} \end{pmatrix} \begin{pmatrix} cd+1 & -1 \\ -cd & 1 \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & p \end{pmatrix} = \begin{pmatrix} * & * \\ p^{-1}c(1-ad) & * \end{pmatrix},$$

which belongs to $\Gamma_0(N)$ because $p \mid N$, $N \mid c$, and $N \mid 1 - ad$ (as $ad - bc = 1$). If $p \nmid N$ then the Chinese remainder theorem implies that there exists a $d' \in \mathbb{Z}$ with $d' \equiv 1 \pmod{N}$ and $d' \equiv -a \pmod{p}$. Necessarily $(c, d') = 1$ because $N \mid c$. Thus there exists δ' of the form $\delta' = \begin{pmatrix} a' & b' \\ c & d' \end{pmatrix} \in \Gamma_1(N)$, and we have

$$\begin{pmatrix} 1 & 0 \\ 0 & p^{-1} \end{pmatrix} \delta' \gamma \begin{pmatrix} 1 & 0 \\ 0 & p \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & p^{-1} \end{pmatrix} \begin{pmatrix} a' & b' \\ c & d' \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & p \end{pmatrix} = \begin{pmatrix} * & * \\ p^{-1}c(a+d') & * \end{pmatrix},$$

which again belongs to $\Gamma_0(N)$ because $p \mid d' + a$ and $N \mid c$. This prove the forward containment, and hence the original reverse containment, thus completing the proof. $\square$

With Lemma 2.18, it is not too hard to see that the diamond and Hecke operators commute:

Proposition 2.19. *For every $d \in (\mathbb{Z}/N\mathbb{Z})^*$ and prime p , the diamond operators $\langle d \rangle$ and Hecke operators T_p on $\mathcal{S}_k(\Gamma_1(N))$ commute:*

$$\langle d \rangle T_p = T_p \langle d \rangle$$

Proof. Let $\alpha = \begin{pmatrix} 1 & 0 \\ 0 & p \end{pmatrix}$. For $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma_0(N)$, we have

$$\gamma \alpha \gamma^{-1} \equiv \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & p \end{pmatrix} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} = \begin{pmatrix} 1 & (p-1)ab \\ 0 & p \end{pmatrix} \pmod{N},$$

because $c \equiv 0 \pmod{N}$, $ad - bc = 1$, and $ad \equiv 1 \pmod{N}$. By Lemma 2.18, $\gamma \alpha \gamma^{-1} \in \Gamma_1(N) \alpha \Gamma_1(N)$ and so we can use this representative in place of α . On the one hand,

$$\Gamma_1(N) \alpha \Gamma_1(N) = \bigcup_{\beta \in \Gamma_1(N) \setminus \Gamma_1(N) \alpha \Gamma_1(N)} \Gamma_1(N) \beta.$$

On the other hand, using $\gamma \alpha \gamma^{-1}$ in place of α and the normality of $\Gamma_1(N)$ in $\Gamma_0(N)$, we have

$$\begin{aligned} \Gamma_1(N) \alpha \Gamma_1(N) &= \Gamma_1(N) \gamma \alpha \gamma^{-1} \Gamma_1(N) \\ &= \gamma \Gamma_1(N) \alpha \Gamma_1(N) \gamma^{-1} \\ &= \gamma \bigcup_{\beta \in \Gamma_1(N) \setminus \Gamma_1(N) \alpha \Gamma_1(N)} \Gamma_1(N) \beta \gamma^{-1} \\ &= \bigcup_{\beta \in \Gamma_1(N) \setminus \Gamma_1(N) \alpha \Gamma_1(N)} \Gamma_1(N) \gamma \beta \gamma^{-1}. \end{aligned}$$

Upon comparing these two decompositions of $\Gamma_1(N) \alpha \Gamma_1(N)$ gives

$$\bigcup_{\beta \in \Gamma_1(N) \setminus \Gamma_1(N) \alpha \Gamma_1(N)} \Gamma_1(N) \beta = \bigcup_{\beta \in \Gamma_1(N) \setminus \Gamma_1(N) \alpha \Gamma_1(N)} \Gamma_1(N) \gamma \beta \gamma^{-1}.$$

Now let $f \in \mathcal{S}_k(\Gamma_1(N))$. Then this equivalence of unions implies

$$\langle d \rangle T_p f = \sum_{\beta \in \Gamma_1(N) \setminus \Gamma_1(N) \alpha \Gamma_1(N)} f \|_k \beta \gamma = \sum_{\beta \in \Gamma_1(N) \setminus \Gamma_1(N) \alpha \Gamma_1(N)} f \|_k \gamma \beta = T_p \langle d \rangle f. \quad \square$$

Using Lemma 2.18 we can obtain an explicit description of the Hecke operator T_p :

Proposition 2.20. *Let $f \in \mathcal{S}_k(\Gamma_1(N))$. Then the Hecke operator T_p acts on f as follows:*

$$(T_p f)(z) = \begin{cases} \sum_{j \pmod{p}} \left(f \Big|_k \begin{pmatrix} 1 & j \\ 0 & p \end{pmatrix} \right) (z) + \left(f \Big|_k \begin{pmatrix} m & n \\ N & p \end{pmatrix} \begin{pmatrix} p & 0 \\ 0 & 1 \end{pmatrix} \right) (z) & \text{if } p \nmid N, \\ \sum_{j \pmod{p}} \left(f \Big|_k \begin{pmatrix} 1 & j \\ 0 & p \end{pmatrix} \right) (z) & \text{if } p \mid N, \end{cases}$$

where m and n are chosen such that $\det \left(\begin{pmatrix} m & n \\ N & p \end{pmatrix} \right) = 1$.

Proof. Let $\alpha = \begin{pmatrix} 1 & 0 \\ 0 & p \end{pmatrix}$ and define

$$\beta_j = \begin{pmatrix} 1 & j \\ 0 & p \end{pmatrix} \quad \text{and} \quad \beta_\infty = \begin{pmatrix} m & n \\ N & p \end{pmatrix} \begin{pmatrix} p & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} pm & n \\ Np & p \end{pmatrix},$$

for j taken modulo p and where m and n are chosen such that $\det \left(\begin{pmatrix} m & n \\ N & p \end{pmatrix} \right) = 1$. It will suffice to show that $\beta_1, \dots, \beta_{p-1}$ and $\beta_1, \dots, \beta_{p-1}, \beta_\infty$ run over complete sets of representatives for $\Gamma_1(N) \backslash \Gamma_1(N) \alpha \Gamma_1(N)$ depending on if $p \mid N$ or not. To accomplish this, we will find a complete set of representatives for $(\alpha^{-1} \Gamma_1(N) \alpha \cap \Gamma_1(N)) \backslash \Gamma_1(N)$ and then use Lemma 1.5. First we require an explicit description of $\alpha^{-1} \Gamma_1(N) \alpha \cap \Gamma_1(N)$. Let

$$\Gamma^0(p) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \text{SL}_2(\mathbb{Z}) : \begin{pmatrix} a & b \\ c & d \end{pmatrix} \equiv \begin{pmatrix} * & 0 \\ * & * \end{pmatrix} \pmod{p} \right\},$$

and define

$$\Gamma_1^0(N, p) = \Gamma_1(N) \cap \Gamma^0(p).$$

We claim $\alpha^{-1} \Gamma_1(N) \alpha \cap \Gamma_1(N) = \Gamma_1^0(N, p)$. First, let $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma_1(N)$ and observe that

$$\alpha^{-1} \gamma \alpha = \begin{pmatrix} 1 & 0 \\ 0 & p^{-1} \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & p \end{pmatrix} = \begin{pmatrix} a & pd \\ p^{-1}c & d \end{pmatrix}.$$

So for the forward containment, if $\alpha^{-1} \gamma \alpha \in \alpha^{-1} \Gamma_1(N) \alpha \cap \Gamma_1(N)$ then $\alpha^{-1} \gamma \alpha \in \Gamma_1(N)$ and thus $p \mid c$ so that $\alpha^{-1} \gamma \alpha \in \text{SL}_2(\mathbb{Z})$. Moreover, the previous computation implies $\alpha^{-1} \gamma \alpha \in \Gamma_1^0(N, p)$ and so $\alpha^{-1} \Gamma_1(N) \alpha \cap \Gamma_1(N) \subseteq \Gamma_1^0(N, p)$ proving the forward containment. For the reverse containment, suppose $\eta = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma_1^0(N, p)$. Then $b = pk$ for some $k \in \mathbb{Z}$. Now observe

$$\eta = \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & p^{-1} \end{pmatrix} \begin{pmatrix} a & k \\ pc & d \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & p \end{pmatrix} = \alpha^{-1} \gamma \alpha,$$

where $\gamma = \begin{pmatrix} a & k \\ pc & d \end{pmatrix}$. As $\eta \in \Gamma_1(N)$ we conclude $\gamma \in \Gamma_1(N)$ as well. Therefore $\Gamma_1^0(N, p) \subseteq \alpha^{-1} \Gamma_1(N) \alpha \cap \Gamma_1(N)$ proving the reverse containment. Now let

$$\beta'_j = \begin{pmatrix} 1 & j \\ 0 & 1 \end{pmatrix} \quad \text{and} \quad \beta'_\infty = \begin{pmatrix} pm & n \\ N & 1 \end{pmatrix},$$

for j taken modulo p and where m and n are as before. Clearly $\beta'_j \in \Gamma_1(N)$ for all j . As $pm - Nn = 1$, we have $pm \equiv 1 \pmod{N}$ so that $\beta'_\infty \in \Gamma_1(N)$ as well. We claim that $\beta'_1, \dots, \beta'_{p-1}$ and $\beta'_1, \dots, \beta'_{p-1}, \beta'_\infty$ run over complete sets of representatives for $(\alpha^{-1}\Gamma_1(N)\alpha \cap \Gamma_1(N)) \setminus \Gamma_1(N)$ depending on if $p \mid N$ or not. Again let $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma_1(N)$ and consider

$$\gamma\beta'_j{}^{-1} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} 1 & -j \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} a & b - aj \\ c & d - cj \end{pmatrix}.$$

As $\gamma\beta'_j{}^{-1} \in \Gamma_1(N)$ because both γ and β'_j are, $\gamma\beta'_j{}^{-1} \in \alpha^{-1}\Gamma_1(N)\alpha \cap \Gamma_1(N) = \Gamma_1^0(N, p)$ for some j if and only if

$$b \equiv aj \pmod{p}.$$

First suppose $p \nmid a$. Then a is invertible modulo p so we may take $j = \bar{a}b \pmod{p}$. Now suppose $p \mid a$. If there is some j satisfying $b \equiv aj \pmod{p}$ then we also have $p \mid b$. But as $ad - bc = 1$, this is impossible and so no such j exists. As $a \equiv 1 \pmod{N}$, we have $p \mid a$ if and only if $p \nmid N$. In this case consider instead

$$\gamma\beta'_\infty{}^{-1} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} 1 & -n \\ -N & pm \end{pmatrix} = \begin{pmatrix} a - Nb & pmb - na \\ c - Nd & pmd - nc \end{pmatrix}.$$

Since $p \mid a$, we have $pmb - na \equiv 0 \pmod{p}$ so that $\gamma\beta'_\infty{}^{-1} \in \alpha^{-1}\Gamma_1(N)\alpha \cap \Gamma_1(N) = \Gamma_1^0(N, p)$. Altogether, we have shown that $\beta'_1, \dots, \beta'_{p-1}$ and $\beta'_1, \dots, \beta'_{p-1}, \beta'_\infty$ run over representatives for $(\alpha^{-1}\Gamma_1(N)\alpha \cap \Gamma_1(N)) \setminus \Gamma_1(N)$ depending on if $p \mid N$ or not. To show they run over complete sets of representatives, we need to show that no two representatives belong to the same coset. To this end, suppose i and j are distinct and consider

$$\beta'_i\beta'_j{}^{-1} = \begin{pmatrix} 1 & i \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & j \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & i - j \\ 0 & 1 \end{pmatrix}.$$

Then $\beta'_i\beta'_j{}^{-1} \in \alpha^{-1}\Gamma_1(N)\alpha \cap \Gamma_1(N) = \Gamma_1^0(N, p)$ if and only if $i - j \equiv 0 \pmod{p}$. This is impossible since i and j are distinct. Hence β'_i and β'_j represent distinct cosets. Now consider

$$\beta'_i\beta'_\infty{}^{-1} = \begin{pmatrix} 1 & i \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & -n \\ -N & pm \end{pmatrix} = \begin{pmatrix} 1 - Ni & pmj - n \\ -N & pm \end{pmatrix}.$$

Now $\beta'_i\beta'_\infty{}^{-1} \in \alpha^{-1}\Gamma_1(N)\alpha \cap \Gamma_1(N) = \Gamma_1^0(N, p)$ if and only if $pmi - n \equiv 0 \pmod{p}$. This is impossible since $pm - Nn = 1$ implies $p \nmid n$. Therefore β'_i and β'_∞ represent distinct cosets. It follows that $\beta'_1, \dots, \beta'_{p-1}$ and $\beta'_1, \dots, \beta'_{p-1}, \beta'_\infty$ run over complete sets of representatives for $(\alpha^{-1}\Gamma_1(N)\alpha \cap \Gamma_1(N)) \setminus \Gamma_1(N)$ depending on if $p \mid N$ or not. As

$$\alpha\beta'_j = \begin{pmatrix} 1 & 0 \\ 0 & p \end{pmatrix} \begin{pmatrix} 1 & j \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & j \\ 0 & p \end{pmatrix} = \beta_j \quad \text{and} \quad \alpha\beta'_\infty = \begin{pmatrix} 1 & 0 \\ 0 & p \end{pmatrix} \begin{pmatrix} pm & n \\ N & 1 \end{pmatrix} = \begin{pmatrix} pm & n \\ Np & p \end{pmatrix} = \beta_\infty,$$

Lemma 1.5 finishes the proof. $\square$

This explicit definition of T_p can be used to compute how the Hecke operators act on the Fourier coefficients of a modular form:

Proposition 2.21. *Let $f \in \mathcal{S}_k(\Gamma_1(N))$ have Fourier coefficients $a_n(f)$. Then for all primes p , we have*

$$(T_p f)(z) = \sum_{n \geq 1} \left(a_{np}(f) + \chi_{N,0}(p) p^{k-1} a_{\frac{n}{p}}(\langle p \rangle f) \right) e^{2\pi i n z},$$

is the Fourier series of $T_p f$ where it is understood that $a_{\frac{n}{p}}(f) = 0$ if $p \nmid n$. Moreover, if $f \in \mathcal{S}_k(N, \chi)$ then $T_p f \in \mathcal{S}_k(N, \chi)$ and

$$(T_p f)(z) = \sum_{n \geq 1} \left(a_{np}(f) + \chi(p) p^{k-1} a_{\frac{n}{p}}(f) \right) e^{2\pi i n z},$$

where it is understood that $a_{\frac{n}{p}}(f) = 0$ if $p \nmid n$.

Proof. In view of Proposition 2.17 and the linearity of the Hecke operators, it suffices to assume $f \in \mathcal{S}_k(N, \chi)$. By Proposition 2.19, $T_p f \in \mathcal{S}_k(N, \chi)$. It remains to verify the second formula. Observe that

$$\left(f \left\|_k \begin{pmatrix} 1 & j \\ 0 & p \end{pmatrix} \right) (z) = \frac{1}{p} \sum_{n \geq 1} a_n(f) e^{\frac{2\pi i n(z+j)}{p}}.$$

Summing over all j modulo p gives

$$\sum_{j \pmod{p}} \left(f \left\|_k \begin{pmatrix} 1 & j \\ 0 & p \end{pmatrix} \right) (z) = \sum_{n \geq 1} a_n(f) e^{\frac{2\pi i n z}{p}} \frac{1}{p} \sum_{j \pmod{p}} e^{\frac{2\pi i n j}{p}}.$$

If $p \nmid N$ then the inner sum vanishes because it is the sum over all p -th roots of unity. If $p \mid N$ then the sum is also p . Therefore

$$\sum_{j \pmod{p}} \left(f \left\|_k \begin{pmatrix} 1 & j \\ 0 & p \end{pmatrix} \right) (z) = \sum_{n \geq 1} a_{np}(f) e^{2\pi i n z}.$$

If $p \mid N$ then Proposition 2.20 implies

$$(T_p f)(z) = \sum_{n \geq 1} a_{np}(f) e^{2\pi i n z}, \tag{21}$$

which is the claimed Fourier series of $T_p f$. If $p \nmid N$ then we have the additional term

$$\begin{aligned} \left(f \left\|_k \begin{pmatrix} m & n \\ N & p \end{pmatrix} \begin{pmatrix} p & 0 \\ 0 & 1 \end{pmatrix} \right) (z) &= \left(\langle p \rangle f \left\|_k \begin{pmatrix} p & 0 \\ 0 & 1 \end{pmatrix} \right) (z) \\ &= p^{k-1} (\langle p \rangle f)(pz) \\ &= \sum_{n \geq 1} p^{k-1} a_n(\langle p \rangle f) e^{2\pi i n p z} \\ &= \sum_{n \geq 1} \chi(p) p^{k-1} a_n(f) e^{2\pi i n p z}, \end{aligned}$$

where the first equality holds because $\begin{pmatrix} m & n \\ N & p \end{pmatrix} \in \Gamma_0(N)$ and the last equality holds because $\langle p \rangle f = \chi(p)f$. In this case, Proposition 2.20 gives

$$(T_p f)(z) = \sum_{n \geq 1} a_{np}(f) + \chi(p)p^{k-1}a_{\frac{n}{p}}(f)e^{2\pi i n z}.$$

Since $\chi(p) = 0$ if $p \mid N$, these two cases can be expressed together as

$$(T_p f)(z) = \sum_{n \geq 1} \left(a_{np}(f) + \chi(p)p^{k-1}a_{\frac{n}{p}}(f) \right) e^{2\pi i n z}.$$

□

We now mention the crucial result about Hecke operators which is that they form a simultaneously commuting family with the diamond operators:

Proposition 2.22. *Let p and q be primes and $d, e \in (\mathbb{Z}/N\mathbb{Z})^*$. The Hecke operators T_p and T_q and diamond operators $\langle d \rangle$ and $\langle e \rangle$ on $\mathcal{S}_k(\Gamma_1(N))$ form a simultaneously commuting family:*

$$T_p T_q = T_q T_p, \quad \langle d \rangle T_p = T_p \langle d \rangle, \quad \text{and} \quad \langle d \rangle \langle e \rangle = \langle e \rangle \langle d \rangle.$$

Proof. Showing the diamond and Hecke operators commute was Proposition 2.19. To show commutativity of the diamond operators, let $\gamma = \begin{pmatrix} * & * \\ * & d \end{pmatrix} \in \Gamma_0(N)$ and $\eta = \begin{pmatrix} * & * \\ * & e \end{pmatrix} \in \Gamma_0(N)$. Then

$$\gamma\eta \equiv \begin{pmatrix} * & * \\ 0 & de \end{pmatrix} \equiv \begin{pmatrix} * & * \\ 0 & ed \end{pmatrix} \equiv \eta\gamma \pmod{N}.$$

So for any $f \in \mathcal{S}_k(\Gamma_1(N))$, we have

$$\langle d \rangle \langle e \rangle f = f|_k \gamma \eta = f|_k \eta \gamma = \langle e \rangle \langle d \rangle f.$$

We now show that the Hecke operators commute. In view of Proposition 2.17 and linearity of the Hecke operators, it suffices to prove this for $f \in \mathcal{S}_k(N, \chi)$. Applying Proposition 2.21 twice, for any $n \geq 1$ we compute

$$\begin{aligned} a_n(T_p T_q f) &= a_{np}(T_q f) + \chi(p)p^{k-1}a_{\frac{n}{p}}(T_q f) \\ &= a_{npq}(f) + \chi(q)q^{k-1}a_{\frac{np}{q}}(f) + \chi(p)p^{k-1}(a_{\frac{nq}{p}}(f) + \chi(q)q^{k-1}a_{\frac{n}{pq}}(f)) \\ &= a_{npq}(f) + \chi(q)q^{k-1}a_{\frac{np}{q}}(f) + \chi(p)p^{k-1}a_{\frac{nq}{p}}(f) + \chi(pq)(pq)^{k-1}a_{\frac{n}{pq}}(f). \end{aligned}$$

The last expression is symmetric in p and q so that $a_n(T_p T_q f) = a_n(T_q T_p f)$ for all $n \geq 1$. Since all of the Fourier coefficients are equal, we get

$$T_p T_q f = T_q T_p f.$$

□

We can use Proposition 2.22 to construct diamond operators $\langle m \rangle$ and Hecke operators T_m for all $m \geq 1$. The **diamond operator** $\langle m \rangle : \mathcal{S}_k(\Gamma_1(N)) \rightarrow \mathcal{S}_k(\Gamma_1(N))$ is defined to be the linear operator given by

$$\langle m \rangle = \begin{cases} \langle m \rangle \text{ with } m \text{ taken modulo } N & \text{if } (m, N) = 1, \\ 0 & \text{if } (m, N) > 1. \end{cases}$$

Now for the Hecke operators. If $m = p_1^{r_1} p_2^{r_2} \cdots p_k^{r_k}$ is the prime decomposition of m then we define the m -th **Hecke operator** $T_m : \mathcal{S}_k(\Gamma_1(N)) \rightarrow \mathcal{S}_k(\Gamma_1(N))$ to be the linear operator given by

$$T_m = \prod_{1 \leq i \leq k} T_{p_i^{r_i}},$$

where T_{p^r} is defined inductively by

$$T_{p^r} = \begin{cases} T_p T_{p^{r-1}} - p^{k-1} \langle p \rangle T_{p^{r-2}} & \text{if } p \nmid N, \\ T_p^r & \text{if } p \mid N, \end{cases}$$

for all $r \geq 2$. Note that when $m = 1$, the product is empty and so T_1 is the identity operator. By Proposition 2.22, the Hecke operators T_m are multiplicative but not completely multiplicative in m . Moreover, they commute with the diamond operators $\langle m \rangle$. Using these definitions, Propositions 2.21 and 2.22, a more general formula for how the Hecke operators T_m act on Fourier coefficients can be derived:

Proposition 2.23. *Let $f \in \mathcal{S}_k(\Gamma_1(N))$ have Fourier coefficients $a_n(f)$. Then for $m \geq 1$ with $(m, N) = 1$, we have*

$$(T_m f)(z) = \sum_{n \geq 1} \left(\sum_{d \mid (n, m)} d^{k-1} a_{\frac{nm}{d^2}}(\langle d \rangle f) \right) e^{2\pi i n z},$$

is the Fourier series of $T_m f$. Moreover, if $f \in \mathcal{S}_k(N, \chi)$ then

$$(T_m f)(z) = \sum_{n \geq 1} \left(\sum_{d \mid (n, m)} \chi(d) d^{k-1} a_{\frac{nm}{d^2}}(f) \right) e^{2\pi i n z}.$$

Proof. In view of Proposition 2.17 and linearity of the Hecke operators, we may assume $f \in \mathcal{S}_k(N, \chi)$. Therefore we only need to verify the second formula. When $m = 1$ the result is obvious and when $m = p$, we have

$$\sum_{d \mid (n, p)} \chi(d) d^{k-1} a_{\frac{np}{d^2}}(f) = a_{np}(f) + \chi(p) p^{k-1} a_{\frac{n}{p}}(f),$$

which is the result obtained from Proposition 2.21. By induction assume that the desired formula holds for $m = 1, p, \dots, p^{r-1}$. Using the definition of T_{p^r} and Proposition 2.21, for any $n \geq 1$ we compute

$$\begin{aligned} a_n(T_{p^r} f) &= a_n(T_p T_{p^{r-1}} f) - \chi(p) p^{k-1} a_n(T_{p^{r-2}} f) \\ &= a_{np}(T_{p^{r-1}} f) + \chi(p) p^{k-1} a_{\frac{n}{p}}(T_{p^{r-1}} f) - \chi(p) p^{k-1} a_n(T_{p^{r-2}} f). \end{aligned}$$

By our induction hypothesis, this last expression is

$$\sum_{d \mid (np, p^{r-1})} \chi(d) d^{k-1} a_{\frac{np^r}{d^2}}(f) + \chi(p) p^{k-1} \sum_{d \mid (\frac{n}{p}, p^{r-1})} \chi(d) d^{k-1} a_{\frac{np^{r-2}}{d^2}}(f) - \chi(p) p^{k-1} \sum_{d \mid (n, p^{r-2})} \chi(d) d^{k-1} a_{\frac{np^{r-2}}{d^2}}(f).$$

Write the first sum as

$$\sum_{d|(np, p^{r-1})} \chi(d) d^{k-1} a_{\frac{np^r}{d^2}}(f) = a_{np^r}(f) + \sum_{d|(n, p^{r-2})} \chi(d) d^{k-1} a_{\frac{np^{r-2}}{d^2}}(f),$$

and observe that the sum on the right-hand side cancels the entire third term above. Therefore our expression reduces to

$$\begin{aligned} a_{np^r}(f) + \chi(p) p^{k-1} \sum_{d|(\frac{n}{p}, p^{r-1})} \chi(d) d^{k-1} a_{\frac{np^{r-2}}{d^2}}(f) &= a_{np^r}(f) + \sum_{d|(\frac{n}{p}, p^{r-1})} \chi(dp) (dp)^{k-1} a_{\frac{np^{r-2}}{d^2}}(f) \\ &= a_{np^r}(f) + \sum_{\substack{d|(n, p^r) \\ d \neq 1}} \chi(d) d^{k-1} a_{\frac{np^r}{d^2}}(f) \\ &= \sum_{d|(n, p^r)} \chi(d) d^{k-1} a_{\frac{np^r}{d^2}}(f), \end{aligned}$$

where in the second line we have performed the change of variables $dp \mapsto d$ in the sum. This proves the claim when $m = p^r$ for all $r \geq 0$. By multiplicativity of the Hecke operators, it suffices to prove the claim when $m = p^r q^s$ for another prime q and some $s \geq 0$. We compute

$$\begin{aligned} a_n(T_{p^r q^s} f) &= a_n(T_{p^r} T_{q^s} f) \\ &= \sum_{d_1|(n, p^r)} \chi(d_1) d_1^{k-1} a_{\frac{np^r}{d_1^2}}(T_{q^s} f) \\ &= \sum_{d_1|(n, p^r)} \chi(d_1) d_1^{k-1} \sum_{d_2|(\frac{np^r}{d_1^2}, q^s)} \chi(d_2) d_2^{k-1} a_{\frac{np^r q^s}{(d_1 d_2)^2}}(f) \\ &= \sum_{d_1|(n, p^r)} \sum_{d_2|(\frac{np^r}{d_1^2}, q^s)} \chi(d_1 d_2) (d_1 d_2)^{k-1} a_{\frac{np^r q^s}{(d_1 d_2)^2}}(f). \end{aligned}$$

Summing over pairs (d_1, d_2) of divisors of (n, p^r) and $(\frac{np^r}{d_1^2}, q^s)$ respectively is the same as summing over divisors d of $(n, p^r q^s)$. Indeed, because p and q are relative prime, any such d is of the form $d = d_1 d_2$ where $d_1 | (n, p^r)$ and $d_2 | (\frac{np^r}{d_1^2}, q^s)$. Therefore the double sum becomes

$$\sum_{d|(n, p^r q^s)} \chi(d) d^{k-1} a_{\frac{np^r q^s}{d^2}}(f).$$

This completes the proof. $\square$

The diamond and Hecke operators turn out to be normal on the subspace of cusp forms. To prove this fact, we will require a lemma:

Lemma 2.24. *Let Γ be a congruence subgroup and let $\alpha \in \mathrm{GL}_2^+(\mathbb{Q})$. Then there exist $\beta_1, \dots, \beta_n \in \mathrm{GL}_2^+(\mathbb{Q})$, where $n = [\Gamma : \alpha^{-1} \Gamma \alpha \cap \Gamma] = [\Gamma : \alpha \Gamma \alpha^{-1} \cap \Gamma]$, and such that*

$$\Gamma \alpha \Gamma = \bigcup_{1 \leq j \leq n} \Gamma \beta_j = \bigcup_{1 \leq j \leq n} \beta_j \Gamma.$$

Proof. By Lemma 1.4 and that congruence subgroups are preserved under intersection, we see that $\alpha\Gamma\alpha^{-1} \cap \Gamma$ is a congruence subgroup. By Lemma 1.17, it follows that $d_{\alpha^{-1}\Gamma\alpha \cap \Gamma} = d_{\alpha\Gamma\alpha^{-1} \cap \Gamma}$. Dividing both sides by d_Γ gives

$$[\Gamma : \alpha^{-1}\Gamma\alpha \cap \Gamma] = [\Gamma : \alpha\Gamma\alpha^{-1} \cap \Gamma].$$

Therefore we can find representatives $\gamma_1, \dots, \gamma_n \in \Gamma$ and $\tilde{\gamma}_1, \dots, \tilde{\gamma}_n \in \Gamma$ such that

$$\Gamma = \bigcup_{1 \leq j \leq n} (\alpha^{-1}\Gamma\alpha \cap \Gamma)\gamma_j = \bigcup_{1 \leq j \leq n} (\alpha^{-1}\Gamma\alpha \cap \Gamma)\tilde{\gamma}_j^{-1}.$$

Upon invoking Lemma 1.5 twice, we obtain

$$\bigcup_{1 \leq j \leq n} (\alpha^{-1}\Gamma\alpha \cap \Gamma)\gamma_j = \bigcup_{1 \leq j \leq n} \Gamma\alpha\gamma_j \quad \text{and} \quad \bigcup_{1 \leq j \leq n} (\alpha^{-1}\Gamma\alpha \cap \Gamma)\tilde{\gamma}_j^{-1} = \bigcup_{1 \leq j \leq n} \Gamma\alpha^{-1}\tilde{\gamma}_j^{-1}.$$

It follows that

$$\Gamma = \bigcup_{1 \leq j \leq n} \Gamma\alpha\gamma_j = \bigcup_{1 \leq j \leq n} \tilde{\gamma}_j\alpha\Gamma.$$

For each j , the orbits $\Gamma\alpha\gamma_j$ and $\tilde{\gamma}_j\alpha\Gamma$ have nonempty intersection. For if they did we would have

$$\Gamma\alpha\gamma_j \subseteq \bigcup_{\substack{1 \leq i \leq n \\ i \neq j}} \tilde{\gamma}_i\alpha\Gamma,$$

and thus

$$\Gamma\alpha\Gamma \subseteq \bigcup_{\substack{1 \leq i \leq n \\ i \neq j}} \tilde{\gamma}_i\alpha\Gamma.$$

This contradicts the previous decomposition of Γ . Therefore we can find representatives $\beta_j \in \Gamma\alpha\gamma_j \cap \tilde{\gamma}_j\alpha\Gamma$ for every j . Then

$$\Gamma = \bigcup_{1 \leq j \leq n} \Gamma\beta_j = \bigcup_{1 \leq j \leq n} \beta_j\Gamma,$$

as desired. □

We can use Lemma 2.24 to compute adjoints:

Proposition 2.25. *Let Γ be a congruence subgroup and let $\alpha \in \text{GL}_2^+(\mathbb{Q})$. Set $\alpha' = \det(\alpha)\alpha^{-1}$. Then the following are true:*

(i) *If $\alpha^{-1}\Gamma\alpha \subseteq \text{SL}_2(\mathbb{Z})$ then for all $f \in \mathcal{S}_k(\Gamma)$ and $g \in \mathcal{S}_k(\alpha^{-1}\Gamma\alpha)$, we have*

$$\langle f|_k\alpha, g \rangle_{\alpha^{-1}\Gamma\alpha} = \langle f, g|_k\alpha' \rangle_\Gamma.$$

(ii) *For all $f, g \in \mathcal{S}_k(\Gamma)$, we have*

$$\langle f[\Gamma\alpha\Gamma]_k, g \rangle = \langle f, g[\Gamma\alpha'\Gamma]_k \rangle.$$

In particular, if $\alpha^{-1}\Gamma\alpha = \Gamma$ then $|_k\alpha^* = |_k\alpha'$, $\|_k\alpha^* = \|_k\alpha'$, and $[\Gamma\alpha\Gamma]_k^* = [\Gamma\alpha'\Gamma]_k$.

Proof. To prove (i) we first compute

$$\begin{aligned}
\langle f|_k\alpha, g \rangle_{\alpha^{-1}\Gamma\alpha} &= \frac{1}{V_{\alpha^{-1}\Gamma\alpha}} \int_{\mathcal{F}_{\alpha^{-1}\Gamma\alpha}} (f|_k\alpha)(z) \overline{g(z)} \operatorname{Im}(z)^k d\mu \\
&= \frac{1}{V_{\alpha^{-1}\Gamma\alpha}} \int_{\mathcal{F}_{\alpha^{-1}\Gamma\alpha}} \det(\alpha)^{\frac{k}{2}} j(\alpha, z)^{-k} f(\alpha z) \overline{g(z)} \operatorname{Im}(z)^k d\mu \\
&= \frac{1}{V_\Gamma} \int_{\mathcal{F}_{\alpha^{-1}\Gamma\alpha}} \det(\alpha)^{\frac{k}{2}} j(\alpha, z)^{-k} f(\alpha z) \overline{g(z)} \operatorname{Im}(z)^k d\mu && \text{Lemma 1.17} \\
&= \frac{1}{V_\Gamma} \int_{\mathcal{F}_\Gamma} \det(\alpha)^{\frac{k}{2}} j(\alpha, \alpha^{-1}z)^{-k} f(z) \overline{g(\alpha^{-1}z)} \operatorname{Im}(\alpha^{-1}z)^k d\mu && z \mapsto \alpha^{-1}z.
\end{aligned}$$

As α' acts as α^{-1} on $\mathbb{H}$ and $\det(\alpha') = \det(\alpha)$, using Equation (5), this latter integral is equivalent to

$$\frac{1}{V_\Gamma} \int_{\mathcal{F}_\Gamma} \det(\alpha')^{\frac{3k}{2}} j(\alpha, \alpha'z)^{-k} f(z) \overline{g(\alpha'z)} \operatorname{Im}(\alpha'z)^k d\mu.$$

Applying the cocycle condition, Equation (5), and the identity $j(\alpha\alpha', z) = \det(\alpha')$, we can further rewrite the integral as

$$\frac{1}{V_\Gamma} \int_{\mathcal{F}_\Gamma} \det(\alpha')^{\frac{k}{2}} \overline{j(\alpha', z)^{-k}} f(z) \overline{g(\alpha'z)} \operatorname{Im}(z)^k d\mu.$$

Reversing the first computation in the start of the proof but applied to this integral shows that that

$$\frac{1}{V_\Gamma} \int_{\mathcal{F}_\Gamma} \det(\alpha')^{\frac{k}{2}} \overline{j(\alpha', z)^{-k}} f(z) \overline{g(\alpha'z)} \operatorname{Im}(z)^k d\mu = \langle f, g|_k\alpha' \rangle_\Gamma,$$

which completes the proof of (i). To prove (ii), Lemma 2.24 implies the coset decomposition $\Gamma\alpha\Gamma = \bigcup_{1 \leq j \leq n} \Gamma\beta_j$, with $n = [\Gamma : \alpha^{-1}\Gamma\alpha \cap \Gamma] = [\Gamma : \alpha\Gamma\alpha^{-1} \cap \Gamma]$, so that we can use the β_j as representatives for the $[\Gamma\alpha\Gamma]_k$ operator. As the β_j also satisfy $\Gamma\alpha\Gamma = \bigcup_{1 \leq j \leq n} \beta_j\Gamma$, upon inverting β_j and noting that $\beta_j \in \Gamma\alpha$, we obtain $\Gamma\alpha^{-1}\Gamma = \bigcup_{1 \leq j \leq n} \Gamma\beta_j^{-1}$. Since scalar multiplication commutes with matrices and the matrices in Γ have determinant 1, we conclude that $\Gamma\alpha'\Gamma = \bigcup_{1 \leq j \leq n} \Gamma\beta'_j$ where $\beta'_j = \det(\beta_j)\beta_j^{-1}$ (also $\det(\beta_j) = \det(\alpha)$). So we can use the β'_j as representatives in the $[\Gamma\alpha'\Gamma]_k$ operator. The statement now follows from (i) and Equation (20). The last statement is obvious (again using Equation (20)). $\square$

We can now prove that the diamond and Hecke operators are normal:

Proposition 2.26. *On $\mathcal{S}_k(\Gamma_1(N))$, the diamond operators $\langle m \rangle$ and Hecke operators T_m are normal for all $m \geq 1$ with $(m, N) = 1$. Moreover, their adjoints are given by*

$$\langle m \rangle^* = \langle \overline{m} \rangle \quad \text{and} \quad T_p^* = \langle \overline{p} \rangle T_p.$$

Proof. By multiplicativity of the diamond and Hecke operators, it suffices to prove the two formulas when $m = p$ for a prime $p \nmid N$. We will prove the adjoint formula for diamond

operators first. Let $\alpha = \begin{pmatrix} \bar{p} & * \\ * & p \end{pmatrix} \in \Gamma_0(N)$ and $\alpha' = \det(\alpha)\alpha^{-1} = \begin{pmatrix} p & * \\ * & \bar{p} \end{pmatrix} \in \Gamma_0(N)$. Then Proposition 2.25 implies

$$\langle p \rangle^* = \langle \bar{p} \rangle.$$

This proves the adjoint formula for the diamond operators and normality follows from multiplicativity. For the Hecke operators, let $\alpha = \begin{pmatrix} 1 & 0 \\ 0 & p \end{pmatrix}$ and set $\alpha' = \det(\alpha)\alpha^{-1} = \begin{pmatrix} p & 0 \\ 0 & 1 \end{pmatrix}$. Also let m and n be such that $\det\left(\begin{pmatrix} m & n \\ N & p \end{pmatrix}\right) = 1$ (by Bézout's identity). Observe that

$$\begin{pmatrix} 1 & n \\ N & pm \end{pmatrix}^{-1} \alpha \begin{pmatrix} p & n \\ N & m \end{pmatrix} = \begin{pmatrix} pm & -n \\ -N & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & p \end{pmatrix} \begin{pmatrix} p & n \\ N & m \end{pmatrix} = \alpha'.$$

Now $\begin{pmatrix} 1 & n \\ N & pm \end{pmatrix} \in \Gamma_1(N)$ (note that $pm \equiv 1 \pmod{N}$ since $pm - Nn = 1$) and $\begin{pmatrix} p & n \\ N & m \end{pmatrix} \in \Gamma_0(N)$. These facts and that $\Gamma_1(N)$ is normal in $\Gamma_0(N)$ together imply

$$\Gamma_1(N)\alpha'\Gamma_1(N) = \Gamma_1(N)\alpha\Gamma_1(N) \begin{pmatrix} p & n \\ N & m \end{pmatrix}.$$

As $m \equiv \bar{p} \pmod{N}$, the above identity and Proposition 2.25 together give

$$T_p^* = \langle \bar{p} \rangle T_p.$$

This proves the adjoint formula for the Hecke operators and normality follows from multiplicativity. $\square$

Note that on $\mathcal{S}_k(\mathrm{SL}_2(\mathbb{Z}))$, all of the diamond operators are the identity and therefore $T_p^* = T_p$ for all primes p . That is, the Hecke operators are self-adjoint (as are the diamond operators since they are the identity). Now suppose f is a non-constant cusp form. Let the eigenvalue of T_m for f be $\lambda_f(m)$. We say that the $\lambda_f(m)$ are the **Hecke eigenvalues** of f . If f is a simultaneous eigenfunction for all diamond operators $\langle m \rangle$ and Hecke operators T_m with $(m, N) = 1$, we call f an **eigenform**. If the condition $(m, N) = 1$ can be dropped, so that f is a simultaneous eigenfunction for all diamond and Hecke operators, we say f is a **Hecke eigenform**. In particular, on $\mathrm{SL}_2(\mathbb{Z}) \backslash \mathbb{H}$ all eigenforms are Hecke eigenforms. Now let f have Fourier coefficients $a_n(f)$. If f is a Hecke eigenform then Proposition 2.23 immediately implies that the first Fourier coefficient of $T_m f$ is $a_m(f)$ and so

$$a_m(f) = \lambda_f(m)a_1(f),$$

for all $m \geq 1$. Therefore we cannot have $a_1(f) = 0$ for this would mean f is constant. So we can normalize f by dividing by $a_1(f)$ which guarantees that this Fourier coefficient is 1. It follows that

$$a_m(f) = \lambda_f(m),$$

for all $m \geq 1$. This normalization is called the **Hecke normalization** of f . The **Petersson normalization** of f is where we normalize so that $\langle f, f \rangle = 1$. In particular, any orthonormal basis of $\mathcal{S}_k(\Gamma_1(N))$ is Petersson normalized. From the spectral theorem we derive an important corollary:

Theorem 2.27. $\mathcal{S}_k(\Gamma_1(N))$ admits an orthonormal basis of eigenforms.

Proof. By Theorem 2.12, $\mathcal{S}_k(\Gamma_1(N))$ is finite dimensional. The claim then follows from the spectral theorem along with Propositions 2.22 and 2.26. $\square$

The Hecke eigenvalues of Hecke eigenforms satisfy certain relations known as the **Hecke relations** for modular forms:

Proposition (Hecke relations, modular). *Let $f \in \mathcal{S}_k(N, \chi)$ be a Hecke eigenform with Hecke eigenvalues $\lambda_f(m)$. Then the Hecke eigenvalues are multiplicative and satisfy*

$$\lambda_f(n)\lambda_f(m) = \sum_{d|(n,m)} \chi(d)d^{k-1}\lambda_f\left(\frac{nm}{d^2}\right) \quad \text{and} \quad \lambda_f(nm) = \sum_{d|(n,m)} \mu(d)\chi(d)d^{k-1}\lambda_f\left(\frac{n}{d}\right)\lambda_f\left(\frac{m}{d}\right),$$

for all $n, m \geq 1$ with $(nm, N) = 1$. Moreover,

$$\lambda_f(p^r) = \lambda_f(p)^r,$$

for all $p \mid N$ and $r \geq 2$.

Proof. If necessary, Hecke normalize f . The multiplicativity of the Hecke eigenvalues now follows from the multiplicativity of the Hecke operators. The first identity follows from computing the n -th Fourier coefficient of $T_m f$ in two different ways. On the one hand, use that f is a Hecke eigenform to get $\lambda_f(n)\lambda_f(m)$. On the other hand, use Proposition 2.23 to obtain $\sum_{d|(n,m)} \chi(d)d^{k-1}\lambda_f\left(\frac{nm}{d^2}\right)$. For the second identity, computing the p -th Fourier coefficient of T_p in two different ways just as before, we have

$$\chi(p)p^{k-1} = \lambda_f(p)^2 - \lambda_f(p^2),$$

provided $(p, N) = 1$. As the left-hand sides is completely multiplicative, the first identity implies that $\lambda_f(n)$ is specially multiplicative. The second identity now follows from ???. The third identity follows from inspecting the first Fourier coefficient of $T_{p^r} f$ and noting that $T_{p^r} = T_p^r$ provided $p \mid N$. $\square$

As an immediate consequence of the Hecke relations, the Hecke operators satisfy analogous relations:

Corollary 2.28. *The Hecke operators are multiplicative and satisfy*

$$T_n T_m = \sum_{d|(n,m)} \chi(d)d^{k-1}T_{\frac{nm}{d^2}} \quad \text{and} \quad T_{nm} = \sum_{d|(n,m)} \mu(d)\chi(d)d^{k-1}T_{\frac{n}{d}}T_{\frac{m}{d}},$$

for all $n, m \geq 1$ with $(nm, N) = 1$.

Proof. This is immediate from Theorem 2.27 and the Hecke relations. $\square$

The identities in Corollary 2.28 can also be established directly. Moreover, the first identity is symmetric in n and m so it can be used to show that the Hecke operators commute.

2.6 Atkin-Lehner Theory

So far, our entire theory of modular forms has started with a fixed congruence subgroup of some level. Atkin-Lehner theory, or the theory of oldforms and newforms, allows us to discuss modular forms in the context of moving between levels. In this setting, we will only deal with congruence subgroups of the form $\Gamma_1(N)$ and $\Gamma_0(N)$. The easiest way lift a modular form from a smaller level to a larger level is to observe that if $M \mid N$ then $\Gamma_1(N) \subseteq \Gamma_1(M)$ so there is a natural inclusion $\mathcal{S}_k(\Gamma_1(M)) \subseteq \mathcal{S}_k(\Gamma_1(N))$. There is a less trivial way of lifting from $\mathcal{S}_k(\Gamma_1(M))$ to $\mathcal{S}_k(\Gamma_1(N))$. For any $d \mid \frac{N}{M}$, let

$$\alpha_d = \begin{pmatrix} d & 0 \\ 0 & 1 \end{pmatrix}.$$

If $f \in \mathcal{S}_k(\Gamma_1(M))$, we consider

$$(f|_k \alpha_d)(z) = d^{k-1} f(dz).$$

It turns out that $|_k \alpha_d$ maps $\mathcal{S}_k(\Gamma_1(M))$ into $\mathcal{S}_k(\Gamma_1(N))$ and more:

Proposition 2.29. *Let M and N be positive integers such that $M \mid N$. For any $d \mid \frac{N}{M}$, $|_k \alpha_d$ maps $\mathcal{S}_k(\Gamma_1(M))$ into $\mathcal{S}_k(\Gamma_1(N))$. In particular, $|_k \alpha_d$ takes $\mathcal{S}_k(M, \chi)$ into $\mathcal{S}_k(N, \chi)$.*

Proof. By Proposition 2.17, it suffices to verify only the latter statement so we may assume $f \in \mathcal{S}_k(M, \chi)$. It is clear that holomorphy is satisfied for $f|_k \alpha_d$. To verify modularity, let $\gamma = \begin{pmatrix} a & b \\ c & d' \end{pmatrix} \in \Gamma_1(N)$ and set $\gamma' = \begin{pmatrix} a & bd \\ d^{-1}c & d' \end{pmatrix} \in \Gamma_1(M)$ (because $c \equiv 0 \pmod{N}$ and $d \mid \frac{N}{M}$). Then

$$\alpha_d \gamma \alpha_d^{-1} = \begin{pmatrix} d & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} a & b \\ c & d' \end{pmatrix} \begin{pmatrix} d^{-1} & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} a & bd \\ d^{-1}c & d' \end{pmatrix} = \gamma',$$

and therefore $\alpha_d \Gamma_1(N) \alpha_d^{-1} \subseteq \Gamma_1(M)$, or equivalently, $\Gamma_1(N) \subseteq \alpha_d^{-1} \Gamma_1(M) \alpha_d$. It is also clear that $\chi(\gamma') = \chi(\gamma)$. Using these facts, we compute

$$\begin{aligned} ((f|_k \alpha_d)|_k \gamma)(z) &= d^{\frac{k}{2}-1} (f|_k \alpha_d \gamma)(z) && \text{Equation (20)} \\ &= d^{\frac{k}{2}-1} (f|_k \gamma' \alpha_d)(z) \\ &= \chi(\gamma') d^{\frac{k}{2}-1} (f|_k \alpha_d)(z) && \text{modularity} \\ &= \chi(\gamma) d^{\frac{k}{2}-1} (f|_k \alpha_d)(z) \\ &= \chi(\gamma) (f|_k \alpha_d)(z). \end{aligned}$$

This verifies modularity of $f|_k \alpha_d$. For the growth condition, let $\sigma_{\mathfrak{a}}$ be a scaling matrix for a cusp $\mathfrak{a}$ of $\Gamma_1(M) \backslash \mathbb{H}$. Then $\alpha_d \sigma_{\mathfrak{a}}$ takes ∞ to an element of $\hat{\mathbb{Q}}$ since $\alpha_d \in \text{GL}_2^+(\mathbb{Q})$. In other words, $\alpha_d \sigma_{\mathfrak{a}} \infty = \mathfrak{b}$ for some cusp $\mathfrak{b}$ of $\Gamma_1(N) \backslash \mathbb{H}$. Then by Equation (20),

$$((f|_k \alpha_d)|_k \sigma_{\mathfrak{a}})(z) = d^{\frac{k}{2}-1} (f|_k \alpha_d \sigma_{\mathfrak{a}})(z),$$

and so the growth condition follows from that of f . In particular, $f|_k \alpha_d$ is a cusp form since f is. $\square$

We can now define oldforms and newforms. For each divisor d of N , set

$$i_d : \mathcal{S}_k \left(\Gamma_1 \left(\frac{N}{d} \right) \right) \oplus \mathcal{S}_k \left(\Gamma_1 \left(\frac{N}{d} \right) \right) \rightarrow \mathcal{S}_k(\Gamma_1(N)) \quad (f, g) \mapsto f + g|_k \alpha_d.$$

This map is well-defined by Proposition 2.29. The subspace of **oldforms** of level N is

$$\mathcal{S}_k^{\text{old}}(\Gamma_1(N)) = \bigoplus_{p|N} \text{Im}(i_p),$$

and the subspace of **newforms** of level N is

$$\mathcal{S}_k^{\text{new}}(\Gamma_1(N)) = \mathcal{S}_k^{\text{old}}(\Gamma_1(N))^\perp.$$

An element of these subspaces is called an **oldform** or **newform** respectively. Note that there are no oldforms of level 1. We will need a useful operator for the study of oldforms and newforms. Let

$$W_N = \begin{pmatrix} 0 & -1 \\ N & 0 \end{pmatrix},$$

and note that $\det(W_N) = N$. This matrix normalizes $\Gamma_0(N)$. Indeed, let $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma_0(N)$ and set $g' = \begin{pmatrix} d & -N^{-1}c \\ -Nb & a \end{pmatrix} \in \Gamma_0(N)$. Then

$$W_N \gamma W_N^{-1} = \begin{pmatrix} 0 & -1 \\ N & 0 \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} 0 & N^{-1} \\ -1 & 0 \end{pmatrix} = \begin{pmatrix} d & -N^{-1}c \\ -Nb & a \end{pmatrix} = \gamma'. \quad (22)$$

It follows that W_N normalizes $\Gamma_0(N)$ as claimed. We define the **Atkin-Lehner operator** ω_N to be the linear operator on $\mathcal{S}_k(\Gamma_1(N))$ given by

$$(\omega_N f)(z) = N^{1-k} (f|_k W_N)(z).$$

As W_N is invertible, so is the Atkin-Lehner operator ω_N . It is not difficult to see how the Atkin-Lehner operator acts on $\mathcal{S}_k(\Gamma_1(N))$:

Proposition 2.30. *The Atkin-Lehner operator ω_N maps $\mathcal{S}_k(\Gamma_1(N))$ into itself. In particular, ω_N takes $\mathcal{S}_k(N, \chi)$ into $\mathcal{S}_k(N, \bar{\chi})$ and preserves the subspaces of oldforms and newforms. Moreover, the adjoint is given by*

$$\omega_N^* = (-1)^k \omega_N.$$

We also have

$$\omega_N^2 = (-1)^k.$$

Proof. In light of Proposition 2.17, the first statement is a consequence of the latter ones. Therefore we may assume $f \in \mathcal{S}_k(N, \chi)$. Holomorphy is obvious. For modularity, let $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma_0(N)$ and set $\gamma' = \begin{pmatrix} d & -N^{-1}c \\ -Nb & a \end{pmatrix} \in \Gamma_0(N)$. Then $W_N \gamma = \gamma' W_N$ by Equation (22) and we also have $\chi(\gamma') = \bar{\chi}(\gamma)$ because $ad \equiv 1 \pmod{N}$. Using these facts, we compute

$$\begin{aligned} ((\omega_N f)|_k \gamma)(z) &= N^{-\frac{k}{2}} (f|_k W_N \gamma)(z) && \text{Equation (20)} \\ &= N^{-\frac{k}{2}} (f|_k \gamma' W_N)(z) \\ &= \chi(\gamma') N^{-\frac{k}{2}} (f|_k W_N)(z) && \text{modularity} \\ &= \bar{\chi}(\gamma) N^{1-k} (f|_k W_N)(z) \\ &= \bar{\chi}(\gamma) (\omega_N f)(z). \end{aligned}$$

This verifies modularity of $\omega_N f$. As for the growth condition, let $\sigma_{\mathfrak{a}}$ be a scaling matrix for a cusp $\mathfrak{a}$. Then $W_N \sigma_{\mathfrak{a}}$ takes ∞ to an element of $\hat{\mathbb{Q}}$ since $W_N \in \mathrm{GL}_2^+(\mathbb{Q})$. In other words, $W_N \sigma_{\mathfrak{a}} \infty = \mathfrak{b}$ for some cusp $\mathfrak{b}$. Then

$$((\omega_N f)|_k \sigma_{\mathfrak{a}})(z) = N^{-\frac{k}{2}} (f|_k W_N \sigma_{\mathfrak{a}})(z),$$

and so the growth condition follows from that of f . In particular, $\omega_N f$ is a cusp form because f is. It total we have shown that $\omega_N f \in \mathcal{S}_k(N, \overline{\chi})$. We now prove the adjoint formula for the Atkin-Lehner operator. Let $W'_N = \det(W_N) W_N^{-1}$. Then $W'_N = -W_N$. Since W_N normalizes $\Gamma_0(N)$ so does $-W_N$. As $-W_N$ and W_N act the same on $\mathbb{H}$, Proposition 2.25 implies that

$$\omega_N^* = (-1)^k \omega_N.$$

This proves the adjoint formula for the Atkin-Lehner operator. We now show that the Atkin-Lehner operator preserves the subspaces of oldforms and newforms. To show that it preserves $\mathcal{S}_k^{\mathrm{old}}(\Gamma_1(N))$, let $h = f + g|_k \alpha_p$ be in the image of i_p , with $p \mid N$, so that $i_p(f, g) = h$. Then it suffices to show

$$\omega_N i_p(f, g) = i_p(\omega_{\frac{N}{p}} p^{k-2} g, \omega_{\frac{N}{p}} p^{1-k} f).$$

This will follow from the formulas

$$\omega_N f = (\omega_{\frac{N}{p}} p^{1-k} f)|_k \alpha_p \quad \text{and} \quad \omega_N (g|_k \alpha_p) = \omega_{\frac{N}{p}} (p^{k-2} g),$$

which are implied by the identities

$$W_N = W_{\frac{N}{p}} \alpha_p \quad \text{and} \quad \alpha_p W_N = I_p W_{\frac{N}{p}}.$$

Thus the Atkin-Lehner operator ω_N preserves the subspace of oldforms. To see that ω_N preserves the subspace of newforms as well, let f and g be a newform and an oldform respectively. The adjoint formula for ω_N and that it preserves the subspace of oldforms gives

$$\langle \omega_N f, g \rangle = (-1)^k \langle f, \omega_N g \rangle = 0.$$

Hence $\omega_N f$ must also be a newform and so ω_N preserves the subspace of newforms. It remains to prove the last formula. For this, observe that $W_N^2 = -I_N$. This fact and Equation (20) together imply

$$(\omega_N^2 f)(z) = N^{-k} (f|_k W_N^2)(z) = N^{-k} (f|_k (-I_N))(z) = (-1)^k f.$$

This completes the proof □

Proposition 2.30 shows that the Atkin-Lehner operator ω_N is self-adjoint and an involution if k is even. In fact, regardless of the parity of k , ω_N is at most of order 4. We now need to understand how the Atkin-Lehner operator interacts with the diamond and Hecke operators:

Proposition 2.31. *On $\mathcal{S}_k(\Gamma_1(N))$, the diamond operators $\langle m \rangle$ and Hecke operators T_m satisfy the adjoint formulas*

$$\langle m \rangle^* = \omega_N \langle m \rangle \omega_N^{-1} \quad \text{and} \quad T_m^* = \omega_N T_m \omega_N^{-1},$$

for all $m \geq 1$.

Proof. By multiplicativity of the diamond and Hecke operators, it suffices to prove the two adjoint formulas when $m = p$ for a prime p . For the diamond operators, the formula is obvious when $p \mid N$ since $\langle p \rangle$ is the zero operator. So suppose $p \nmid N$, let $\alpha = \begin{pmatrix} \bar{p} & * \\ * & p \end{pmatrix} \in \Gamma_0(N)$ and set $\alpha' = \det(\alpha)\alpha^{-1} = \begin{pmatrix} p & * \\ * & \bar{p} \end{pmatrix} \in \Gamma_0(N)$. Equation (22) shows that $\alpha' = W_N \begin{pmatrix} \bar{p} & * \\ * & p \end{pmatrix} W_N^{-1}$ for some $\begin{pmatrix} \bar{p} & * \\ * & p \end{pmatrix} \in \Gamma_0(N)$. Then Proposition 2.25 gives

$$\langle p \rangle^* = \omega_N \langle p \rangle \omega_N^{-1}.$$

This is the desired adjoint formula for the diamond operators when $p \mid N$. Thus the adjoint formula holds for all the diamond operators. For the Hecke operators, let $\alpha = \begin{pmatrix} 1 & 0 \\ 0 & p \end{pmatrix}$ and set $\alpha' = \det(\alpha)\alpha^{-1} = \begin{pmatrix} p & 0 \\ 0 & 1 \end{pmatrix}$. Equation (22) shows that $\alpha' = W_N \alpha W_N^{-1}$. Then Proposition 2.25 gives

$$T_p^* = \omega_N T_p \omega_N^{-1}.$$

This is the desired adjoint formula for the Hecke operators. $\square$

It turns out that the spaces of oldforms and newforms are invariant under the diamond and Hecke operators:

Proposition 2.32. *On $\mathcal{S}_k(\Gamma_1(N))$, the diamond operators $\langle m \rangle$ and Hecke operators T_m preserve the subspaces of oldforms and newforms for all $m \geq 1$.*

Proof. By multiplicativity of the diamond and Hecke operators, it suffices to prove this when m is prime. Moreover, if $N = 1$ the result is trivial because there are no oldforms. Therefore we may assume $N > 1$ and hence consider a prime p with $p \mid N$. We will first show that the diamond and Hecke operators preserve $\mathcal{S}_k^{\text{old}}(\Gamma_1(N))$. Let $h = f + g\|_k \alpha_p$ be in the image of i_p so that $i_p(f, g) = h$ and let q be a prime. For the diamond operators, it suffices to show

$$\langle q \rangle i_p(f, g) = i_p(\langle q \rangle f, \langle q \rangle g).$$

This will follow from the formulas

$$\langle q \rangle f = \langle q \rangle f \quad \text{and} \quad \langle q \rangle (g\|_k \alpha_p) = (\langle q \rangle g)\|_k \alpha_p,$$

where the diamond operators on the left-hand and right-hand sides are on levels N and $\frac{N}{p}$ respectively. Both formulas are trivial if $q \mid N$ because then $\langle q \rangle$ is the zero operator. So suppose $q \nmid N$. If $\alpha = \begin{pmatrix} * & * \\ * & q \end{pmatrix} \in \Gamma_0(N)$ then $\alpha \in \Gamma_0\left(\frac{N}{p}\right)$ and the first formula follows. The second formula follows from the identity

$$\begin{pmatrix} * & * \\ * & q \end{pmatrix} \begin{pmatrix} p & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} p & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} * & * \\ * & q \end{pmatrix}.$$

Therefore the diamond operators preserve $\mathcal{S}_k^{\text{old}}(\Gamma_1(N))$. The case for the Hecke operators is slightly more involved. Consider the Hecke operator T_q for a prime p with $q \neq p$. For this Hecke operator, it suffices to show

$$T_q i_p(f, g) = i_p(T_q f, T_q g).$$

This will follow from the formulas

$$T_q f = T_q f \quad \text{and} \quad T_q(g|_k \alpha_p) = (T_q g)|_k \alpha_p,$$

where the Hecke operators on the left-hand and right-hand sides are on levels N and $\frac{N}{p}$ respectively. The first formula follows immediately from Proposition 2.21 since this shows that the action of the Hecke operators on the Fourier coefficients is identical. For the second formula, by Proposition 2.17 we may assume that $g \in \mathcal{S}\left(\frac{N}{p}, \chi\right)$. Then $g|_k \alpha_p \in \mathcal{S}_k(N, \chi)$ (recall Proposition 2.29) and the second formula follows by comparing Fourier coefficients using Proposition 2.21. Thus the Hecke operators T_q , save for $q = p$, preserve $\mathcal{S}_k^{\text{old}}(\Gamma_1(N))$. Lastly, we consider the Hecke operator T_p . For this Hecke operator, it suffices to show

$$T_p i_p(f, g) = i_p(T_p f + p^{k-1}g, -\langle p \rangle f).$$

This will follow from the formulas

$$T_p f = T_p f - (\langle p \rangle f)|_k \alpha_p \quad \text{and} \quad T_p(g|_k \alpha_p) = p^{k-1}g,$$

where the diamond and Hecke operators on the left-hand and right-hand sides are on levels N and $\frac{N}{p}$ respectively. Since $p \mid N$, Proposition 2.20 gives

$$T_p f = \sum_{j \pmod{p}} f \Big|_k \begin{pmatrix} 1 & j \\ 0 & p \end{pmatrix} = T_p f - (\langle p \rangle f)|_k \alpha_p,$$

where the second equality follows regardless of whether $p \mid \frac{N}{p}$ or not because either $\langle p \rangle$ is the zero operator on level $\frac{N}{p}$ (in the case $p \mid \frac{N}{p}$) or it annihilates the extra term in the Hecke operator T_p on level $\frac{N}{p}$ (in the case $p \nmid \frac{N}{p}$). Therefore the first formula holds. Similarly, since $p \mid N$, Proposition 2.20 gives

$$T_p(g|_k \alpha_p)(z) = \sum_{j \pmod{p}} g \Big|_k \begin{pmatrix} p & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & j \\ 0 & p \end{pmatrix} = \sum_{j \pmod{p}} g \Big|_k \begin{pmatrix} p & pj \\ 0 & p \end{pmatrix} = p^{k-1}g,$$

where the last equality follows because g is 1-periodic (as $h_\infty = 1$). This shows that the second formula holds. Thus the Hecke operator T_p preserves $\mathcal{S}_k^{\text{old}}(\Gamma_1(N))$. We have now shown that all of the diamond and Hecke operators preserve $\mathcal{S}_k^{\text{old}}(\Gamma_1(N))$. To show that they also preserve $\mathcal{S}_k^{\text{new}}(\Gamma_1(N))$, let f and g be a newform and an oldform respectively. Then for any $m \geq 1$, Proposition 2.31 gives

$$\langle \langle m \rangle f, g \rangle = \langle f, \omega_N \langle m \rangle \omega_N^{-1} g \rangle = 0 \quad \text{and} \quad \langle T_m f, g \rangle = \langle f, \omega_N T_m \omega_N^{-1} g \rangle = 0,$$

where the second equality in either identity holds because the diamond and Hecke operators preserve $\mathcal{S}_k^{\text{old}}(\Gamma_1(N))$ and so does the Atkin-Lehner operator ω_N (and hence its inverse as well) by Proposition 2.30. It follows that the diamond and Hecke operators preserve $\mathcal{S}_k^{\text{new}}(\Gamma_1(N))$ and this completes the proof. $\square$

As a corollary, we deduce that these subspaces admit orthogonal bases of eigenforms:

Corollary 2.33. $\mathcal{S}_k^{\text{old}}(\Gamma_1(N))$ and $\mathcal{S}_k^{\text{new}}(\Gamma_1(N))$ admit orthonormal bases of eigenforms.

Proof. This follows immediately from Theorem 2.27 and Proposition 2.32. $\square$

Something quite amazing happens for the subspace in $\mathcal{S}_k^{\text{new}}(\Gamma_1(N))$; the eigenforms are eigenfunctions for all of the diamond and Hecke operators so that they are Hecke eigenforms. We require a preliminary result whose proof is quite involved but it is not beyond the scope of this text (see [?] for a proof):

Lemma 2.34. *If $f \in \mathcal{S}_k(\Gamma_1(N))$ has Fourier coefficients $a_n(f)$ and is such that $a_n(f) = 0$ for all $n \geq 1$ whenever $(n, N) = 1$ then*

$$f = \sum_{p|N} p^{k-1} f_p \|_k \alpha_p,$$

for some $f_p \in \mathcal{S}_k\left(\Gamma_1\left(\frac{N}{p}\right)\right)$.

The important observation to make about Lemma 2.34 is that if $f \in \mathcal{S}_k(\Gamma_1(N))$ is such that its n -th Fourier coefficients vanish when n is relatively prime to the level then f must be an oldform. With this lemma we can prove the main theorem about $\mathcal{S}_k^{\text{new}}(\Gamma_1(N))$. The introduction of some language will be useful for the statement and its proof. We say that f is a **primitive Hecke eigenform** if it is a nonzero Hecke normalized Hecke eigenform in $\mathcal{S}_k^{\text{new}}(\Gamma_1(N))$. We can now prove the main result about newforms which is that Hecke eigenforms exist:

Theorem 2.35. *Let $f \in \mathcal{S}_k^{\text{new}}(\Gamma_1(N))$ be an eigenform. Then the following hold:*

- (i) f is a Hecke eigenform.
- (ii) If g is any cusp form with the same Hecke eigenvalues at all primes then $g = cf$ for some nonzero $c \in \mathbb{C}$.

Moreover, the primitive Hecke eigenforms in $\mathcal{S}_k^{\text{new}}(\Gamma_1(N))$ form an orthogonal basis and each such eigenform lies in an eigenspace $\mathcal{S}_k(N, \chi)$.

Proof. First suppose $f \in \mathcal{S}_k(\Gamma_1(N))$ is an eigenform with Fourier coefficients $a_n(f)$. For $m \geq 1$ with $(m, N) = 1$, there exists $\lambda_f(m), \mu_f(m) \in \mathbb{C}$ such that $T_m f = \lambda_f(m) f$ and $\langle m \rangle f = \mu_f(m) f$. Actually, $\langle m \rangle f = \mu_f(m) f$ holds for all $m \geq 1$ because $\langle m \rangle$ is the zero operator if $(m, N) > 1$ and in this case we can take $\mu_f(m) = 0$. If we set $\chi(n) = \mu_f(n)$ then χ is a Dirichlet character modulo N . This follows because multiplicativity of $\langle m \rangle$ implies the same for χ and χ is N -periodic since $\langle m \rangle$ is N -periodic ($\langle m \rangle$ is defined by m taken modulo N). But then $\langle m \rangle f = \chi(m) f$ so that $f \in \mathcal{S}_k(N, \chi)$. As f is an eigenform, we also have $a_m(f) = \lambda_f(m) a_1(f)$ provided $(m, N) = 1$. So if $a_1(f) = 0$ then Lemma 2.34 implies $f \in \mathcal{S}_k^{\text{old}}(\Gamma_1(N))$. With this fact in hand, we can prove the statements.

- (i) The claim is trivial if f is zero, so assume otherwise. If $f \in \mathcal{S}_k^{\text{new}}(\Gamma_1(N))$ then $f \notin \mathcal{S}_k^{\text{old}}(\Gamma_1(N))$ and so by what we have shown $a_1(f) \neq 0$. Therefore we may Hecke normalize f so that $a_1(f) = 1$ and $a_m(f) = \lambda_f(m)$. Now set $g_m = T_m f - \lambda_f(m)f$ for any $m \geq 1$. By Proposition 2.32, $g_m \in \mathcal{S}_k^{\text{new}}(\Gamma_1(N))$. Moreover, g_m is an eigenform and its first Fourier coefficient is zero. But then $g_m \in \mathcal{S}_k^{\text{old}}(\Gamma_1(N))$ too and so $g_m = 0$ because $\mathcal{S}_k^{\text{new}}(\Gamma_1(N))$ and $\mathcal{S}_k^{\text{old}}(\Gamma_1(N))$ are orthogonal subspaces. This means $T_m f = \lambda_f(m)f$ for any $m \geq 1$. Therefore f is a primitive Hecke eigenform and so is a Hecke eigenform before Hecke normalization.
- (ii) Suppose g has the same Hecke eigenvalues at all primes. By multiplicativity of the Hecke operators, g is a Hecke eigenform. After Hecke normalization, f and g have the same Fourier coefficients and so are identical. It follows that before Hecke normalization $f = cg$ for some nonzero $c \in \mathbb{C}$.

Note that our initial remarks together with (i) show that each primitive Hecke eigenform f belongs to some eigenspace $\mathcal{S}_k(N, \chi)$. By Corollary 2.33, $\mathcal{S}_k^{\text{new}}(\Gamma_1(N))$ admits an orthogonal basis of eigenforms which by (i) are Hecke eigenforms. As $\mathcal{S}_k^{\text{new}}(\Gamma_1(N))$ is finite dimensional (because $\mathcal{S}_k(\Gamma_1(N))$ is), it follows that all of the primitive Hecke eigenforms form an orthogonal basis for $\mathcal{S}_k^{\text{new}}(\Gamma_1(N))$ if we can show that they are linearly independent. So suppose to the contrary that we have a nontrivial linear relation

$$\sum_{1 \leq i \leq r} c_i f_i = 0,$$

for some primitive Hecke eigenforms f_i , nonzero constants c_i , and with r minimal. Note that $r \geq 2$ for else we do not have a nontrivial linear relation. Letting $m \geq 1$ applying the operator $T_m - \lambda_{f_1}(m)$ to our nontrivial linear relation gives

$$\sum_{2 \leq i \leq r} c_i (\lambda_{f_i}(m) - \lambda_{f_1}(m)) f_i = 0,$$

which has one less term. Since r was chosen to be minimal, this implies $\lambda_{f_i}(m) - \lambda_{f_1}(m) = 0$ for all i . But m was arbitrary, so $f_i = f_1$ for all i by (ii). Hence $r = 1$ which is a contradiction. $\square$

Statement (i) in Theorem 2.35 implies that primitive Hecke eigenforms satisfy the Hecke relations for all $n, m \geq 1$. Statement (ii) is known as **multiplicity one** for modular forms and can be interpreted as saying that a basis of newforms for $\mathcal{S}_k^{\text{new}}(\Gamma_1(N))$ contains one element per set of eigenvalues for the Hecke operators. We will now discuss conjugate cusp forms. For any $f \in \mathcal{S}_k(N, \chi)$, we define the **conjugate** f^σ of f by

$$f^\sigma(z) = \overline{f(-\bar{z})}.$$

Note that if f has Fourier coefficients $a_n(f)$ then f^σ has Fourier coefficients $\overline{a_n(f)}$. It turns out that f^σ is indeed a cusp form and behaves well with respect to the Hecke operators:

Proposition 2.36. *If $f \in \mathcal{S}_k(N, \chi)$ then $f^\sigma \in \mathcal{S}_k(N, \bar{\chi})$. Moreover,*

$$T_m f^\sigma = (T_m f)^\sigma,$$

for all $m \geq 1$ with $(m, N) = 1$. In particular, if f is an eigenform with Hecke eigenvalues $\lambda_f(m)$ then f^σ is too with Hecke eigenvalues $\overline{\lambda_f(m)}$.

Proof. Holomorphy is clear so we need next need to verify modularity. For this, let $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma_0(N)$ and set $\gamma' = \begin{pmatrix} a & -b \\ -c & d \end{pmatrix} \in \Gamma_0(N)$. Then $-\overline{\gamma z} = \gamma'(-\overline{z})$ and $j(\gamma', -\overline{z}) = j(\gamma, \overline{z})$ for all $z \in \mathbb{H}$. We also have $\chi(\gamma') = \chi(\gamma)$. Using these fact, we compute

$$\begin{aligned}
(f^\sigma|_k \gamma)(z) &= j(\gamma, z)^{-k} f^\sigma(\gamma z) \\
&= \overline{j(\gamma, \overline{z})^{-k} f(-\overline{\gamma z})} \\
&= \overline{j(\gamma, \overline{z})^{-k} f(\gamma'(-\overline{z}))} \\
&= \overline{\chi(\gamma) f(-\overline{z})} && \text{modularity} \\
&= \overline{\chi(\gamma)} f^\sigma(z).
\end{aligned}$$

This proves modularity of f^σ . The growth condition follows immediately from that of f and thus f^σ is a cusp form since f is. This proves the first statement. The second statement is immediate from Proposition 2.23. For the last statement, suppose $f \in \mathcal{S}_k(N, \chi)$ is an eigenform with Hecke eigenvalues $\lambda_f(m)$. Then by what we have already shown,

$$(T_m f^\sigma)(z) = ((T_m f)^\sigma)(z) = \overline{T_m f(-\overline{z})} = \overline{\lambda_f(m) f(-\overline{z})} = \overline{\lambda_f(m)} f^\sigma(z),$$

which completes the proof. $\square$

In conjunction with Theorem 2.35, Proposition 2.36 implies that the primitive Hecke eigenforms in $\mathcal{S}_k^{\text{new}}(\Gamma_1(N))$ are conjugate invariant and if $f \in \mathcal{S}_k(N, \chi)$ is such an eigenform then $f^\sigma \in \mathcal{S}_k(N, \overline{\chi})$ is as well. The crucial fact we need is how $\omega_N f$ is related to f^σ when f is a primitive Hecke eigenform:

Proposition 2.37. *If $f \in \mathcal{S}_k(N, \chi)$ is a primitive Hecke eigenform then*

$$\omega_N f = \omega_N(f) f^\sigma,$$

where $f^\sigma \in \mathcal{S}_k(N, \overline{\chi})$ is a primitive Hecke eigenform and $\omega_N(f) \in \mathbb{C}$ is nonzero with $|\omega_N(f)| = 1$.

Proof. Let f have Hecke eigenvalues $\lambda_f(m)$. On the one hand, Theorem 2.35 and Proposition 2.36 together imply that f^σ is a primitive Hecke eigenform with Hecke eigenvalues $\overline{\lambda_f(m)}$. On the other hand, Proposition 2.31 implies $\omega_N T_m = T_m^* \omega_N$ for all $m \geq 1$. Then

$$\langle T_m \omega_N f, \omega_N f \rangle = \langle \omega_N f, T_m^* \omega_N f \rangle = \langle \omega_N f, \omega_N T_m f \rangle = \langle \omega_N f, \lambda_f(m) \omega_N f \rangle = \overline{\lambda_f(m)} \langle \omega_N f, \omega_N f \rangle,$$

and it follows that $T_m \omega_N f = \overline{\lambda_f(m)} \omega_N f$. In other words, $\omega_N f$ is a Hecke eigenform with Hecke eigenvalues $\overline{\lambda_f(m)}$. Then multiplicity one gives

$$\omega_N f = \omega_N(f) f^\sigma,$$

for some nonzero $\omega_N(f) \in \mathbb{C}$. Actually, by Proposition 2.30 we see that ω_N is at most of order 4 so necessarily $|\omega_N(f)| = 1$. $\square$

2.7 The Ramanujan-Petersson Conjecture

We will now discuss a famous conjecture about the size of the Hecke eigenvalues of primitive Hecke eigenforms. Historically the conjecture was born from conjectures made about the **modular discriminant** Δ given by

$$\Delta = \frac{1}{1728}(E_4^3 - E_6^2),$$

which is a primitive Hecke eigenform in $\mathcal{S}_{12}(1)$ (see [?] for a proof). Therefore it is natural to begin our discussion here. It can be shown that the Fourier series of the modular discriminant is

$$\Delta(z) = \sum_{n \geq 1} \tau(n) e^{2\pi i n z},$$

where the $\tau(n)$ are integers with $\tau(1) = 1$ and $\tau(2) = -24$ (see [?] for a proof). The function $\tau : \mathbb{Z}_{\geq 1} \rightarrow \mathbb{Z}$ is called **Ramanujan's τ function**. Ramanujan himself studied this function in his 1916 paper (see [?]), and computed $\tau(n)$ for $1 \leq n \leq 30$. From these computations he conjectured that τ should satisfy the following three properties:

- (i) If $(n, m) = 1$ then $\tau(nm) = \tau(n)\tau(m)$.
- (ii) $\tau(p^n) = \tau(p^{n-1})\tau(p) - p^{11}\tau(p^{n-2})$ for all primes p .
- (iii) $|\tau(p)| \leq 2p^{\frac{11}{2}}$ for all primes p .

Note that (i) means τ is multiplicative. Moreover, (i) and (ii) are strikingly similar to the properties satisfied by the Hecke operators. In fact, (i) and (ii) are follow from the fact that Δ is a Hecke eigenform. Property (iii) turned out to be drastically more difficult to prove and is essentially the Ramanujan-Petersson conjecture for the modular discriminant Δ . To state the Ramanujan-Petersson conjecture for modular forms, suppose $f \in \mathcal{S}_k(N, \chi)$ is a primitive Hecke eigenform with Hecke eigenvalues $\lambda_f(m)$. For each prime p , consider the polynomial

$$1 - \lambda_f(p)p^{-\frac{k-1}{2}}p^{-s} + \chi(p)p^{-2s}.$$

We call this the p -th **Hecke polynomial** of f . We call the roots $\alpha_1(p)$ and $\alpha_2(p)$ the p -th **Hecke roots** of f . From this quadratic, we have

$$\alpha_1(p) + \alpha_2(p) = \lambda_f(p)p^{-\frac{k-1}{2}} \quad \text{and} \quad \alpha_1(p)\alpha_2(p) = \chi(p).$$

Then the more general **Ramanujan-Petersson conjecture** for modular forms is following statement:

Theorem (Ramanujan-Petersson conjecture, modular). *Let $f \in \mathcal{S}_k(N, \chi)$ be a primitive Hecke eigenform with Hecke eigenvalues $\lambda_f(m)$ and let $\alpha_1(p)$ and $\alpha_2(p)$ be the p -th Hecke roots of f . Then for all primes p , we have*

$$|\lambda_f(p)| \leq 2p^{\frac{k-1}{2}}.$$

Moreover, if $p \nmid N$ then

$$|\alpha_1(p)| = |\alpha_2(p)| = 1.$$

In the 1970's Deligne proved the Ramanujan-Petersson conjecture (see [?] and [?] for the full proof). The argument is significantly beyond the scope of this text, and in actuality follows from Deligne's work on the Weil conjectures (except in the case $k = 1$ which requires a modified argument). This requires understanding classical algebraic topology and ℓ -adic cohomology in addition to number theory. As such, the proof of the Ramanujan-Petersson conjecture has been one of the biggest advances in number theory in recent decades. From the Hecke relations and the Ramanujan-Petersson conjecture, we have the more general bound $\lambda_f(m) \ll \sigma_0(m)m^{\frac{k-1}{2}}$. Then ?? gives the slightly weaker estimate $\lambda_f(m) \ll_\varepsilon m^{\frac{k-1}{2}+\varepsilon}$.

2.8 Twists of Modular Forms

We can also twist modular forms by Dirichlet characters. Let $f \in \mathcal{S}_k(N, \chi)$ have Fourier series

$$f(z) = \sum_{n \geq 1} a_n(f) e^{2\pi i n z},$$

and let ψ be a Dirichlet character modulo M . We define the **twist** $f \otimes \psi$ of f by ψ by the Fourier series

$$(f \otimes \psi)(z) = \sum_{n \geq 1} a_n(f) \psi(n) e^{2\pi i n z}.$$

It turns out that $f \otimes \psi$ is a modular form. The following proposition proves this and more in the case ψ is primitive:

Proposition 2.38. *Suppose $f \in \mathcal{S}_k(N, \chi)$ and ψ is a primitive Dirichlet character of conductor q . Then $f \otimes \psi \in \mathcal{S}_k(Nq^2, \chi\psi^2)$.*

Proof. By ??, we can write

$$\begin{aligned} (f \otimes \psi)(z) &= \sum_{n \geq 1} a_n(f) \psi(n) e^{2\pi i n z} \\ &= \sum_{n \geq 1} a_n(f) \left(\frac{1}{\tau(\bar{\psi})} \sum_{r \pmod{q}} \bar{\psi}(r) e^{\frac{2\pi i r n}{q}} \right) e^{2\pi i n z} \\ &= \frac{1}{\tau(\bar{\psi})} \sum_{r \pmod{q}} \bar{\psi}(r) \sum_{n \geq 1} a_n(f) e^{2\pi i n (z + \frac{r}{q})} \\ &= \frac{1}{\tau(\bar{\psi})} \sum_{r \pmod{q}} \bar{\psi}(r) f\left(z + \frac{r}{q}\right). \end{aligned}$$

From this last expression, holomorphy is immediate since the sum is finite. For modularity, let $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma_0(Nq^2)$ and set $\gamma_r = \begin{pmatrix} 1 & \frac{r}{q} \\ 0 & 1 \end{pmatrix}$ for every r modulo q . Then for r and r' taken modulo q , we compute

$$\gamma_r \gamma_{r'}^{-1} = \begin{pmatrix} 1 & \frac{r}{q} \\ 0 & 1 \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} 1 & -\frac{r'}{q} \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} a + \frac{cr}{q} & b - \frac{ar' - dr}{q} - \frac{crr'}{q^2} \\ c & d - \frac{cr'}{q} \end{pmatrix}.$$

Since $c \equiv 0 \pmod{Nq^2}$, if we choose r' (for each r) such that $ar' \equiv dr \pmod{q}$ then $\gamma_r \gamma \gamma_r^{-1} \in \Gamma_0(N)$. Such a choice exists and is unique by Bézout's identity because a and d are relatively prime to q as $ad \equiv 1 \pmod{Nq^2}$. Making this choice and setting $\eta_r = \gamma_r \gamma \gamma_{r'}^{-1}$, modularity implies

$$f\left(\gamma z + \frac{r}{q}\right) = f(\gamma_r \gamma z) = f(\eta_r \gamma_{r'} z) = \chi(\eta_r) j(\eta_r, \gamma_{r'} z)^k f(\gamma_{r'} z) = \chi(\eta_r) j(\eta_r, \gamma_{r'} z)^k f\left(z + \frac{r'}{q}\right).$$

Moreover,

$$\chi(\eta_r) j(\eta_r, \gamma_{r'} z) = \chi\left(d - \frac{cr'}{q}\right) \left(c\gamma_{r'} z + d - \frac{cr'}{q}\right) = \chi(d)(cz + d) = \chi(\gamma) j(\gamma, z).$$

These two computations together imply

$$f\left(\gamma z + \frac{r}{q}\right) = \chi(\gamma) j(\gamma, z)^k f\left(z + \frac{r'}{q}\right).$$

Now, as $ar' \equiv dr \pmod{q}$ and $ad \equiv 1 \pmod{q}$, we have

$$\bar{\psi}(r) = \bar{\psi}(adr') = \psi^2(d) \bar{\psi}(r') = \psi^2(\gamma) \bar{\psi}(r').$$

Putting everything together,

$$\begin{aligned} (f \otimes \psi)(\gamma z) &= \frac{1}{\tau(\bar{\psi})} \sum_{r \pmod{q}} \bar{\psi}(r) f\left(\gamma z + \frac{r}{q}\right) \\ &= \chi(\gamma) j(\gamma, z)^k \frac{1}{\tau(\bar{\psi})} \sum_{r' \pmod{q}} \bar{\psi}(r) f\left(z + \frac{r'}{q}\right) \\ &= \chi \psi^2(\gamma) j(\gamma, z)^k \frac{1}{\tau(\bar{\psi})} \sum_{r' \pmod{q}} \bar{\psi}(r') f\left(z + \frac{r'}{q}\right) \\ &= \chi \psi^2(\gamma) j(\gamma, z)^k (f \otimes \psi)(z). \end{aligned}$$

from which modularity of $f \otimes \psi$ follows. For the growth condition, let $\sigma_{\mathbf{a}}$ be a scaling matrix for a cusp $\mathbf{a}$ of $\Gamma_0(Nq^2) \backslash \mathbb{H}$. As $\gamma_r \in \mathrm{GL}_2^+(\mathbb{Q})$, $\gamma_r \sigma_{\mathbf{a}}$ takes ∞ to an element of $\mathbb{Q}$. Thus $\gamma_r \sigma_{\mathbf{a}} \infty = \mathbf{b}$ for some cusp $\mathbf{b}$ of $\Gamma_0(N) \backslash \mathbb{H}$. Then as $j(\gamma_r, \sigma_{\mathbf{a}} z) = 1$, our previous work implies

$$((f \otimes \psi)|_k \sigma_{\mathbf{a}})(z) = \frac{1}{\tau(\bar{\psi})} \sum_{r \pmod{q}} \bar{\psi}(r) j(\gamma_r \sigma_{\mathbf{a}}, z)^{-k} f(\gamma_r \sigma_{\mathbf{a}} z),$$

and so the growth condition follows from that of f . Thus $f \otimes \psi$ is a cusp form since f is. $\square$

The generalization of Proposition 2.38 to all characters is slightly more involved. To this end, define operators U_p and V_p on $\mathcal{S}_k(\Gamma_1(N))$ to be the linear operators given by

$$(U_p f)(z) = \sum_{n \geq 1} a_{np}(f) e^{2\pi i n z},$$

and

$$(V_p f)(z) = \sum_{n \geq 1} a_n(f) e^{2\pi i n p z},$$

if f has Fourier series

$$f(z) = \sum_{n \geq 1} a_n(f) e^{2\pi i n z}.$$

We will show that both U_p and V_p map $\mathcal{S}_k(\Gamma_1(N))$ into $\mathcal{S}_k(\Gamma_1(Np))$ and more:

Lemma 2.39. *For any prime p , U_p and V_p map $\mathcal{S}_k(\Gamma_1(N))$ into $\mathcal{S}_k(\Gamma_1(Np))$. In particular, U_p and V_p map $\mathcal{S}_k(N, \chi)$ into $\mathcal{S}_k(Np, \chi\chi_{p,0})$.*

Proof. In light of Proposition 2.17, the first statement follows from the second. As $N \mid Np$, $\Gamma_1(Np) \subseteq \Gamma_1(N)$ so that $f \in \mathcal{S}_k(\Gamma_1(Np))$ if $f \in \mathcal{S}_k(\Gamma_1(N))$. Now suppose $f \in \mathcal{S}_k(N, \chi)$. Similarly, $N \mid Np$ implies $\Gamma_0(Np) \subseteq \Gamma_0(N)$ so that $f \in \mathcal{S}_k(Np, \chi\chi_{p,0})$ for the modulus Np character $\chi\chi_{p,0}$. Therefore we may assume $f \in \mathcal{S}_k(Np, \chi\chi_{p,0})$. Now consider U_p . As $p \mid Np$, Equation (21) implies $U_p = T_p$ is the p -th Hecke operator on $\mathcal{S}_k(\Gamma_1(Np))$ and the claim follows from the definition of the Hecke operators and Proposition 2.21. Now consider V_p . We have

$$(V_p f)(z) = f(pz),$$

and the claim follows by regarding $f \in \mathcal{S}_k(Np, \chi\chi_{p,0})$ and that I_p lies in the center of $\mathrm{SL}_2(\mathbb{Z})$. $\square$

We can now generalize Proposition 2.38 to all characters:

Proposition 2.40. *Suppose $f \in \mathcal{S}_k(N, \chi)$ and ψ is a Dirichlet character modulo M . Then $f \otimes \psi \in \mathcal{S}_k(NM^2, \chi\psi^2)$.*

Proof. Let $\tilde{\psi}$ be the primitive character of conductor q inducing ψ . Then $\psi = \tilde{\psi}\psi_{\frac{M}{q},0}$. As $\psi_{\frac{M}{q},0} = \prod_{p \mid \frac{M}{q}} \psi_{p,0}$, it suffices to prove the claim in the cases when ψ is primitive and when $\psi = \psi_{p,0}$. The primitive case follows from Proposition 2.38. So suppose $\psi = \psi_{p,0}$. Then

$$f \otimes \psi_{p,0} = f - V_p U_p f.$$

Now by Lemma 2.39, $V_p U_p f \in \mathcal{S}_k(Np^2, \chi\psi_{p,0}^2)$ (where we have written $\psi_{p,0}^2$ in place of $\chi_{p,0}$). Since we also have $f \in \mathcal{S}_k(Np^2, \chi\psi_{p,0}^2)$ (because $N \mid Np^2$ so that $\Gamma_1(Np^2) \subseteq \Gamma_1(N)$ and $\Gamma_0(Np^2) \subseteq \Gamma_0(N)$ and again writing $\psi_{p,0}^2$ in place of $\chi_{p,0}$), it follows that $f \otimes \psi_{p,0} \in \mathcal{S}_k(Np^2, \chi\psi_{p,0}^2)$. This proves the claim in the case $\psi = \psi_{p,0}$ and thus completes the proof. $\square$

In particular, Proposition 2.40 shows that $f \otimes \psi$ is well-defined for any Dirichlet character ψ .